NEOPLASIA

Chapter 8: p.163

Learning Objectives

- Define cancer
- Identify the cancers with the highest incidence and mortality rates in Canada
- Compare and contrast **malignant** and **benign** growths
- Learn how to name different types of growths
- Describe the cellular characteristics found in malignant growths
- Understand the **genetic mutations** necessary for cancer to develop
- Identify some causes of cancer (including infections)

Basic Terminology

Neoplasia

Tumour literally means "swelling" and is now primarily used to denote abnormal growth of tissue (not necessarily *cancer*)

Neoplasia is new, uncontrolled growth of cells that lacks normal regulation over cell division, creating a **neoplasm** (benign or malignant)

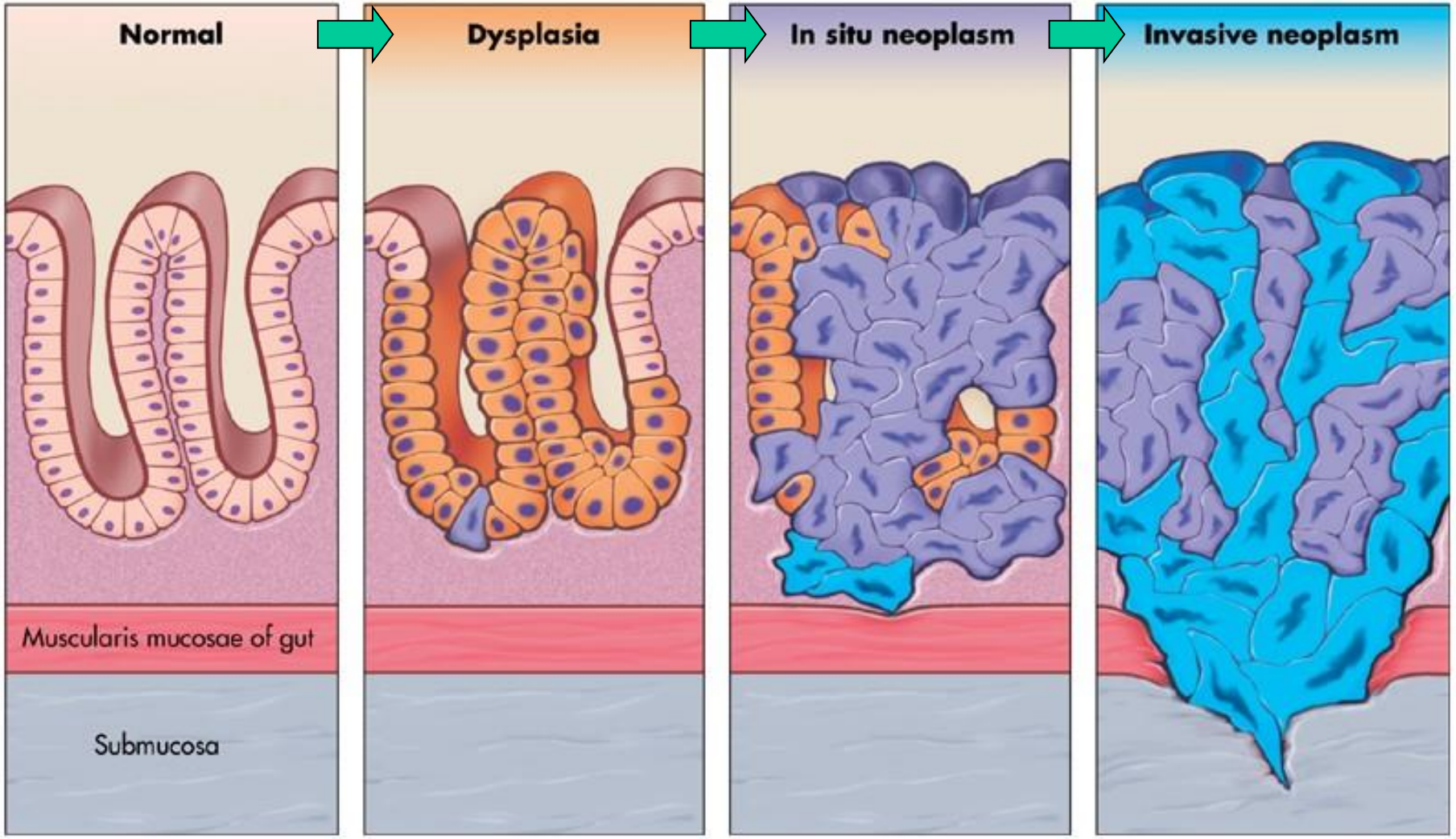
Neoplasia

Cancer is a *class of diseases* characterized by altered differentiation and **uncontrolled division of cells**

Caused by *an accumulation* of genetic mutations (to be discussed later)

The "outcome" of this would be a **malignant growth**

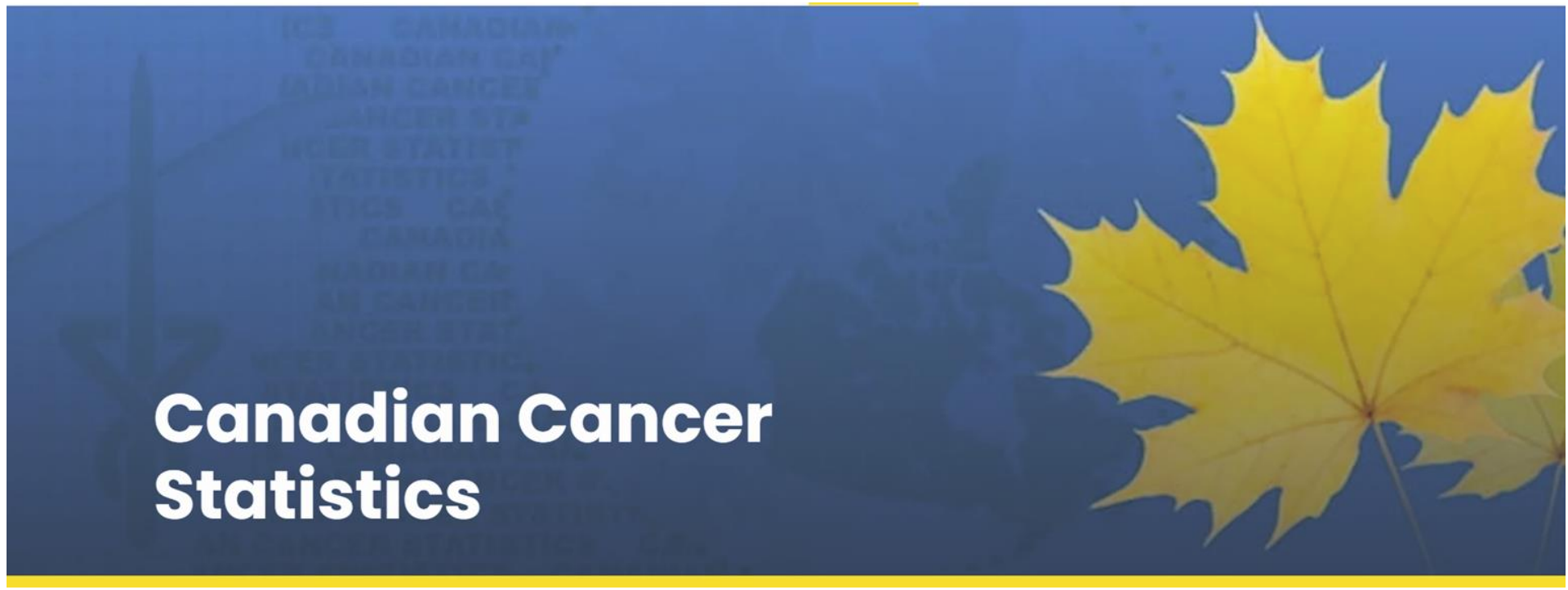
hyperplasia



Modified from Stevens A, Lowe J: *Pathology: illustrated review in color,* ed 2, London, 2000, Mosby.

Mosby items and derived items copyright © 2004, 2000 by Mosby, Inc.

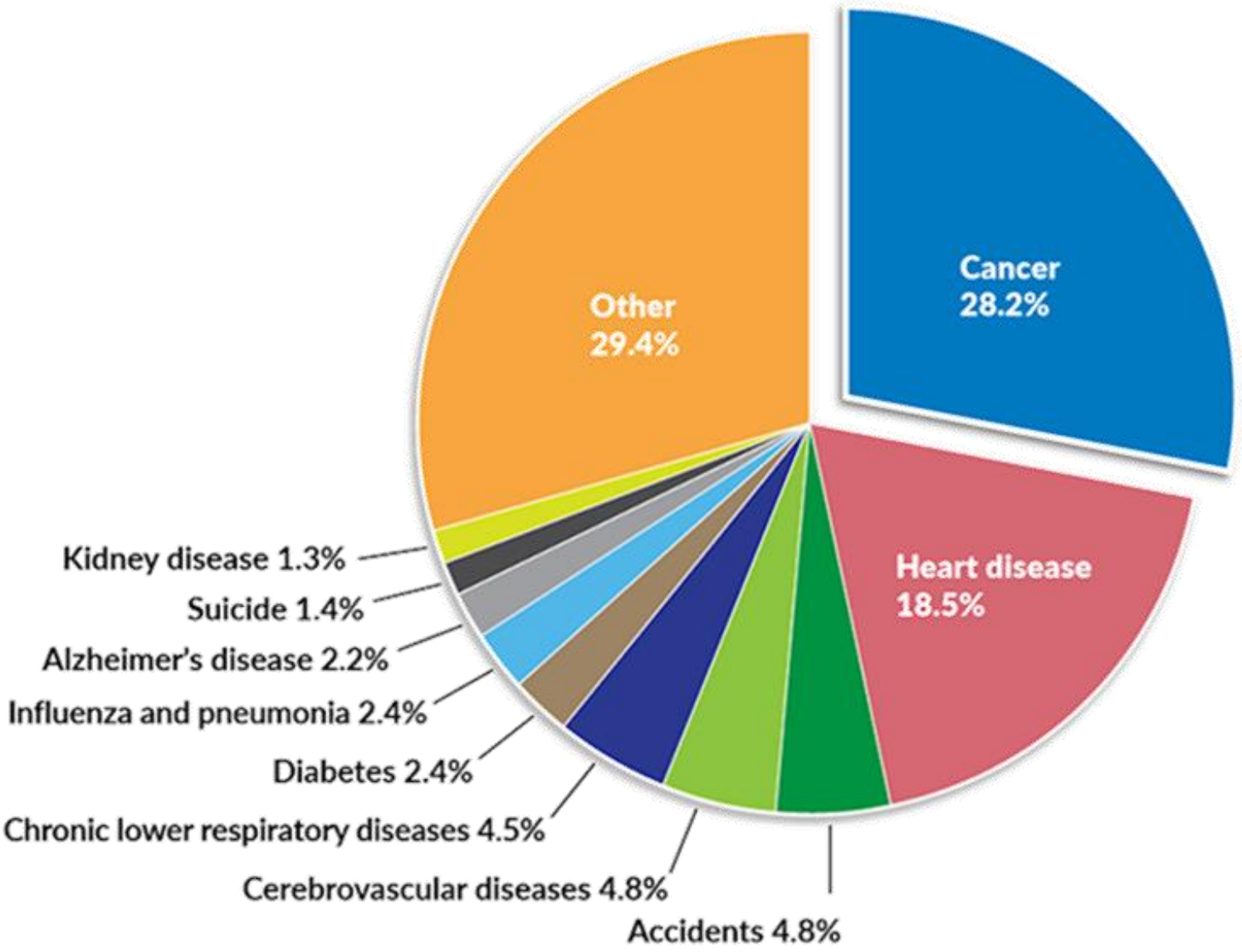
Canadian Cancer Statistics



Canadian Cancer Society
Société canadienne du cancer

Canadian Cancer Society

Government of Canada / Gouvernement du Canada



Canadian Cancer Society

Government of Canada / Gouvernement du Canada

To be deleted:

Canadian Cancer Society

Government of Canada / Gouvernement du Canada

FIGURE 1.1 Lifetime probability of developing cancer, Canada (excluding Quebec*), 2017

To be deleted:

Canadian Cancer Society

Government of Canada / Gouvernement du Canada

Cancer-specific stats 2022

Snapshot of incidence, mortality and survival estimates by cancer type*

Males
118,200
New cases

Prostate	20.3%
Lung and bronchus	12.5%
Colorectal	11.6%
Bladder	8.0%
Non-Hodgkin lymphoma	5.2%
Head and neck	4.6%
Kidney and renal pelvis	4.4%
Melanoma	4.0%
Leukemia	3.4%
Pancreas	3.1%
Stomach	2.2%
Liver	2.2%
Multiple myeloma	1.9%
Esophagus	1.6%
Brain/CNS	1.5%
Thyroid	1.5%
Testis	1.0%
Hodgkin lymphoma	0.5%
Breast	0.2%
All other cancers	10.3%

Females
110,900
New cases

Breast	25.0%
Lung and bronchus	13.3%
Colorectal	10.0%
Uterus (body, NOS)	7.2%
Non-Hodgkin lymphoma	4.5%
Thyroid	4.4%
Melanoma	3.6%
Bladder	2.7%
Pancreas	2.7%
Ovary	2.7%
Leukemia	2.4%
Kidney and renal pelvis	2.3%
Head and neck	1.8%
Multiple myeloma	1.4%
Cervix	1.3%
Stomach	1.3%
Brain/CNS	1.2%
Liver	0.7%
Esophagus	0.5%
Hodgkin lymphoma	0.4%

Both sexes combined	Incidence			Mortality			Survival
	Rank	Cases	Trend	Rank	Deaths	Trend	5-year (%)
All cancers	—	**233,900**	↓	—	**85,100**	↓	**64**
Lung and bronchus	1	30,000	↓	1	20,700	↓	22
Breast	2	28,900	↓	4	5,500	↓	89
Prostate	3	24,600	↓	5	4,600	↓	91
Colorectal	4	24,300	↓	2	9,400	↓	67
Bladder	5	13,300	→	8,9	2,500	→	77
Non-Hodgkin lymphoma	6	11,400	→	7	3,000	↓	69
Melanoma	7	9,000	↑	18	1,200	↓	89
Uterus (body, NOS)	8,9	8,100	→	17	1,500	↑	82
Kidney and renal pelvis	8,9	8,100	→	13,14	1,950	↓	73
Head and neck	10	7,500	↑	11	2,100	→	64
Pancreas	11	6,900	→	3	5,700	→	10
Leukemia	12,13	6,700	↓	6	3,100	↓	61
Thyroid	12,13	6,700	↓	20	250	→	97
Stomach	14	4,100	↓	12	2,000	↓	29
Multiple myeloma	15	4,000	↑	15,16	1,650	↓	50
Liver	16	3,500	→	15,16	1,650	→	22
Brain/CNS	17	3,200	↓	8,9	2,500	↓	22
Ovary	18	3,000	↓	13,14	1,950	↓	44
Esophagus	19	2,500	↓	10	2,400	→	16
Cervix	20	1,450	↓	19	380	→	74
Testis	21	1,200	↑	22	35	↓	97
Hodgkin lymphoma							

Make note of the **FIVE** most common cancers for both incidence and prevalence in Canada (outside of non-melanoma skin cancer)? **Can you find them?**

FIGURE 1.3 Percentage of new cases and age-specific incidence rates for all cancers, by age group and sex, Canada (excluding Quebec*), 2015–2017

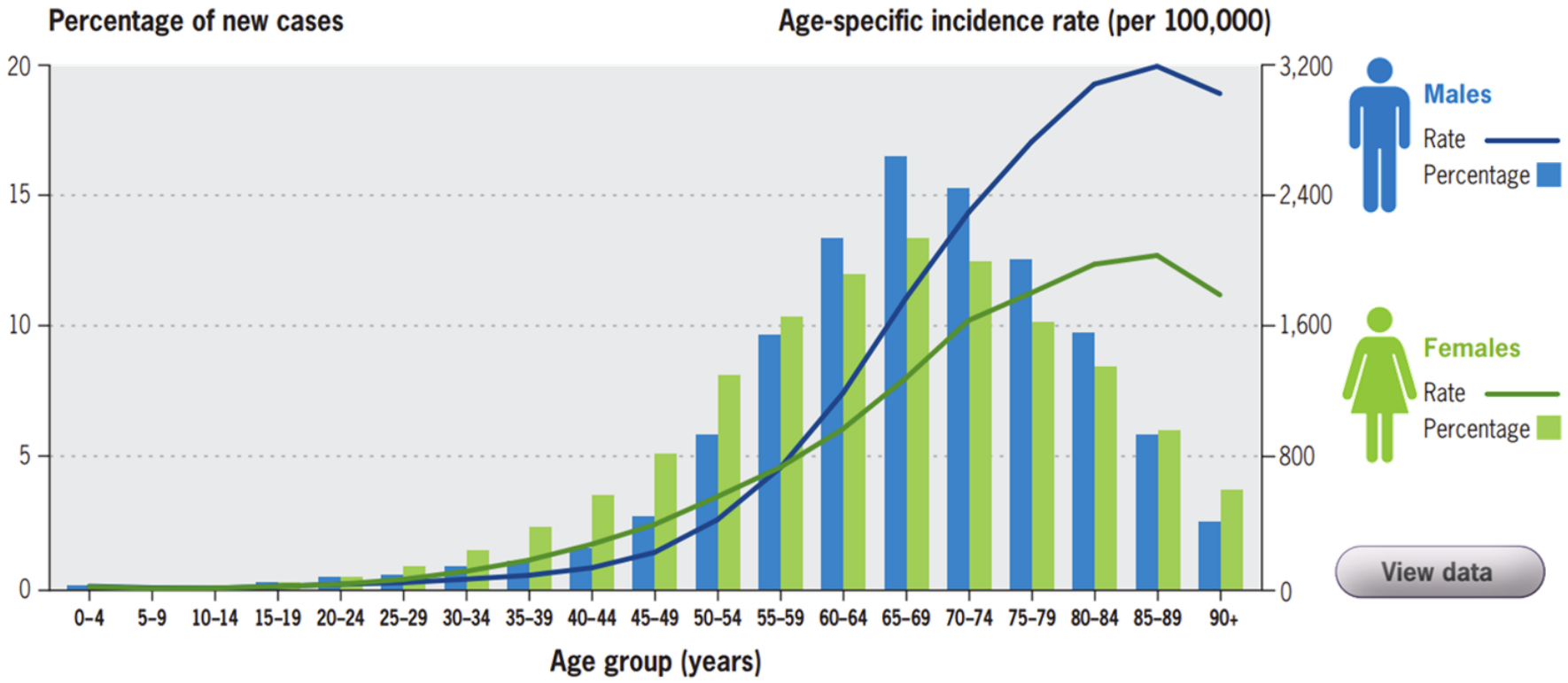
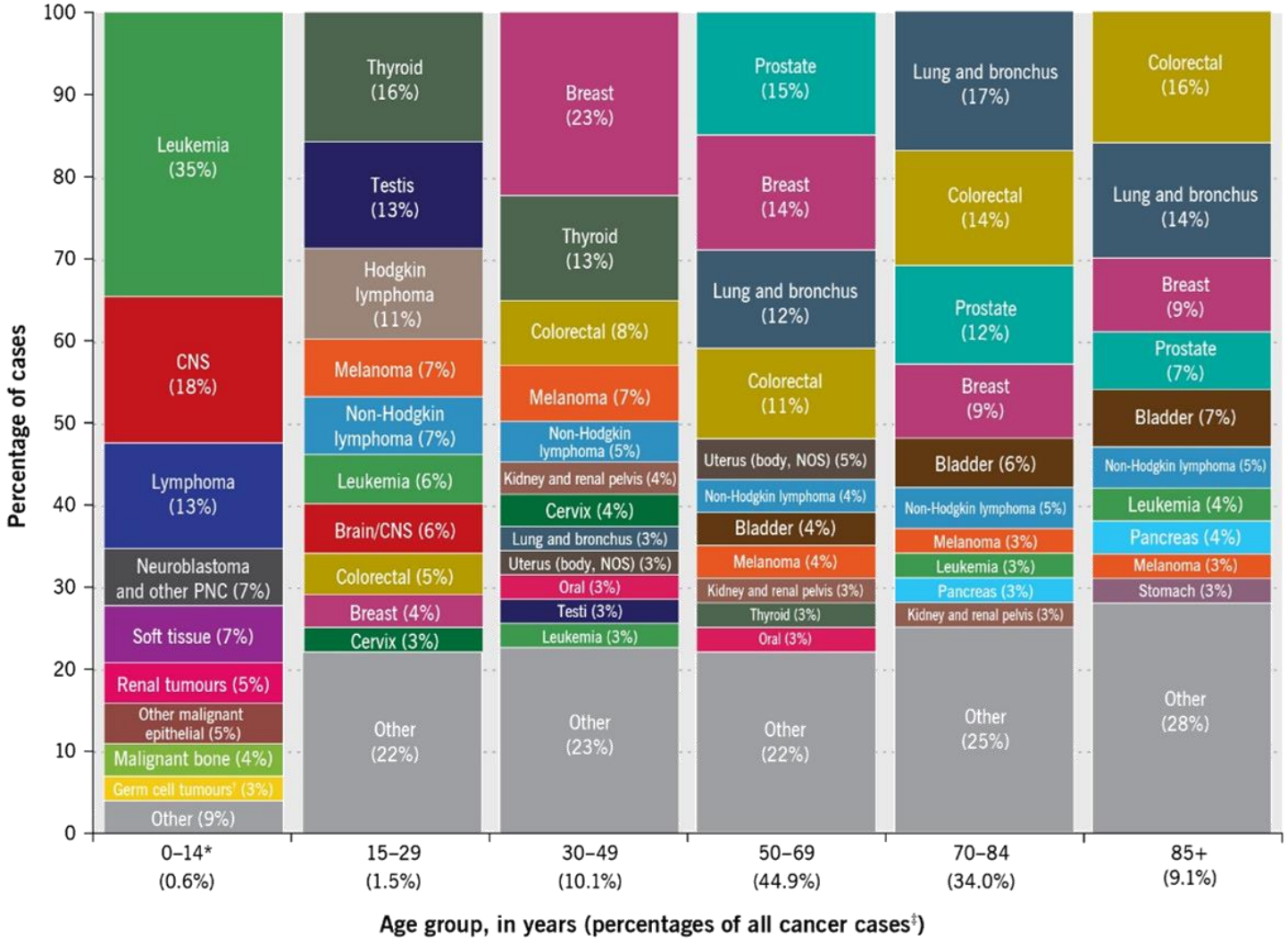


FIGURE 1.4 Distribution of new cancer cases for selected cancers, by age group, Canada (excluding Quebec), 2011–2015



CNS=central nervous system; PNC=peripheral nervous cell tumours; NOS=not otherwise specified

Classifying Neoplasms

"Benign" (NOT "CANCER")

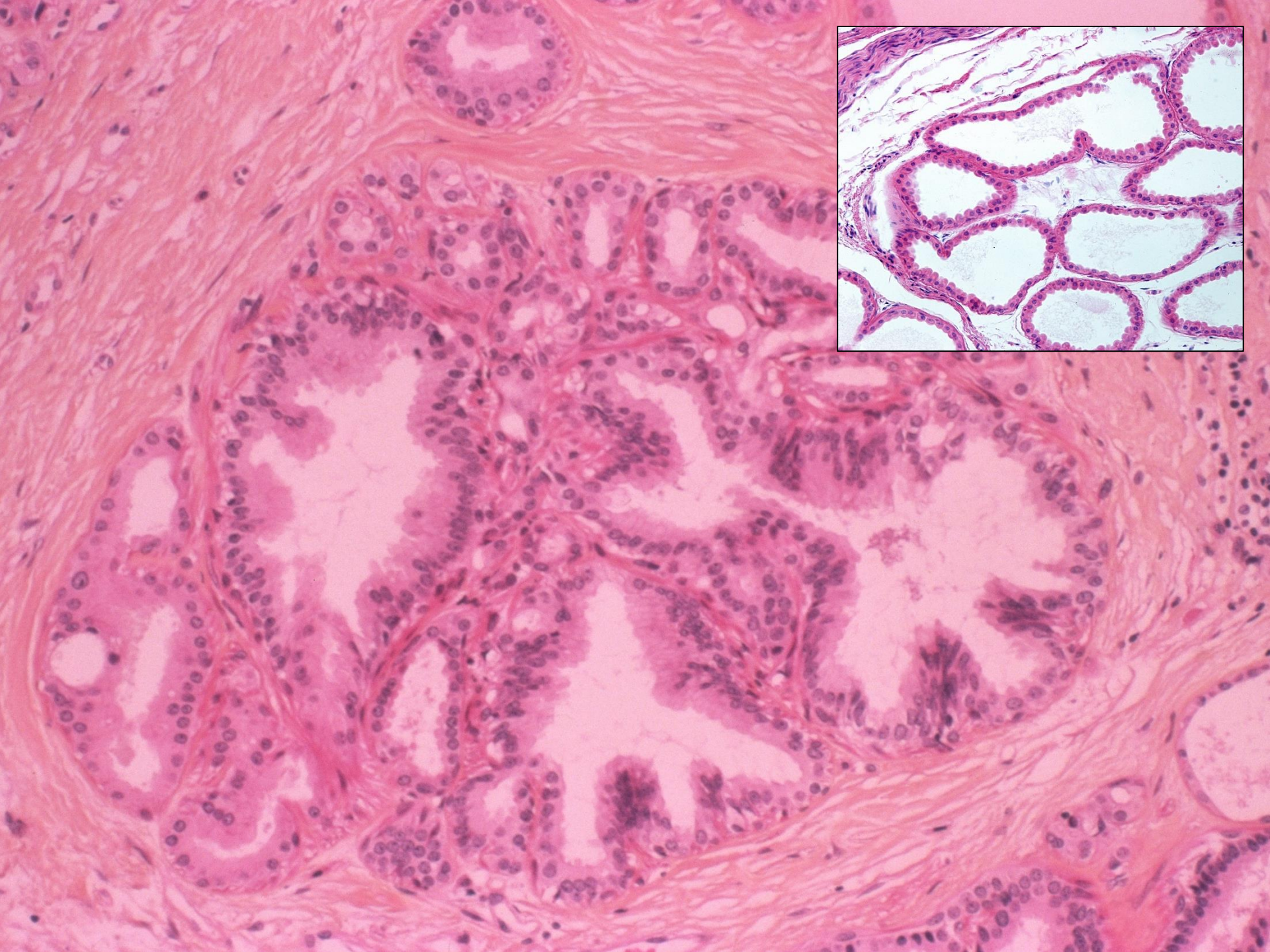
- tumours that resemble the tissue of origin
- lost the ability to control proliferation

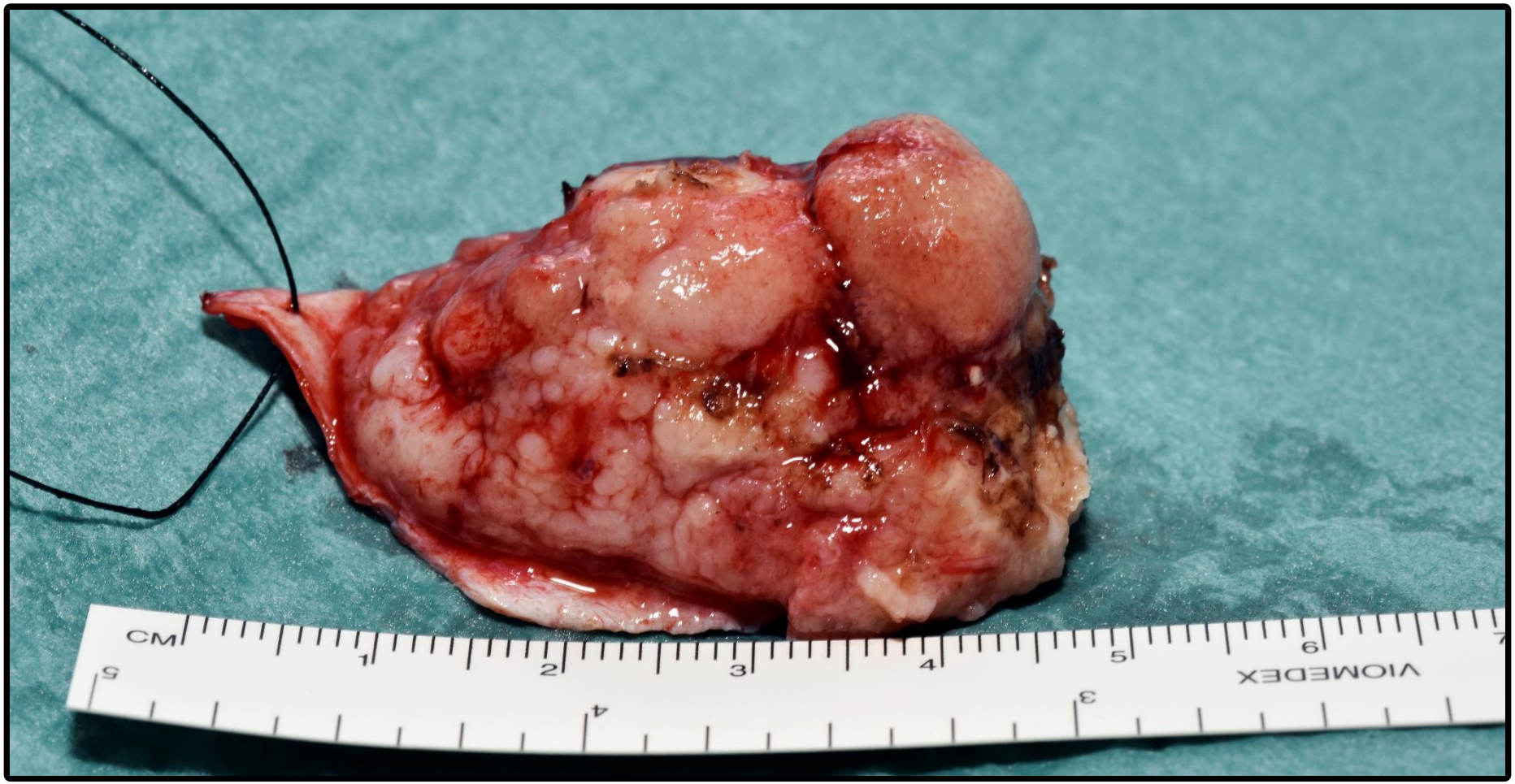
"Malignant"

- undifferentiated tumours that ***no longer appear or function*** like the original tissue
- lost the ability to control proliferation and differentiation
- tumours are aggressive and invasive

Benign Neoplasms

- **Cells may still perform their normal function**
- Tissue becomes autonomous and does not respond well to regular control mechanisms (*negative feedback*)
  - ex. islet cell tumours - produce excessive amounts of insulin
- May develop a rim of compressed connective tissue (**fibrous capsule**)

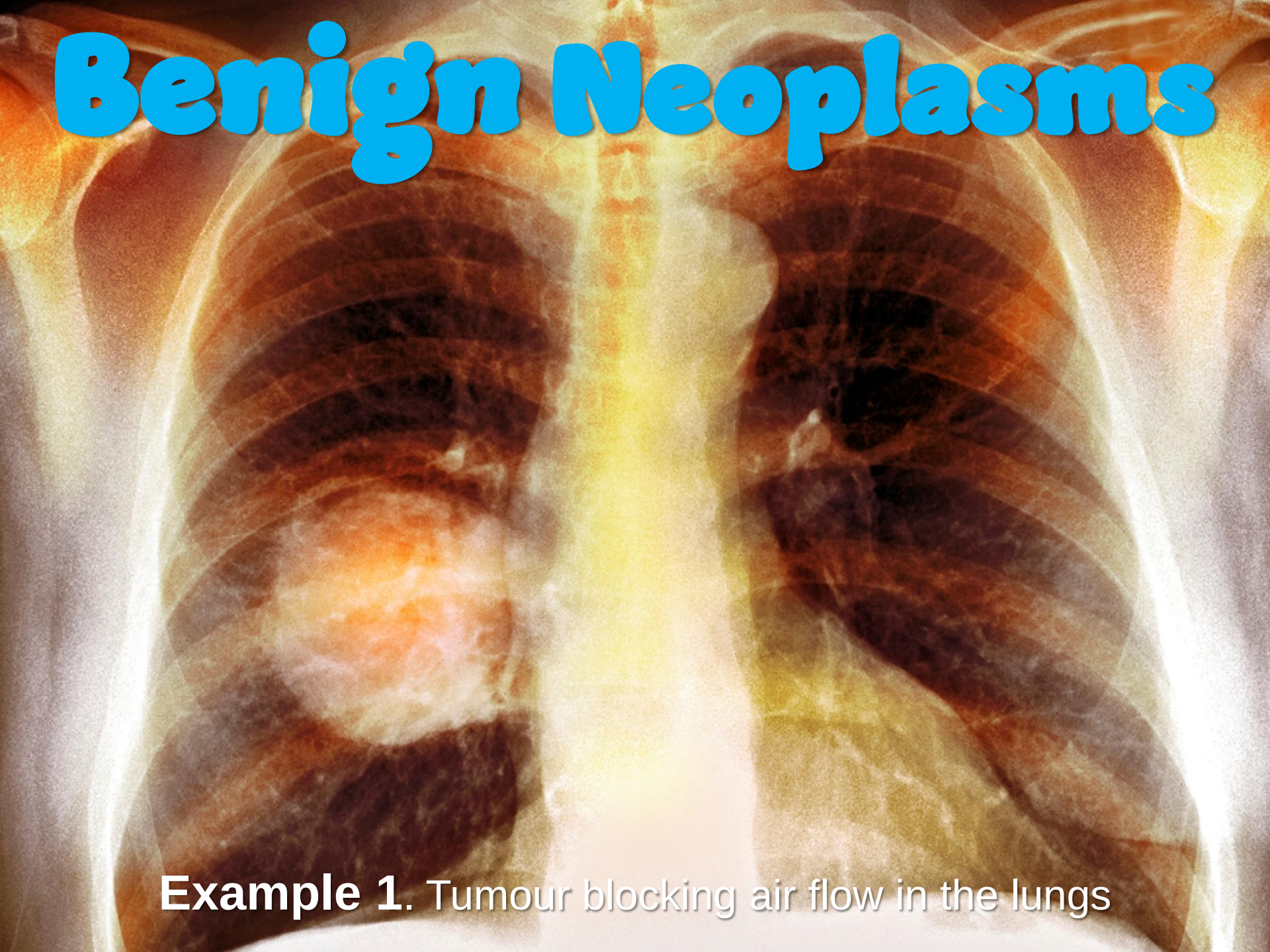




Critical Thinking: ϟ Brainstorm ϟ

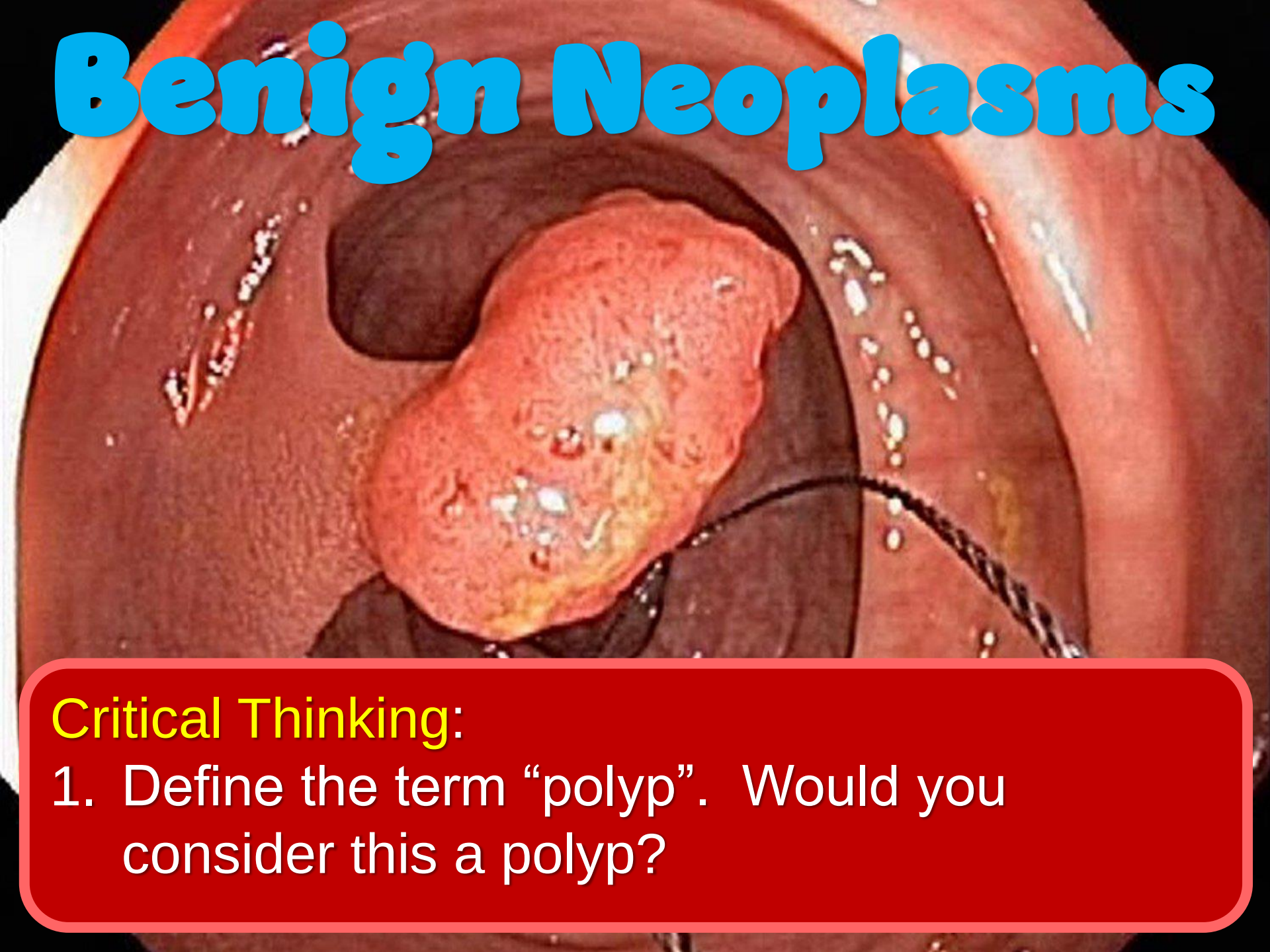
1. Pause the video and come up with **FOUR** concerns regarding benign tumours

Benign Neoplasms



Example 1. Tumour blocking air flow in the lungs

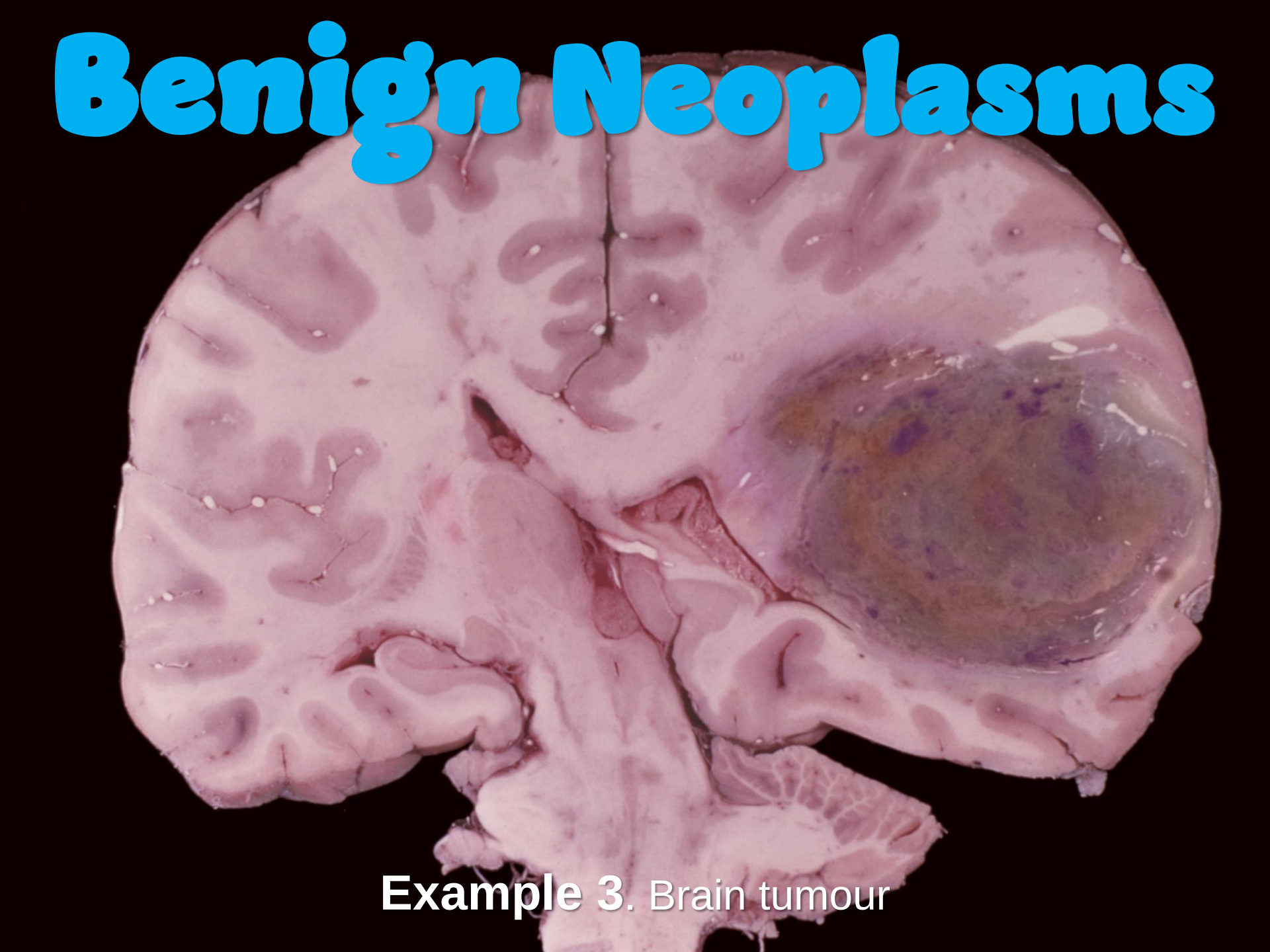
Benign Neoplasms



Critical Thinking:

1. Define the term “polyp”. Would you consider this a polyp?

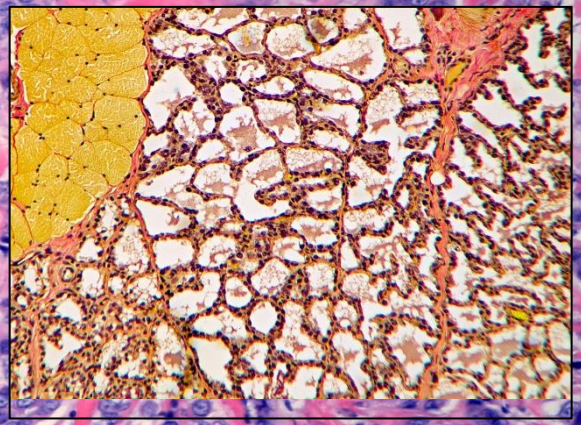
Benign Neoplasms



Example 3. Brain tumour

Malignant Neoplasms

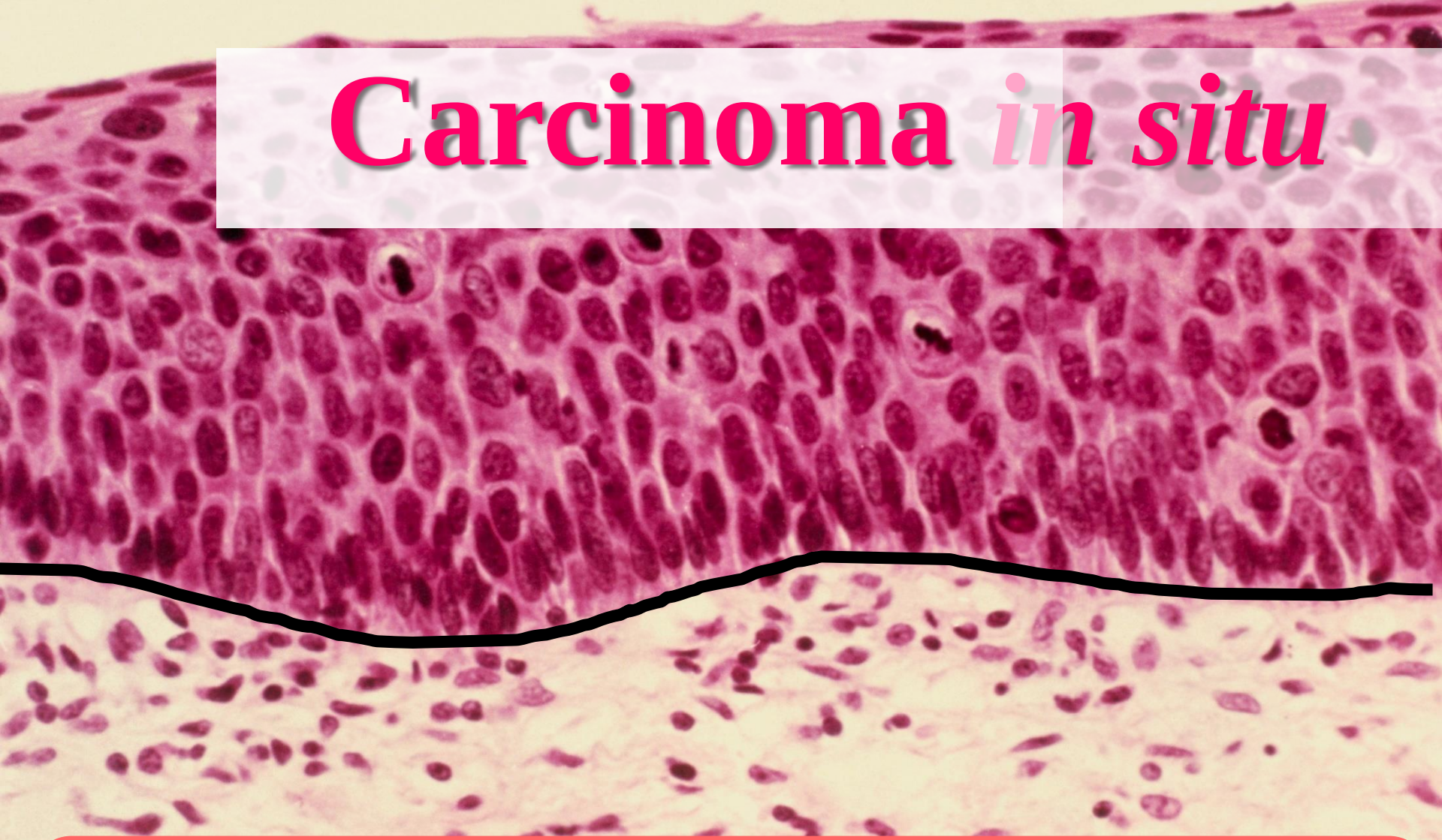
- *Block in differentiation pathway* traps cells in immature, highly **proliferative** state
  - typically lose normal function
  - may secrete other chemicals that cause local or systemic changes/damage (*more on this later*)
  - similar local effects as benign neoplasms due to rapid rate of growth (compress tissue and blood vessels)
  - ***more aggressive*** and _____ than benign tumours



Malignant Neoplasms

Malignant Neoplasms

Carcinoma *in situ*



Critical Thinking:

1. Is this a malignant growth?

Critical Thinking:

1. Is this a carcinoma *in situ*?

IMPORTANT

Benign

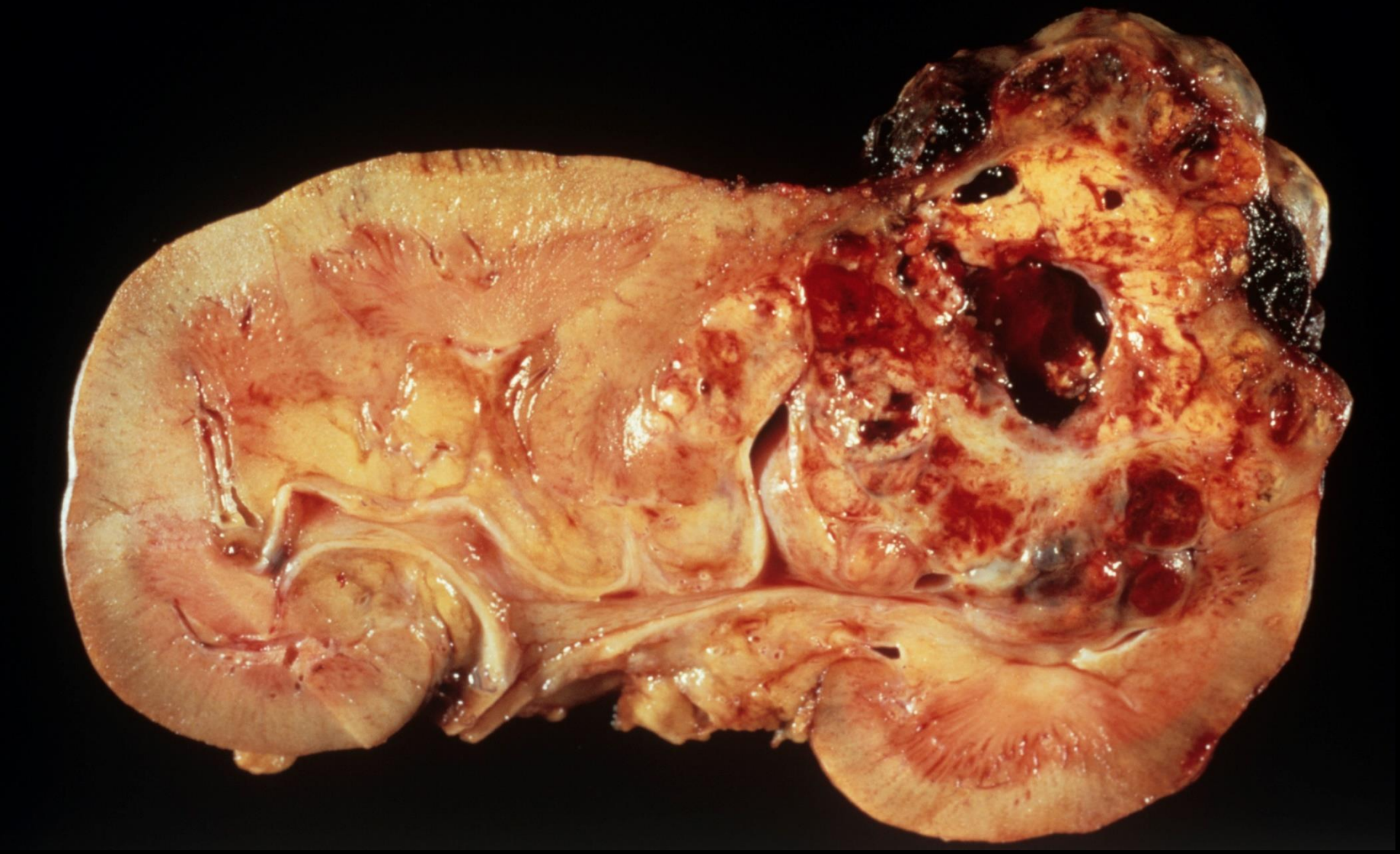
- usually encapsulated
- no tissue invasion
- no metastases
- resemble normal tissue (differentiated)
- normal nuclei
- slow growth
- few new blood vessels
- little recurrence
- rarely necrotic areas

Malignant

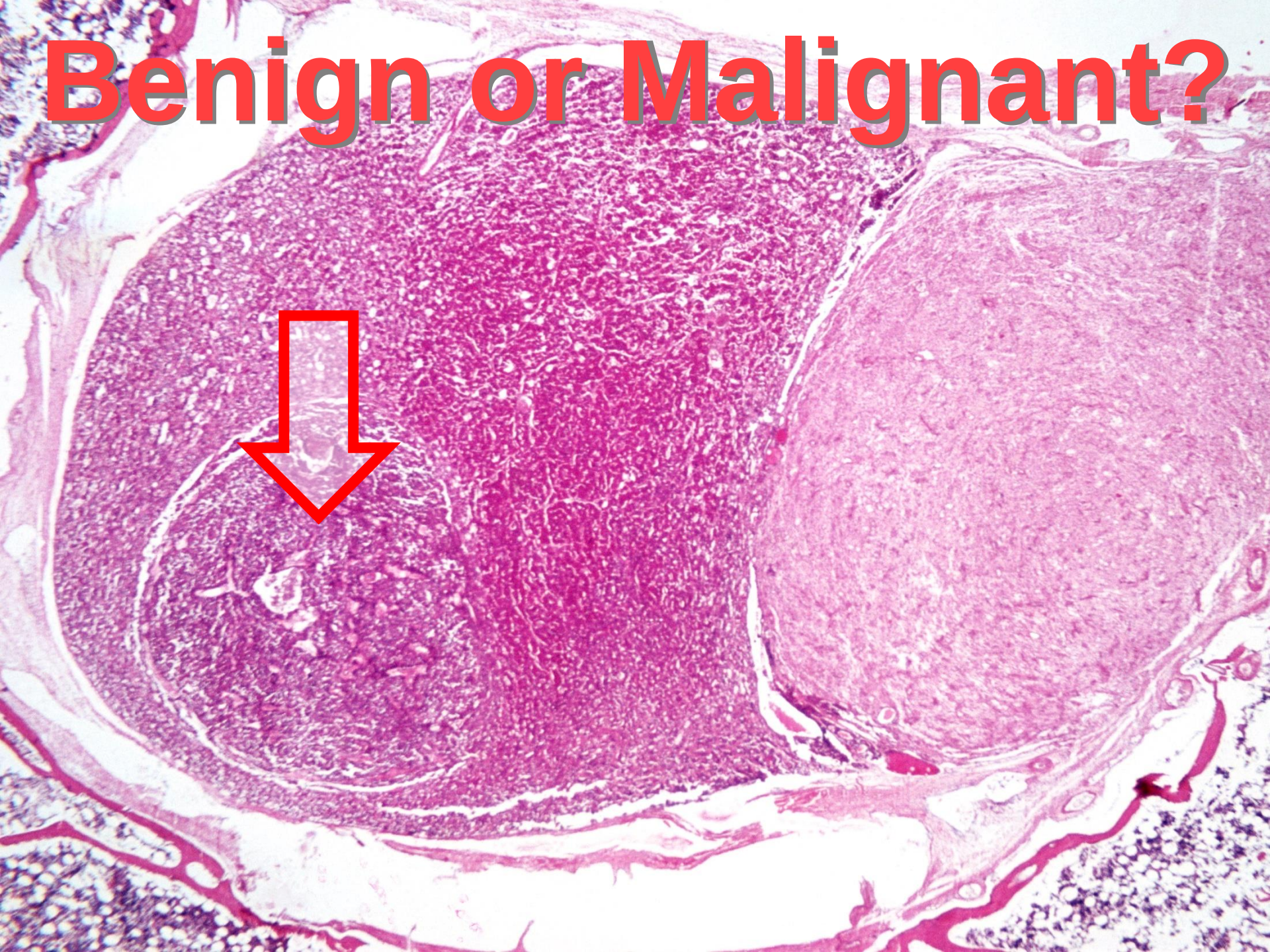
- ill defined borders
- tissue invasion
- metastases
- **anaplasia** (undifferentiated)
- pleomorphic
- nuclear irregularities
- fast growth
-
- recurrence if removed
- necrotic areas common

Benign or Malignant?

Benign or Malignant?

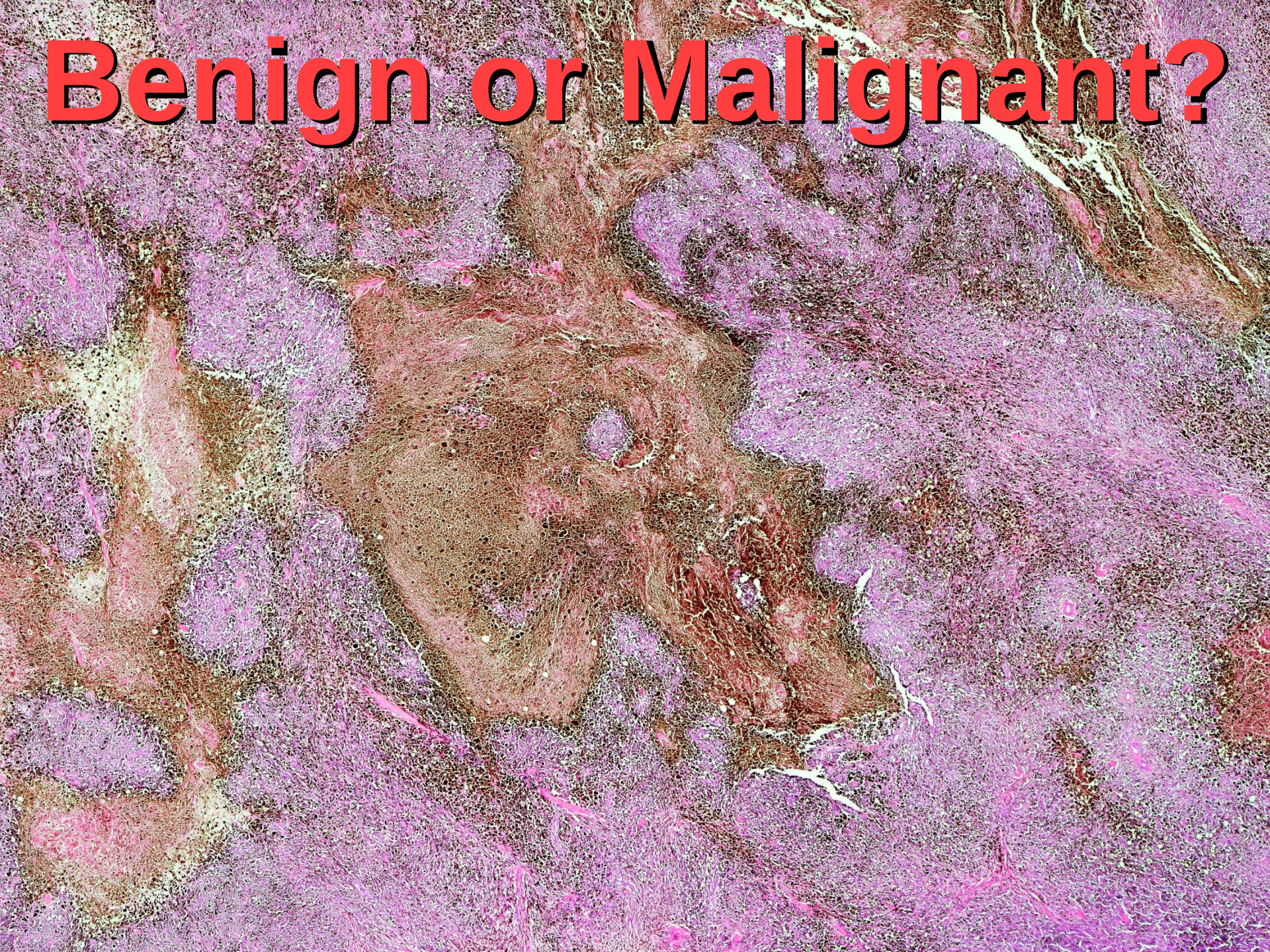


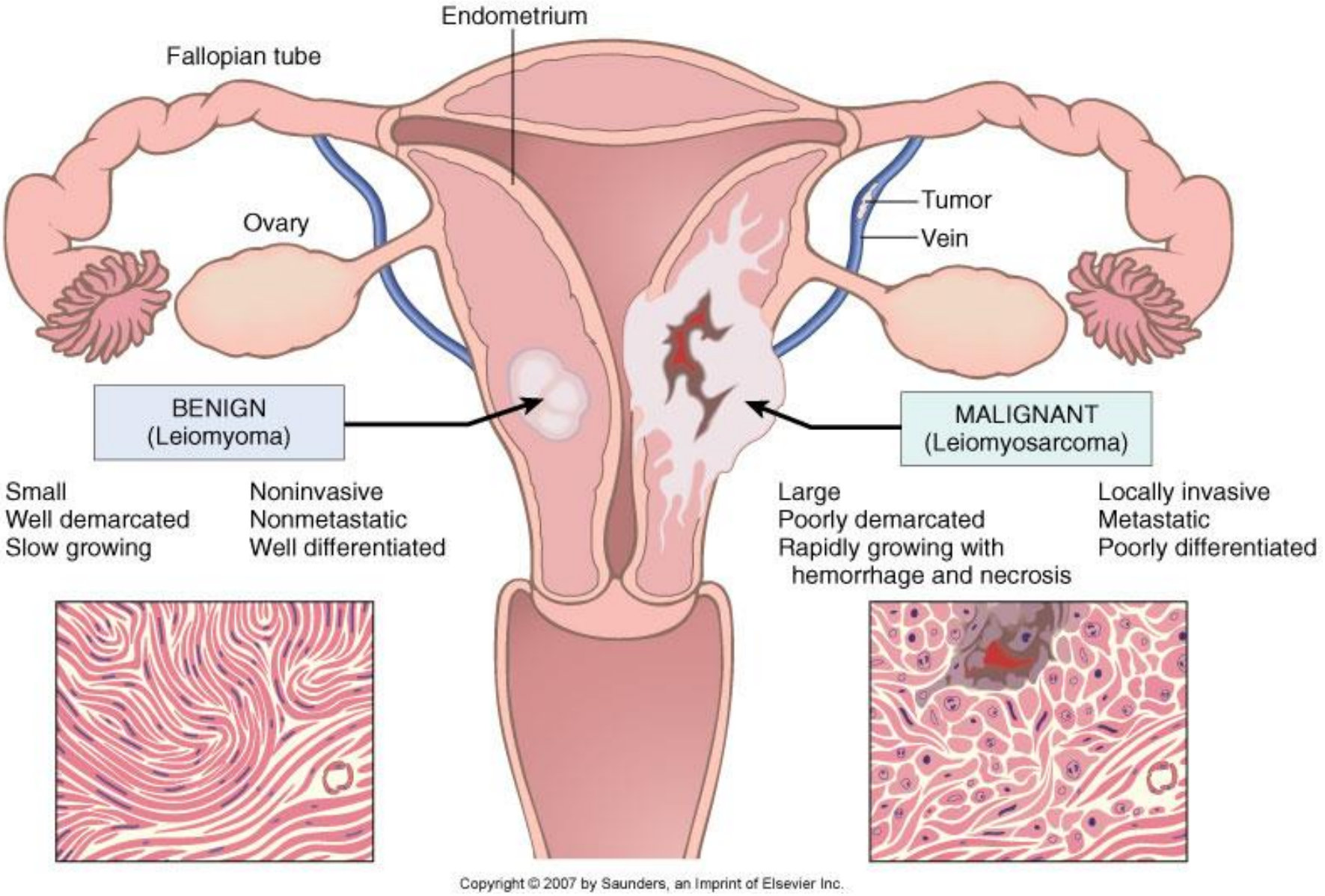
Benign or Malignant?



Benign or Malignant?

Benign or Malignant?





Comparison between a **benign** tumour of the myometrium (leiomyoma) and a **malignant** tumour of similar origin (leiomyosarcoma).

Naming Neoplasms

Prefix	Tissue of Origin
“adeno-”	Gland or duct
“fibro-”	Fibrous tissue
“lipo-”	Fat
“chondro-”	Cartilage
“osteo-”	Bone
“leiomyo-”	Smooth muscle
“rhabdomyo-”	Skeletal/cardiac muscle
“neuro-”	Neuron
“glio-”	Glial tissue

- **Benign**
  - "-oma"
  - ex. osteoma (bone)

- **Malignant**
  - "-carcinoma" (**epithelial**)
  - approx. of cancers
  - "-sarcoma" (**mesenchymal**)
  - nerve, muscle, connective tissue
  - ex. osteosarcoma

TISSUE TYPE	BENIGN TUMORS	MALIGNANT TUMORS
Epithelial		
Surface	Papilloma	Squamous cell carcinoma
Glandular	Adenoma	Adenocarcinoma
Connective		
Fibrous	Fibroma	Fibrosarcoma
Adipose	Lipoma	Liposarcoma
Cartilage	Chondroma	Chondrosarcoma
Bone	Osteoma	Osteosarcoma
Blood vessels	Hemangioma	Hemangiosarcoma
Muscle		
Smooth	Leiomyoma	Leiomyosarcoma
Striated	Rhabdomyoma	Rhabdomyosarcoma
Neural Tissue		
Nerve cell	Neuroma	Neuroblastoma
Glial tissue	Glioma	Glioblastoma, astrocytoma, medulloblastoma, oligodendroglioma
Nerve sheaths	Neurilemmoma	Neurilemmal sarcoma
Meninges	Meningioma	Meningeal sarcoma
Hematologic (next semester)		
Granulocytic		Myelocytic leukemia
Erythrocytic		Erythrocytic leukemia
Plasma cells		Multiple myeloma
Lymphocytic		Lymphocytic leukemia or lymphoma
Monocytic		Monocytic leukemia

Table 8.1 p.170

Naming Exceptions

- **melan*oma*** (**malignant** melanocytes)
- **leukemia** (bone marrow **malignancy**)
- **lymph*oma*** (**malignant** lymph node)
- (not a neoplasm)

LOST? CONFUSED?

GO WATCH A **BONUS VIDEO** ON NAMING NEOPLASMS (POSTED ON OUR BB SITE)

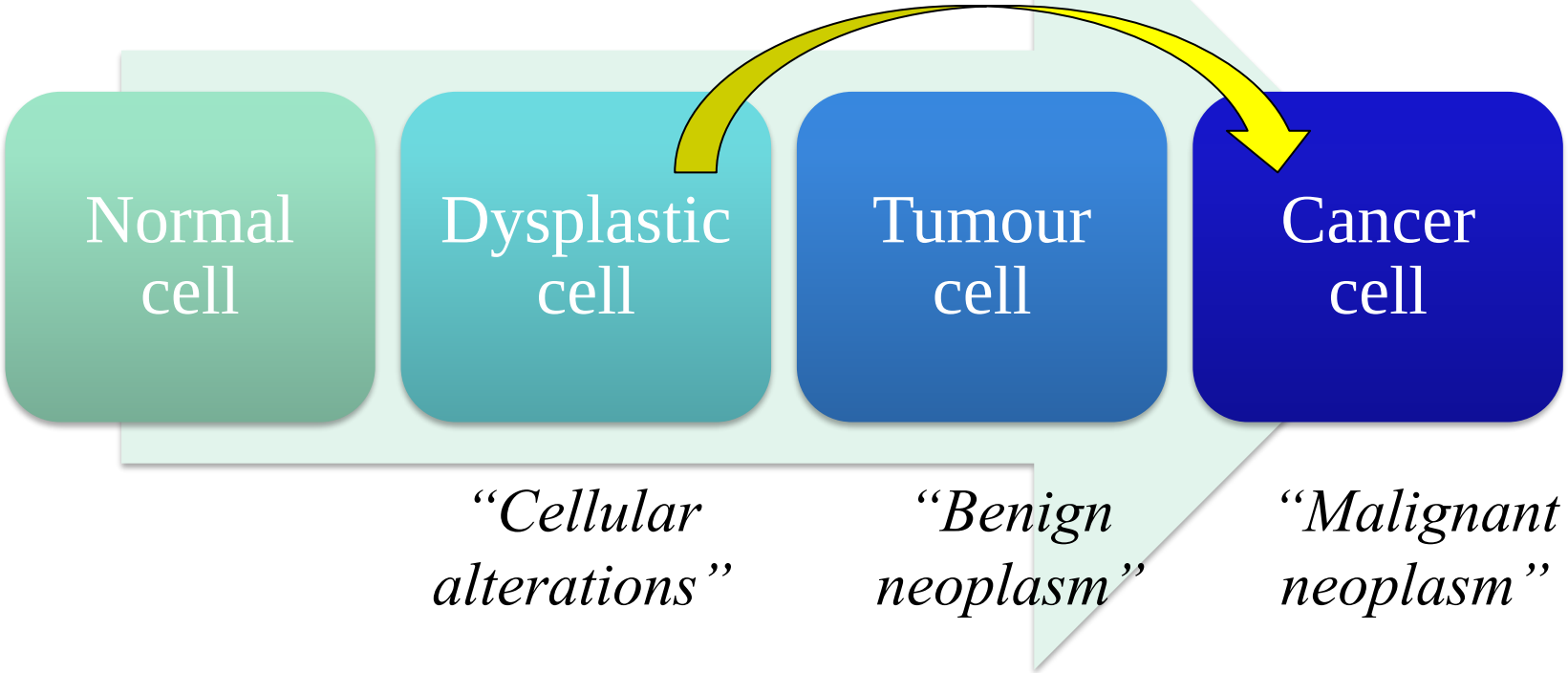
There's a "self-test" at the end!

CHARACTERISTICS *of* CANCER CELLS

Chapter 8: p.171

Cell Transformation

- Process by which a normal cell becomes a cancer cell (also known as **carcinogenesis**)
  - results from **genetic mutations** (discussed later)
  - anatomical and physiological abnormalities beyond simply an increase in proliferation

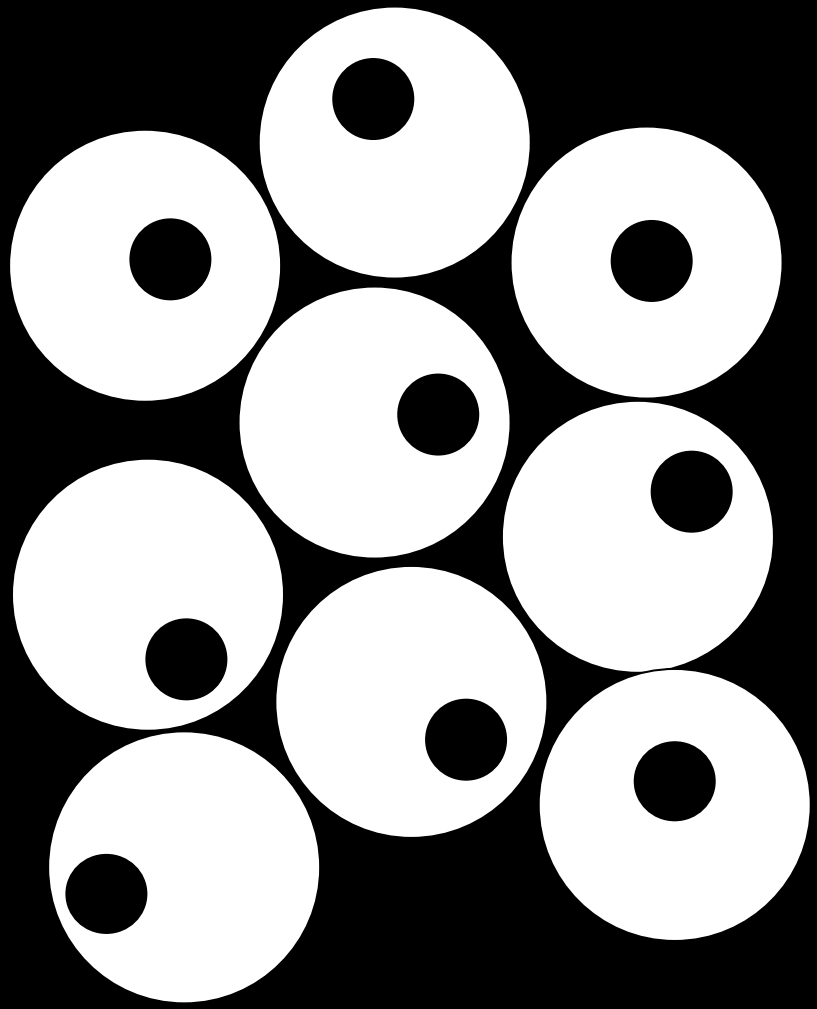


Appearance of Cancer Cells

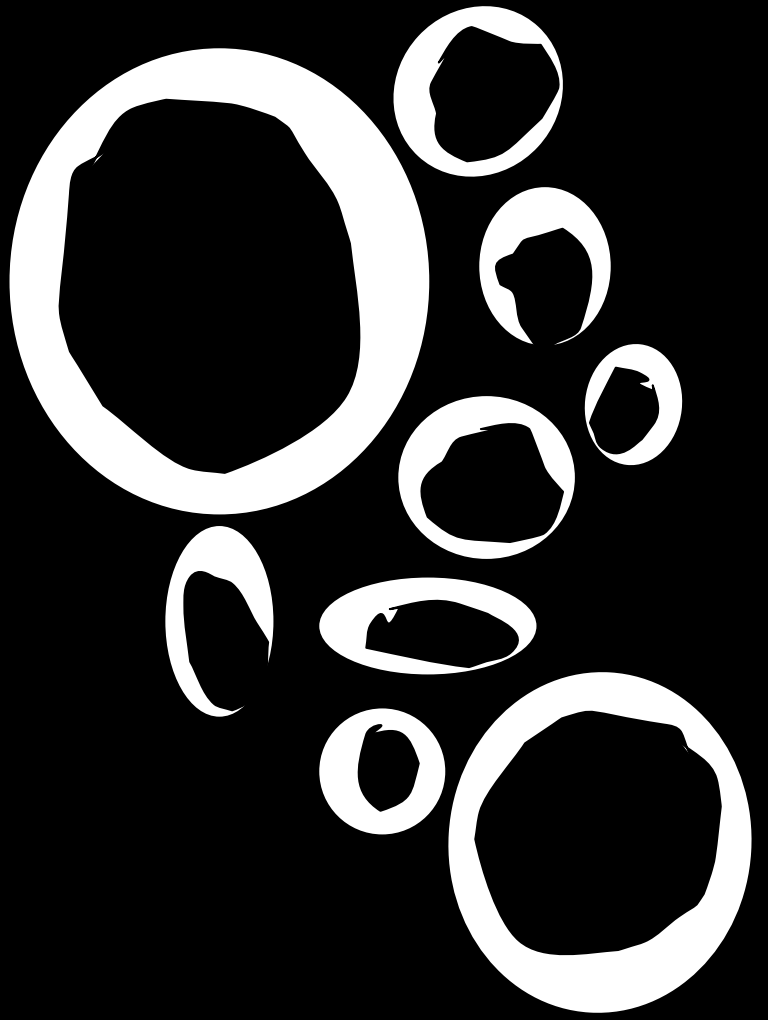
"Pre-Cancerous" Alterations (Dysplasia):

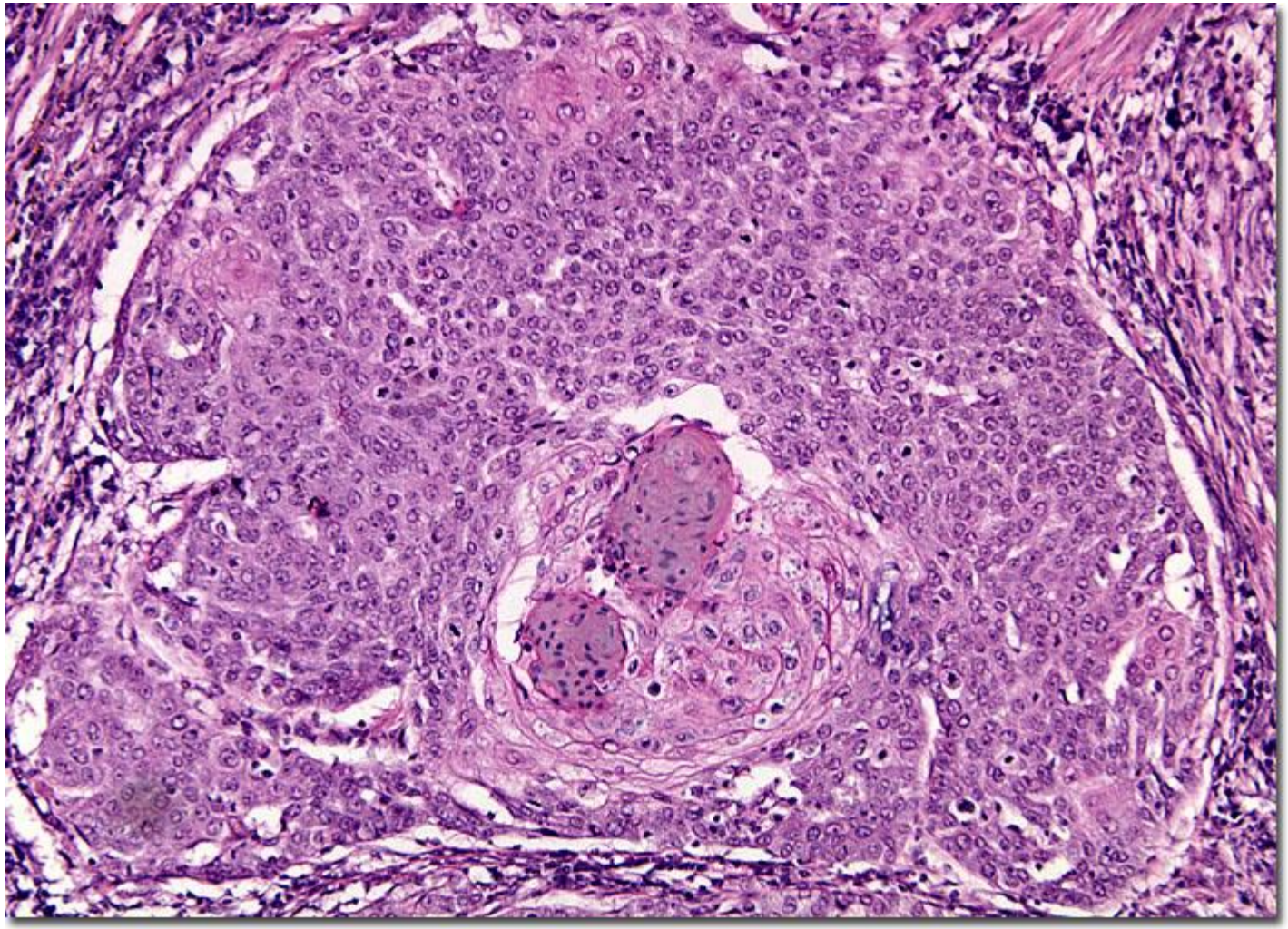
- **High rate of cell division**
- **Anaplasia and pleiomorphism**
  - loss of cellular features and organelles
  - small, irregularly shaped cells
- **Nuclear changes:**
  - large, irregularly shaped
  - increased mitotic events

Normal Tissue



Dysplastic Tissue





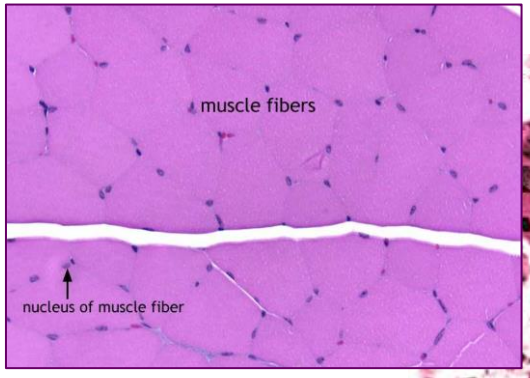
squamous cell carcinoma

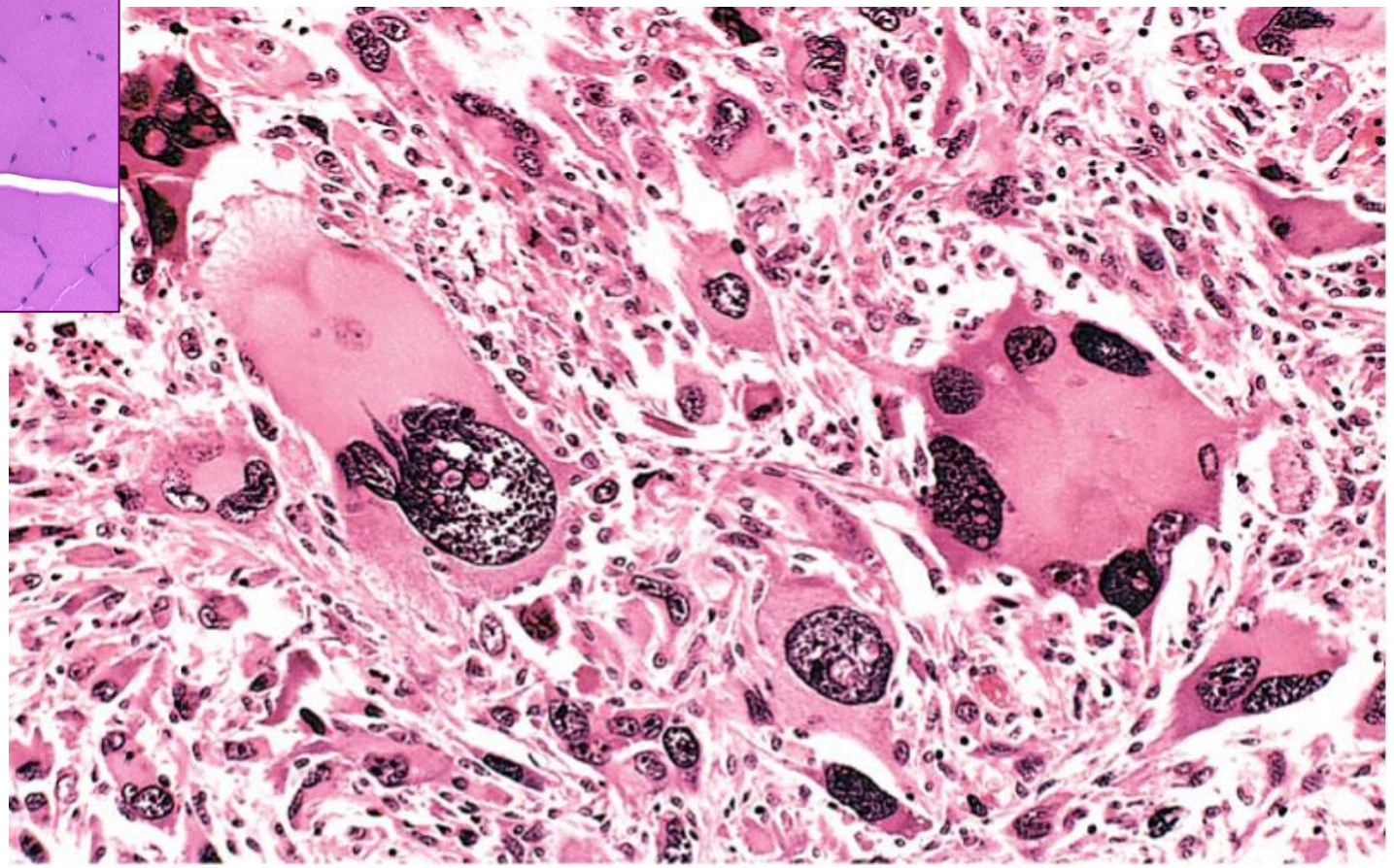
- various-shaped cells packed together without organization, many undergoing mitosis (dark black DNA)

10 Characteristics of Cancer Cells

- Loss of (anaplasia)
- Genetic instability – quick to mutate
- Growth factor independence
- Loss of density-dependent inhibition
- **Autonomy** (lose connections with surroundings)
- Limitless replicative potential (cells don’t age!)
- Avoidance of cell death (apoptosis)
- Promote **angiogenesis** (blood vessels formation)
- Production of enzymes and hormones
- Cytoskeletal changes to facilitate motility

Anaplasia



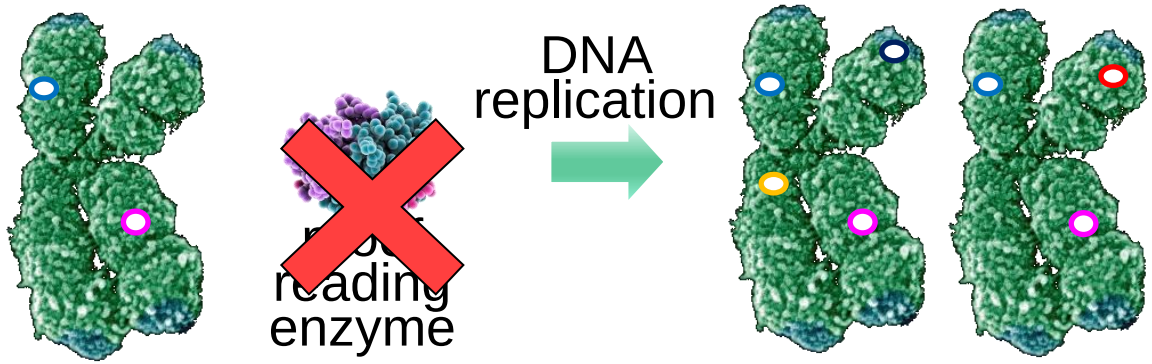


(Courtesy of Dr. Trace Worrell, Department of Pathology, University of Texas Southwestern Medical School, Dallas, Texas.)

Anaplastic rhabdomyosarcoma. Note the marked cellular and nuclear pleomorphism.

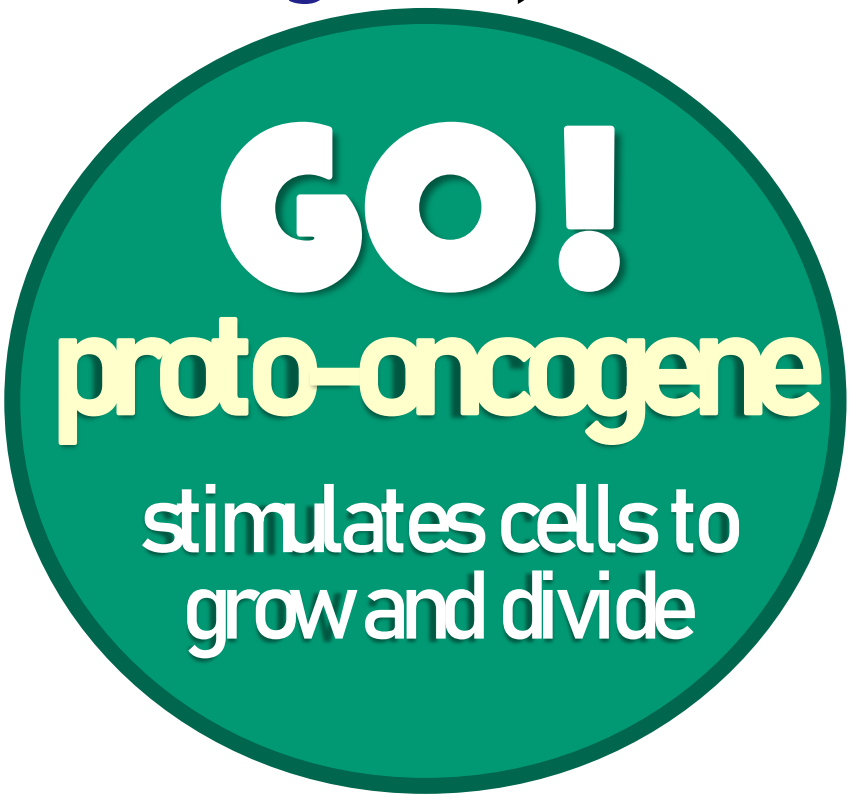
Genetic Instability

- DNA repair proteins normally "proof-read" the genome to ensure no errors
- During the transformation process, a mutation may deactivate the proof-reading enzymes, allowing **mutations to occur and accumulate more quickly**



Growth Factor Independence

There are two main categories of genes that regulate growth and cell division (***cell cycle control genes***):



Growth Factor Independence

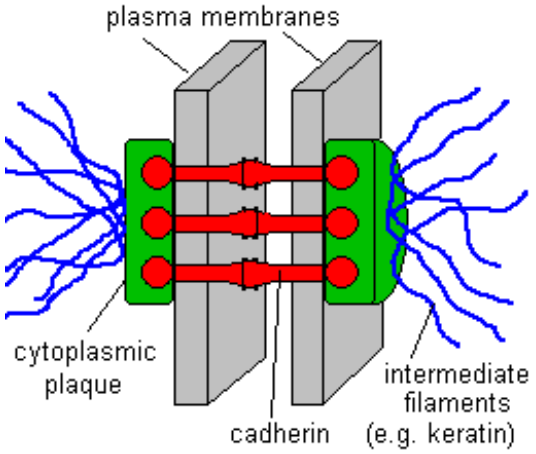
- Cancer develops from mutations in ***cell cycle control genes*** which regulate proliferation
  - cells are able to divide without growth factors (or they produce their own)
  - cells are unable to stop dividing, even after injury or mutation

Cell Density-Dependent Inhibition

- Normally, cells stop growing and dividing when in close proximity to neighbouring cells
  - also know as *contact inhibition*
- Cancer cells lose this ***density-dependent inhibition*** and reproduce regardless of the surrounding cells

Autonomy

- Cancer cells lose much of their contact with neighbouring cells
  - **cell-to-cell communication is lost** (including cytokines and their receptors)
  - connections to neighbouring cells are lost (are cell membrane molecules that bind adjacent cells)
  - a loss of cadherins allows cells to migrate and be invasive



Immunofluorescent light micrograph of a population of epithelial cells, some of which are cancerous
Blue = DNA
Red = cytoskeleton
Green = cadherin

Anchorage Independence

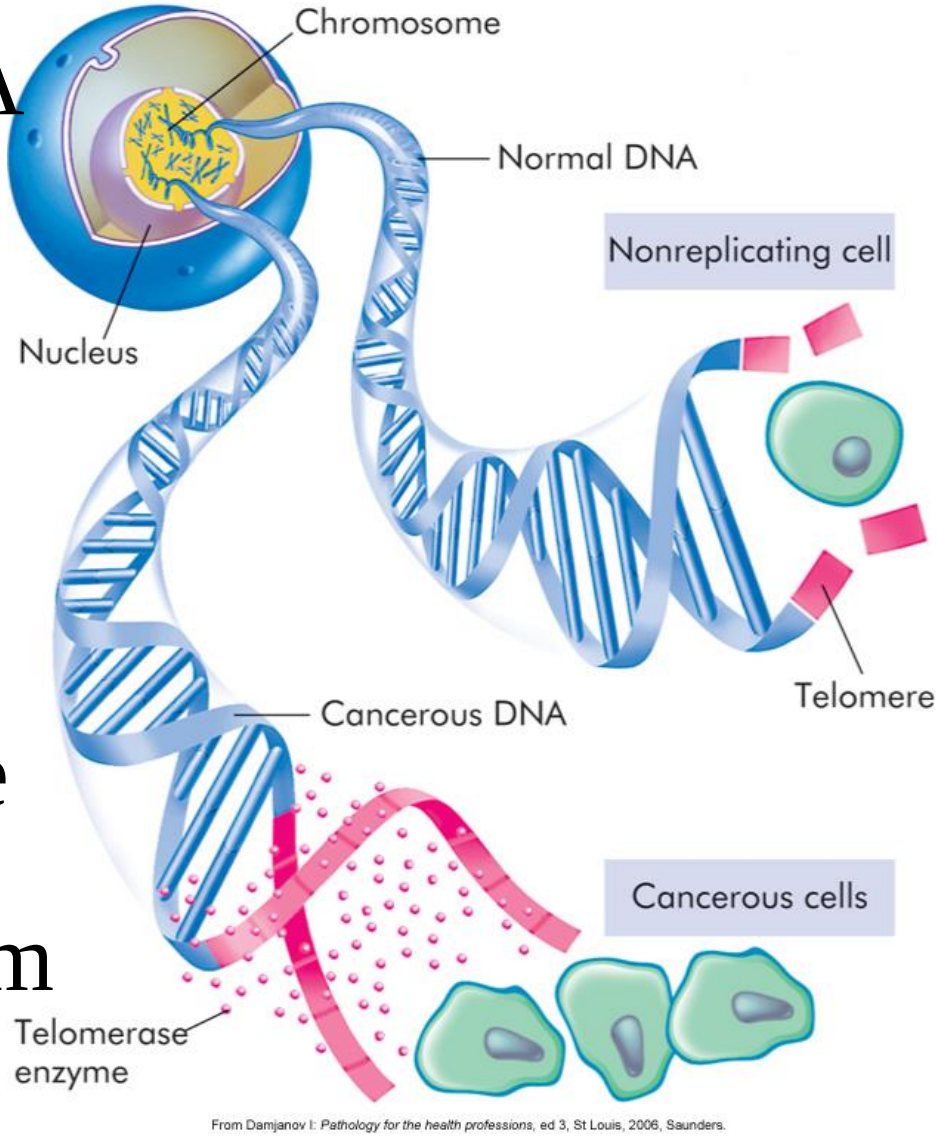
- Normal epithelial cells undergo apoptosis if detached from the neighbouring cell or basement membrane
- Cancerous epithelial cells lose their ***anchorage dependence*** for growth and cell division

DID YOU KNOW?

- Most cells of the body cannot grow and divide forever. WHY NOT?
- How many divisions occur before cells stop dividing or undergo apoptosis?

Limitless Replication

- ***Telomerase*** adds DNA back to the telomere
- This enzyme is only found in a few stem cells
- A mutation in the telomerase gene may allow telomerase to be produced in cancer cells, thus making them "**immortal**" cells

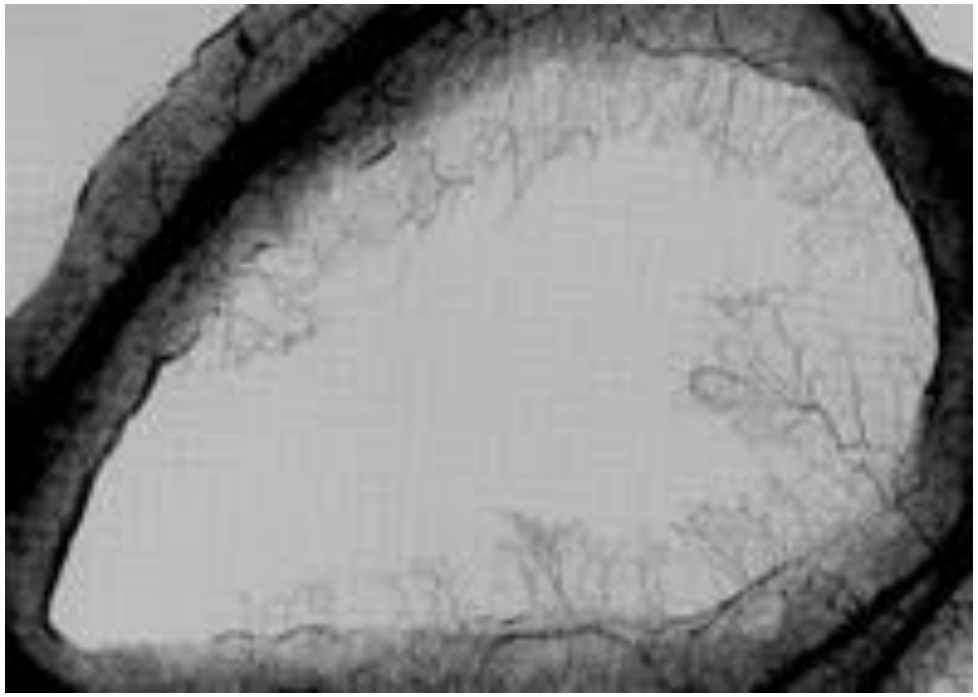


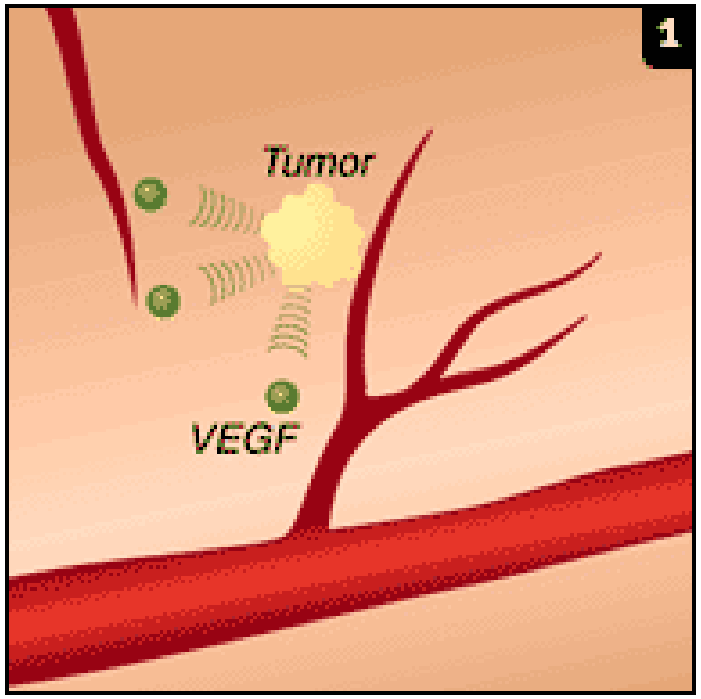
Apoptosis Avoidance

- Several genes are responsible for the programmed cell death pathway (apoptosis)
- Mutations in any number of these genes may

Angiogenesis

- Some cells release angiogenic factors to promote new blood vessel growth
  - one of the most potent is vascular endothelial growth factor ( )





Angiogenesis

- Mutations may allow cancer cells to **produce**
- This will increase growth and may facilitate systemic spread
- Without these mutations cancers will remain small (~2mm) and not clinically relevant

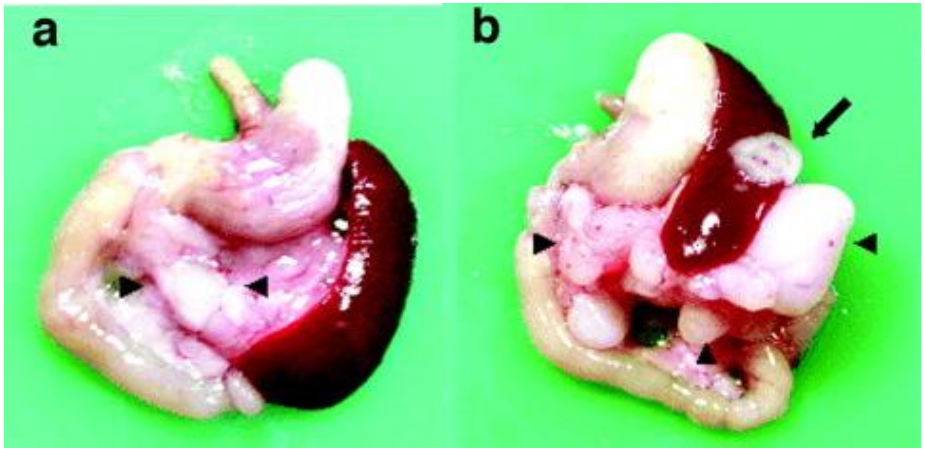
Production of Enzymes

- Cancers cells can <u>secrete enzymes and other substances</u> not normally produced by the original cell
  - ex. (collagenase) that *breakdown the extracellular matrix* thus increasing invasiveness
  - may produce **hormones** and enzymes that do not benefit the cancer, but could impact the patient

Increased Motility

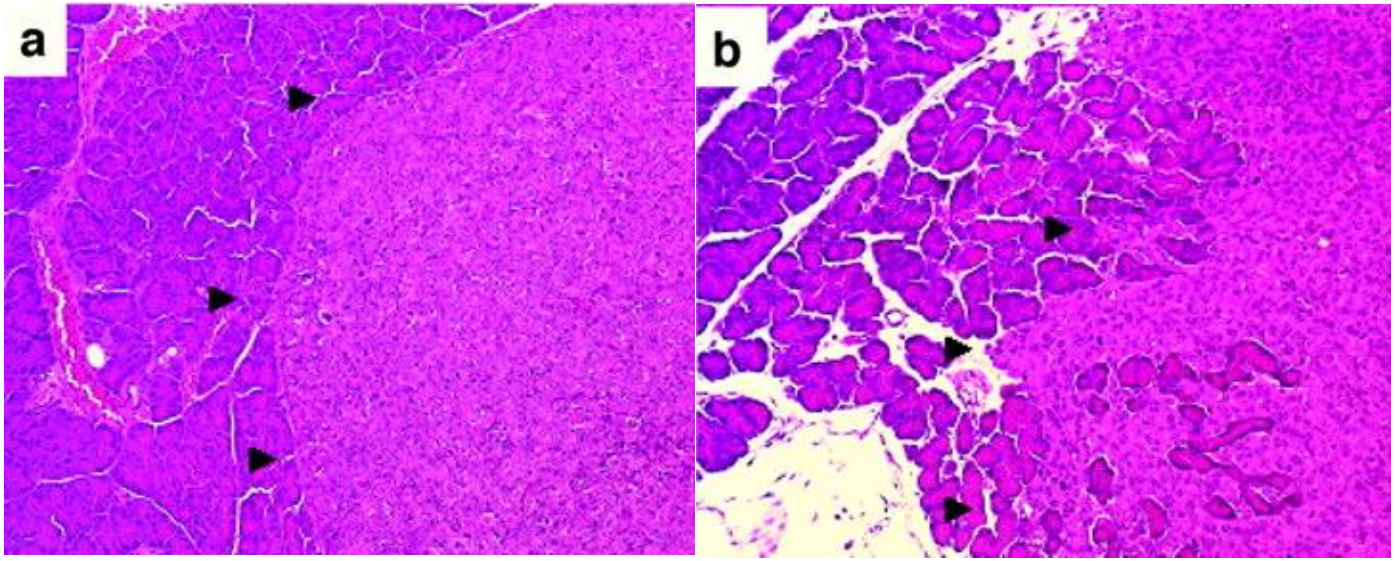
- Cancer cells typically have a variety of cytoskeletal changes
- Secretion of (**AMF**) allows cancer cells to stimulate own movement

Are there cells in the body with the ability to move?



Study Results:

AMF increases aggressiveness of cancer and leads to more secondary tumours



a – low AMF
b – high AMF

Tsutsumi, Soichi & Yanagawa, Takashi & Shimura, Tatsuo & Kuwano, Hiroyuki & Raz, Avraham. (2004). Autocrine motility factor signaling enhances pancreatic cancer metastasis. *Clinical Cancer Research. 10.* 7775-84. DOI: 10.1158/1078-0432.CCR-04-1015.

Invasion and Metastasis

Chapter 8: p.173

Direct Invasion

- Local spread of a cancer cells by direct extension of the primary tumour
- **First step in *metastasis***
- Malignant cells break through their basement membrane or surrounding connective tissue

Important Mechanisms:

Increase the number of cancer cells

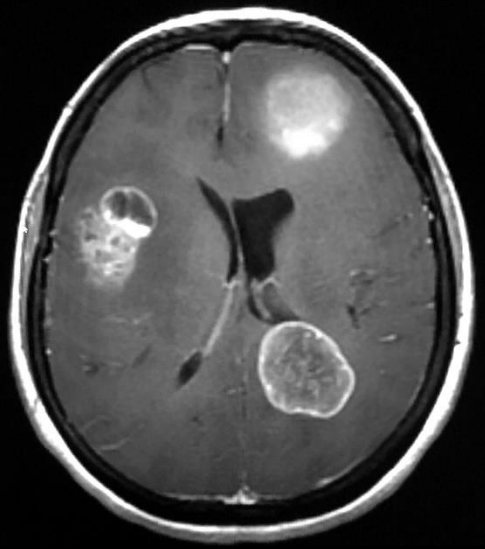
a. growth factor independence
b. loss of density-dependent inhibition
c. release of angiogenic factors
d. apoptosis avoidance
e. limitless replicative ability

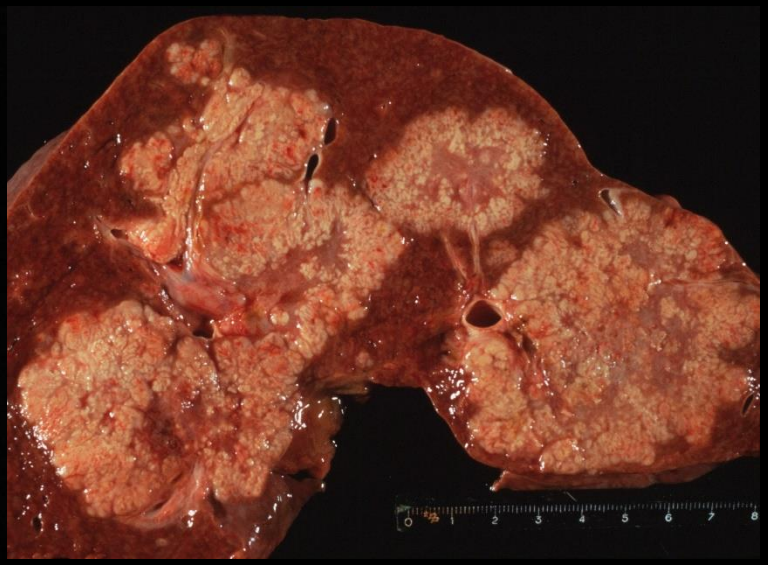
Ability to move

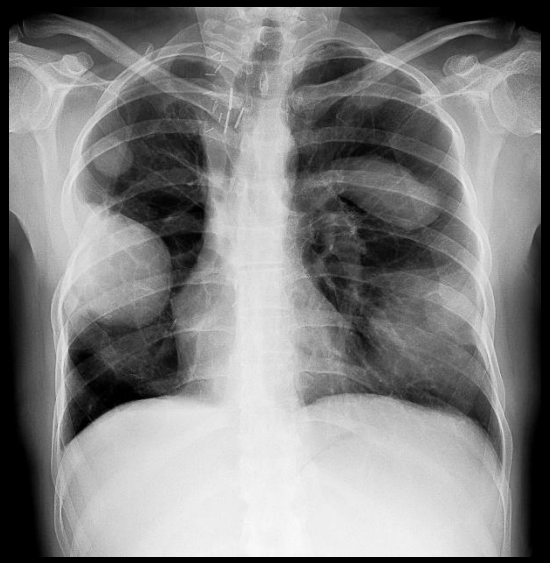
a. cytoskeletal changes to allow movement
b. autonomy – loss of connections to neighbours
c. anchorage independence
d. production of enzymes (collagenase)

Metastasis

- Hallmark feature of malignant growths
- Spread of cancer from its to a distant **secondary site** in the body (liver, lung)
- Secondary tumours may be discovered first

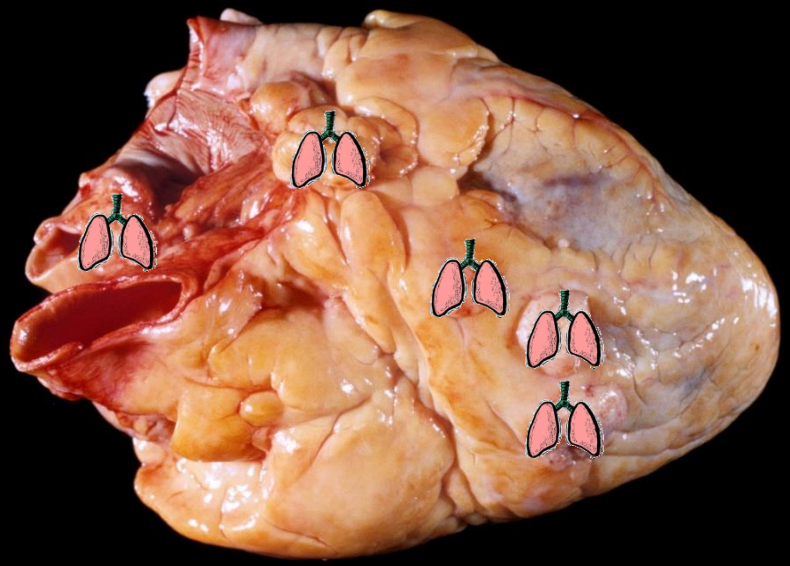






Metastasis

- Metastases often
  of the primary neoplasm
  - *important for diagnosis and prognosis*
- **Life-threatening**



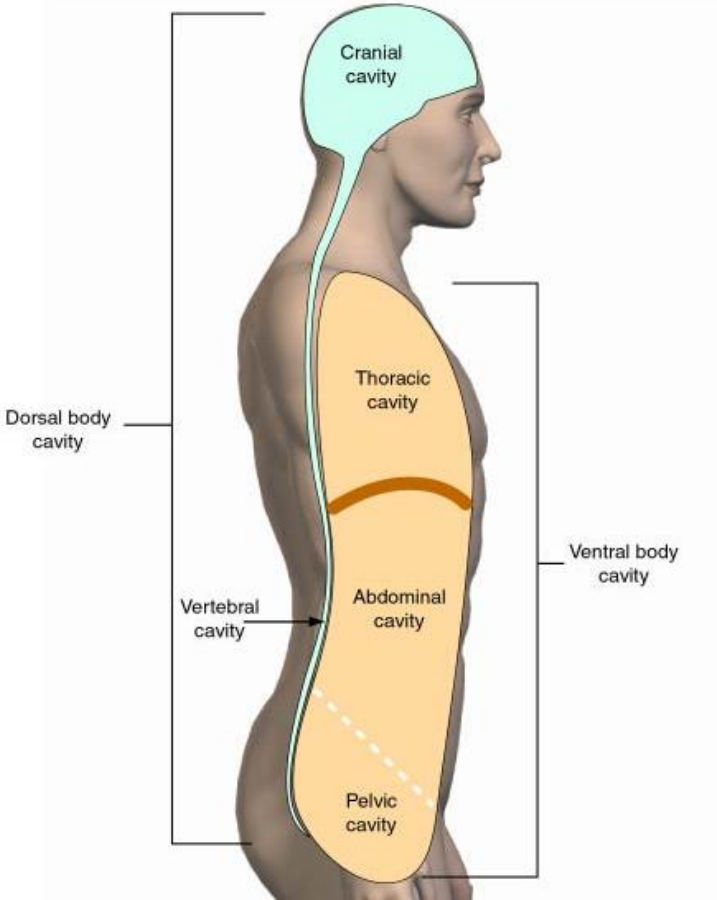
Metastasis

- **Pathways to spread:**
  - seeding (body cavity)
  - lymphatics
  - blood

Which is most common?

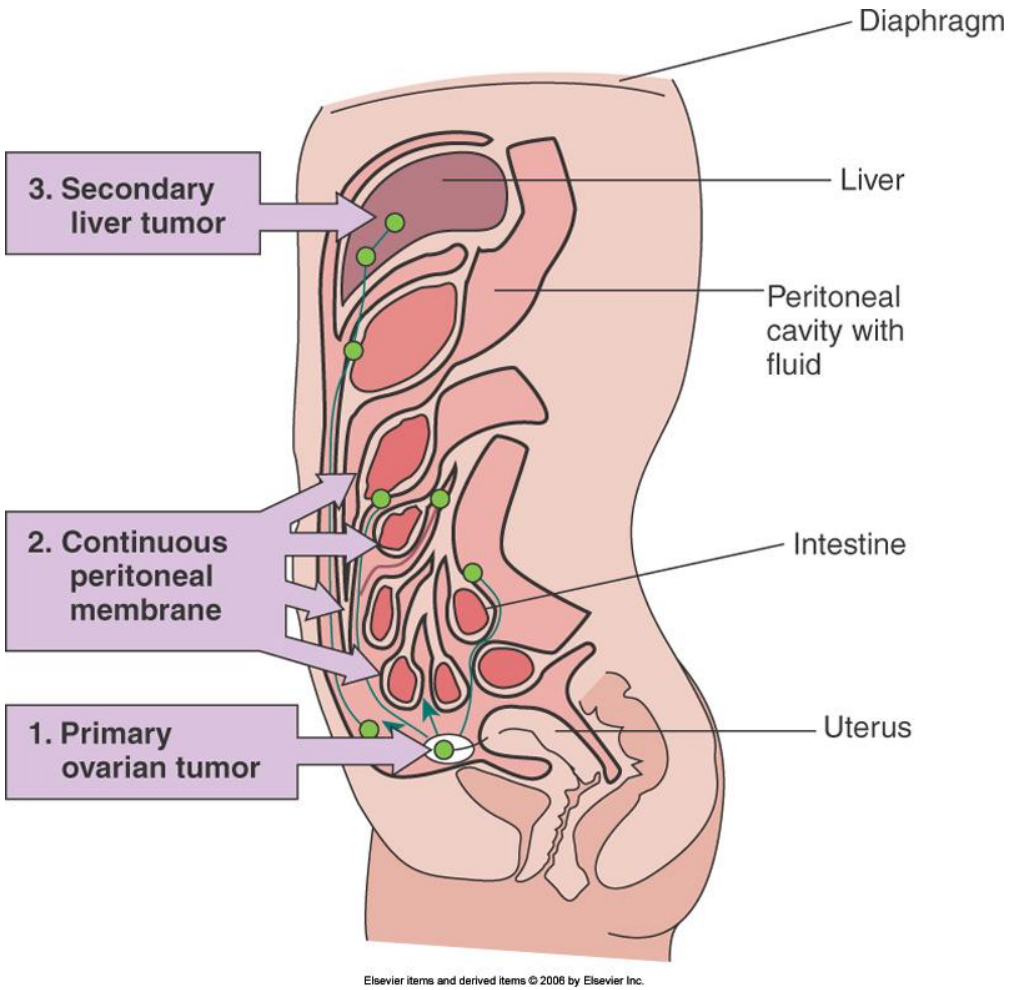
Seeding of Cancer Cells

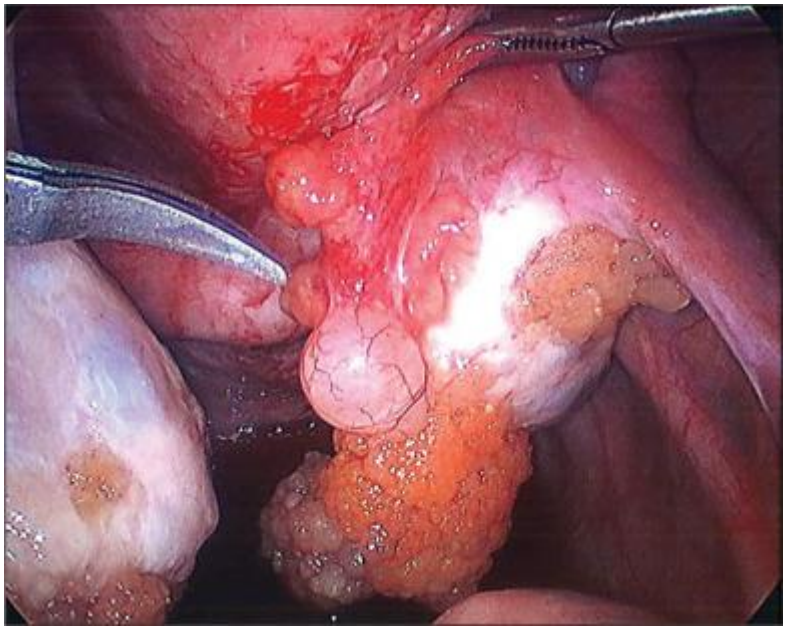
- Cancer cells shed from the primary tumour and move through cavity fluid to form secondary tumours at distant sites



- Typically within the **peritoneal cavity**, but may occur in pleural, pericardical or joint cavities
- A concern when surgically removing a malignant tumour

Seeding of Cancer Cells

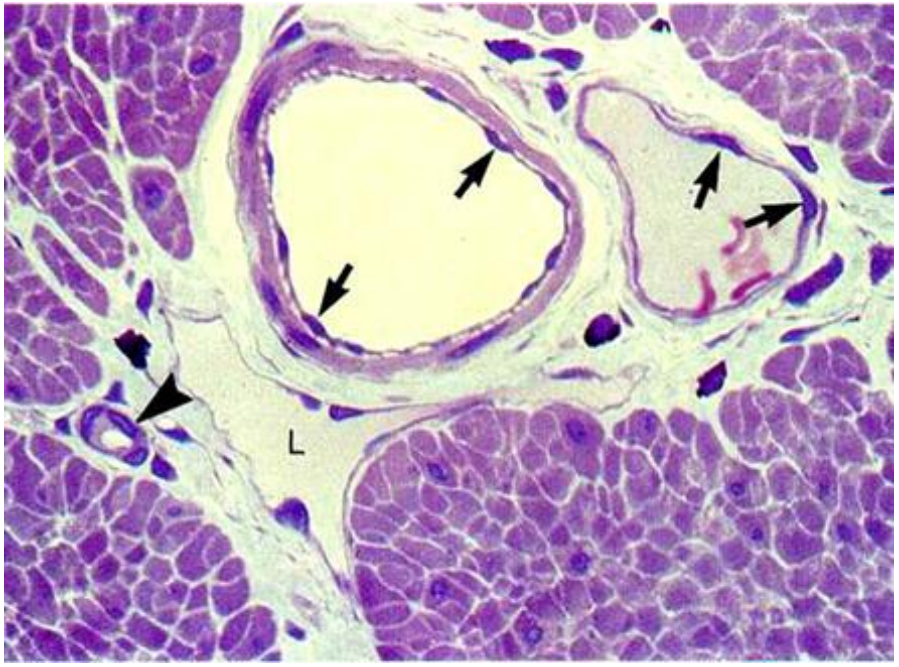


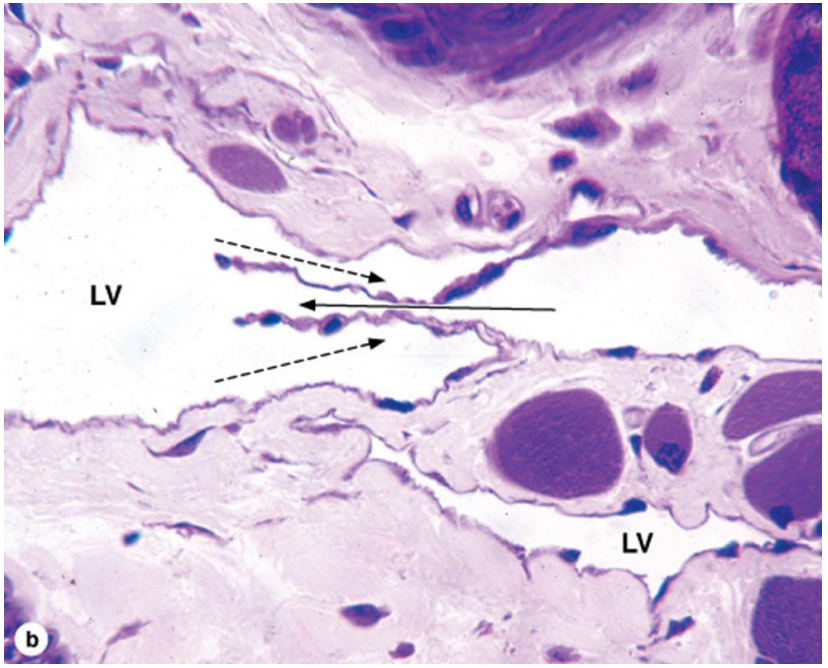


Ovarian cancer

Lymphatic Spread

- **route** for distant metastases
  - lymphatics usually skirt periphery of tumour
  - little mechanical resistance to penetrate between endothelial cells of lymph capillaries





Lymphatic Spread

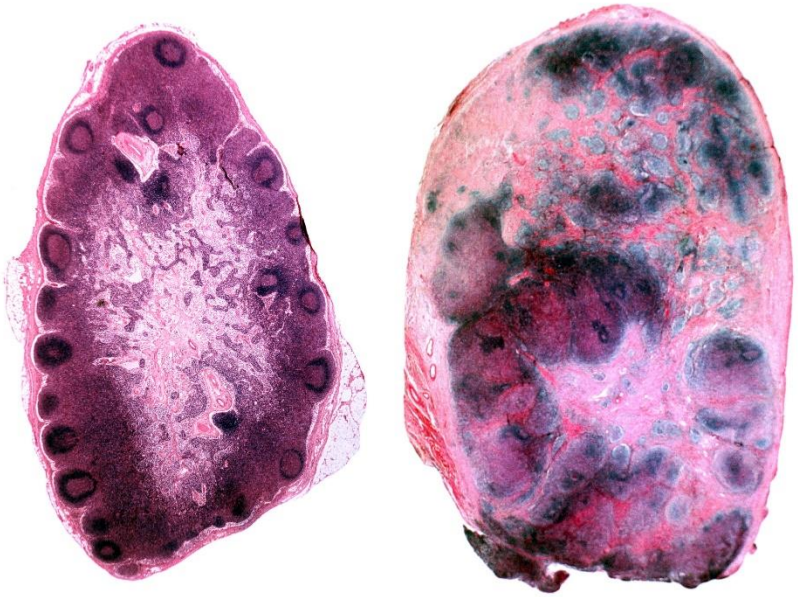
- Cancer cells travel to regional **lymph nodes**
  - = *initial node draining tumour area*
  - radioactive tracers can be used to map drainage area for biopsies (important for **staging** – more on this later)

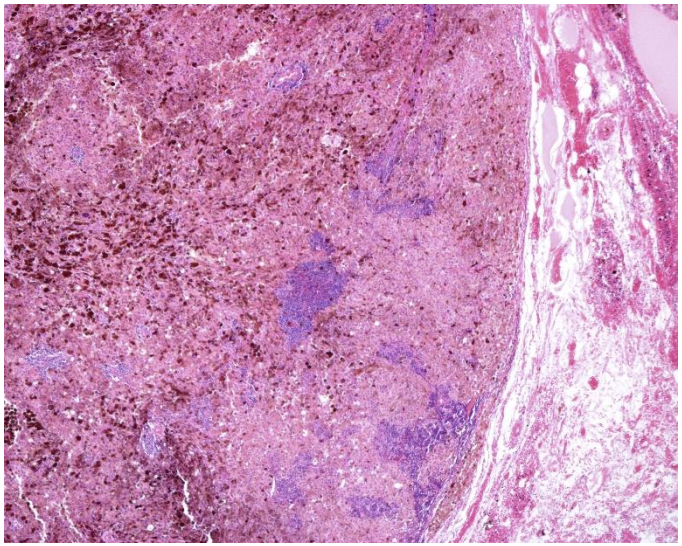
Where does lymph drain into circulating blood?

Where are the first small blood vessels after this union?

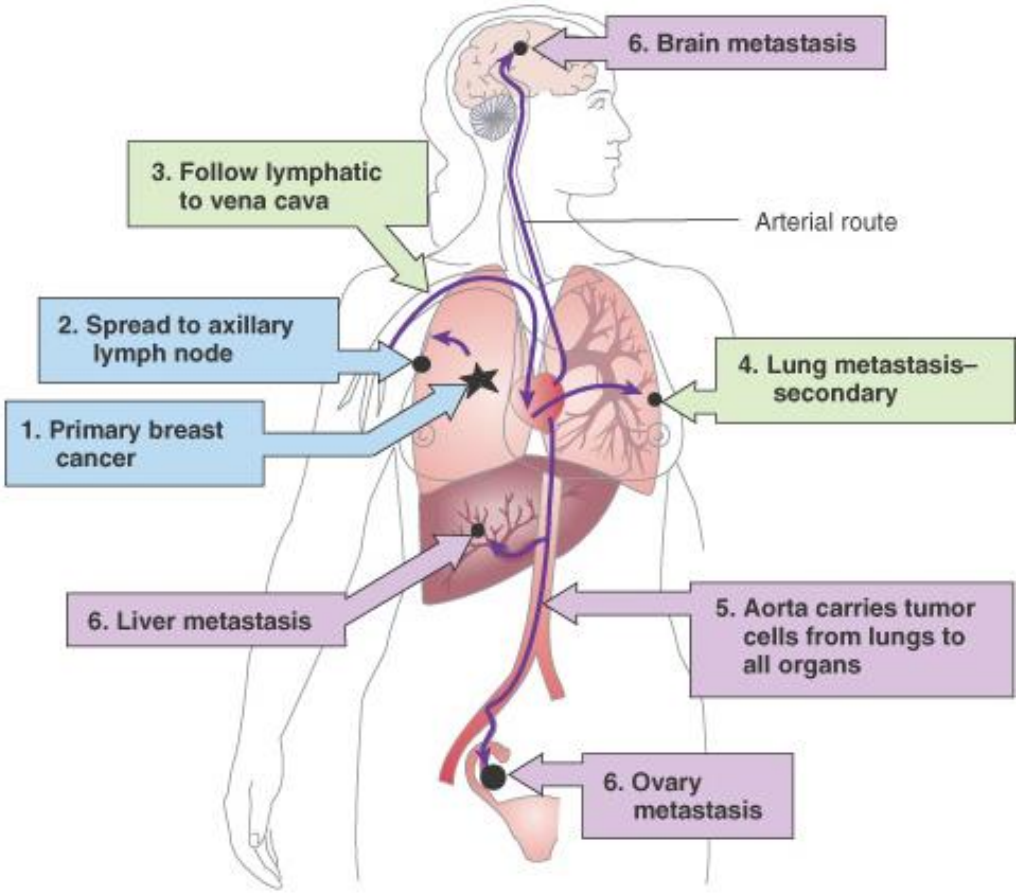
Lymphatic Spread

- Cancer cells lodged in lymph nodes:
  - are **killed** by local inflammatory mediators
  - become
  - grow into noticeable lump
  - escape the node eventually entering vasculature





Lymphatic Spread



Typical metastatic route of **breast cancer** **(almost always through the lymphatic system)**

Hematologic Spread

- Highly vascularized tumours can **grow** quickly and **spread** via blood more easily
- Direct entrance to the bloodstream is typically through the

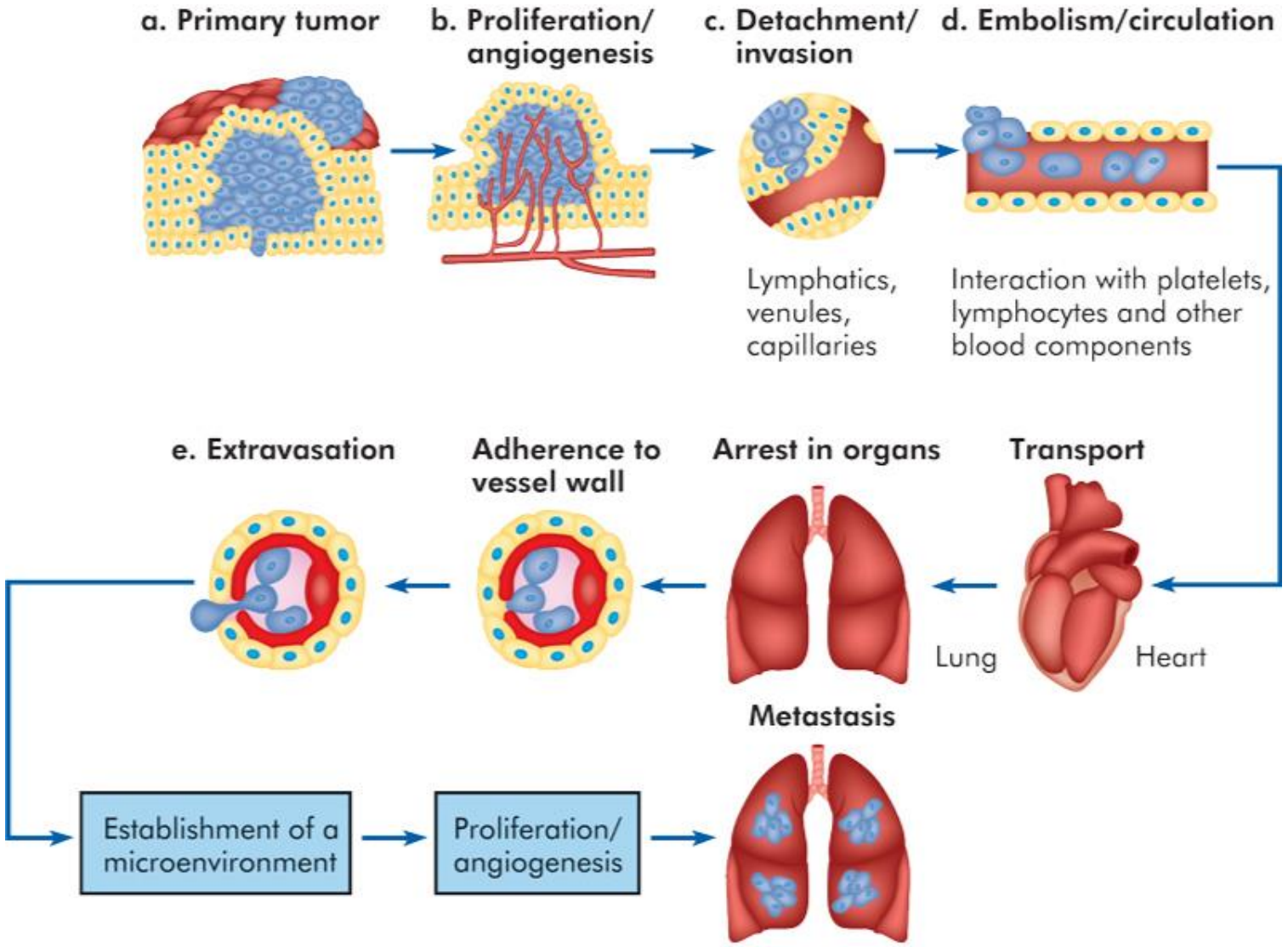
Before entering general circulation, where does venous blood from the GI tract, pancreas and spleen go?

If cancer cells enter systemic venous circulation, where is the first capillary network they would enter?

So cancer cells are getting into circulation, now what?

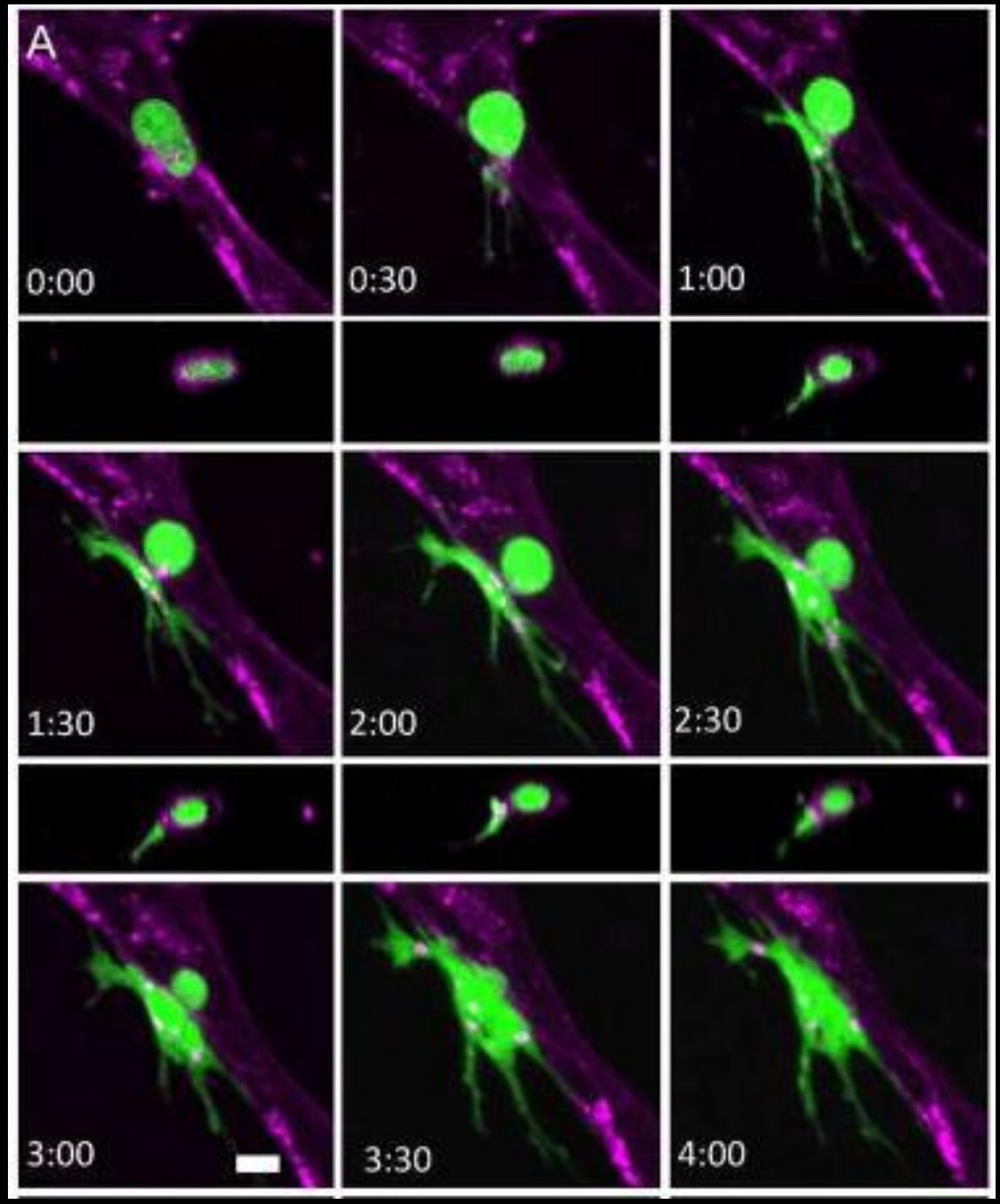
- Malignant cells continually divide and shed into blood stream
  - **millions** may enter into the blood stream without producing secondary tumours (fewer than 1 in 10,000 survive)
  - cancer cells need to **avoid immune cells, survive the blood stream**, get lodged into a capillary network, escape the vasculature and finally grow and reproduce

Metastatic Distribution

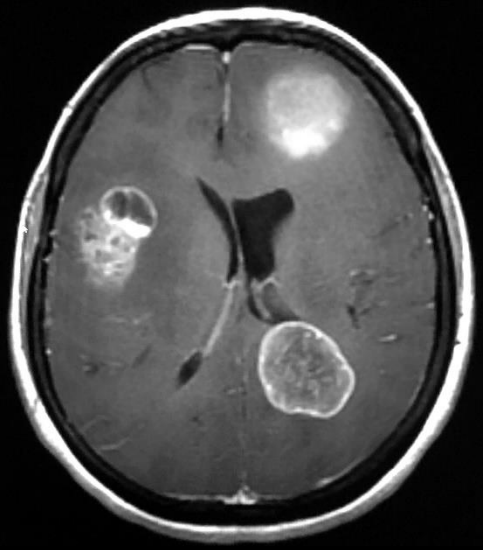


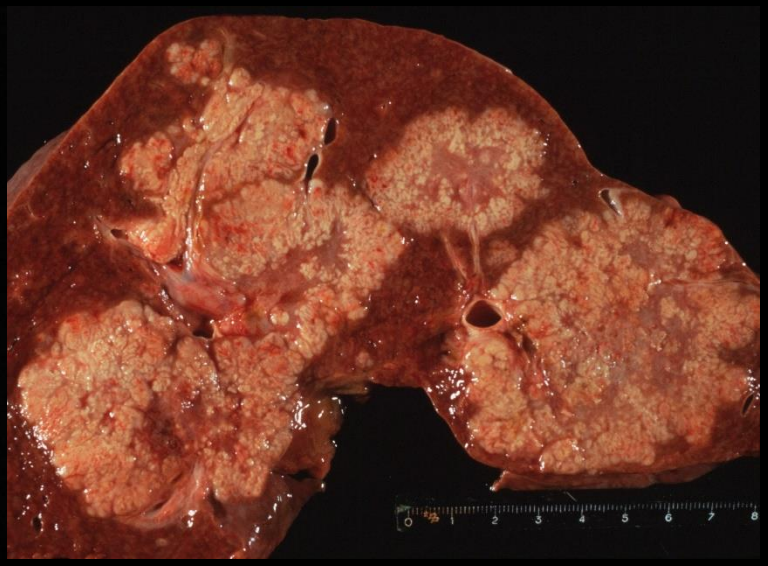
From Fidler IT: The pathogenesis of cancer metastasis: the "seed and soil" hypothesis revisited. *Nat Rev Ca* 3:453–458, 2003.

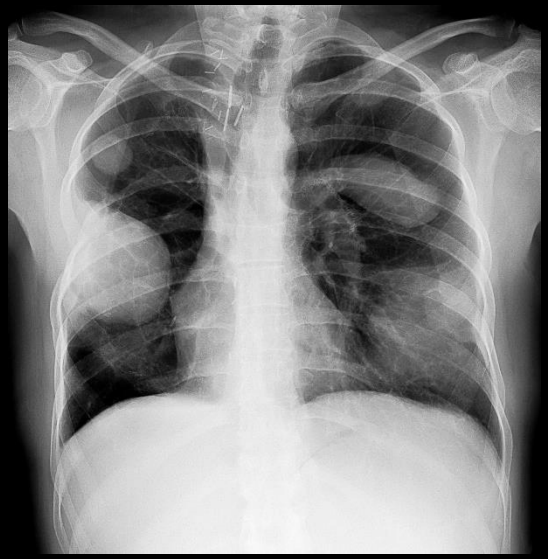
Tumor cells become trapped and travel through the endothelium (four hours)



Once in circulation, where do cancer cells go?







Metastatic Distribution

- Some metastases take place in the **first capillary** network downstream of veins that drain the primary tumour (*lungs*, *liver*)
- Others show (selection), in which they typically metastasize to specific organs
- Most likely the result of the *environment surrounding* the secondary organ
  - chemotatic factors
  - cytokines
  - growth factors
  - hormones

Metastatic Distribution

- Many cancers exhibit **distinct patterns of metastases (tropism)**

Primary Site	Usual Site of Spread
Colorectal	Liver, lungs, bone, CNS
Prostate	Bone, lung, liver, CNS
Lung	Liver, adrenal glands, bone, CNS, kidney
Melanoma	Liver, lung, CNS, bone
Breast	Bone, lungs, liver, CNS

Etiology of Cancer

Genetic and Molecular Basis

Chapter 8: p.176

Principles of Carcinogenesis

Remember, all cells with a nucleus have *access* to the same genes (they just don't normally use them all)

Principles of Carcinogenesis

- are needed to acquire all the characteristics of a cancer cell
- In simplest terms, there are two broad types of mutations needed:
  - "**turn on**" a gene to make a protein that the original cell was not making
    (ex. telomerase, collagenase, VEGF)
  - "**turn off**" a gene that the cell was using that could prevent carcinogenesis
    (ex. tumour suppressors, cadherins, apoptotic factors)

Principles of Carcinogenesis

Keep in mind:

1. Genetic mutations happen **randomly**, most of which will have limited effect on the cell and *would not lead to cancer*
2. Our cells are

Critical Thinking: How would the fact that our cells are diploid affect "turning on" or "turning off" genes?

Principles of Carcinogenesis

Cell Cycle Control Genes

- To become cancerous, a **proto-oncogene** needs to be *mutated* (**turned on**) to become an oncogene (it will now constantly signal the cell to divide)

- *Ex.* *(proto-oncogene)*
  - ras protein is normally involved in initiating cell division *when an appropriate growth factor is present*
  - when correctly mutated, it is possible for Ras protein to ALWAYS initiate cell division

Principles of Carcinogenesis

Ex.

normal

Ras protein instructs cell to divide *only when growth factor is present*

mutated

This mutated form of Ras is now an , its protein will signal cell division *without growth factor present*

Principles of Carcinogenesis

Cell Cycle Control Genes

- To become cancerous, **tumour suppressor genes** need to be *mutated* (**turned off**), thus unable to stop a cell from dividing when damaged

- *Ex. TP53 (tumour suppressor gene)*
  - codes for that stops cell division *under the appropriate conditions*
  - when **BOTH** copies are correctly mutated, p53 protein is unable to stop the cell from dividing

Principles of Carcinogenesis

Ex. *TP53*

normal

p53 stops the cell cycle following significant DNA damage (mutations)

mutated

What will happen?

Principles of Carcinogenesis

Ex. *TP53*

normal

p53 stops the cell cycle following significant DNA damage (mutations)

mutated x2

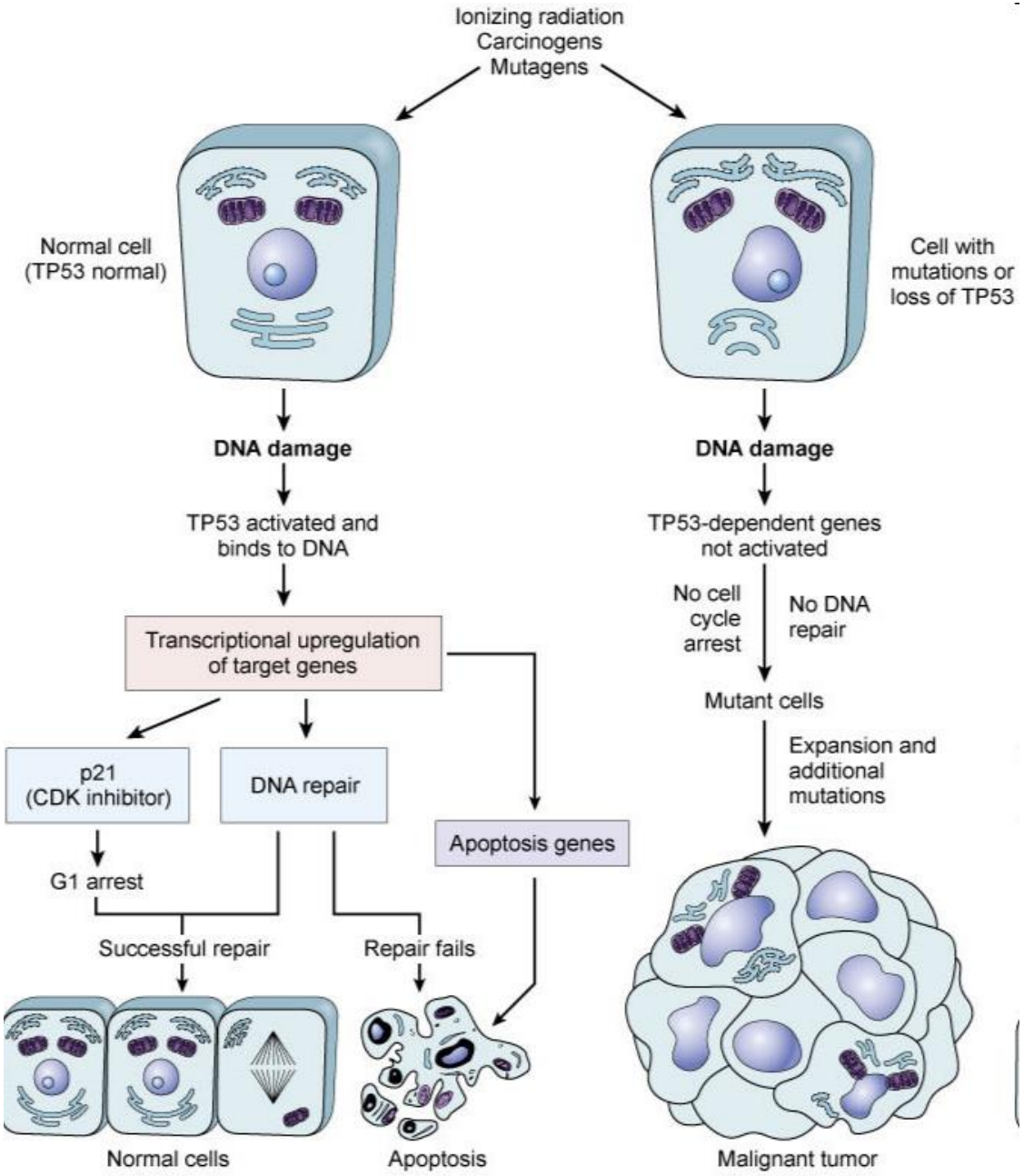
If only one copy of the gene is mutated, **the normal version of the gene will produce p53 that can stop cell division**

Principles of Carcinogenesis

Ex. *TP53* WAIT! There's more!

Functions of p53:

1. Tumour suppressor
2. Involved in the DNA repair process
3. Initiates apoptosis following irreversible DNA damage

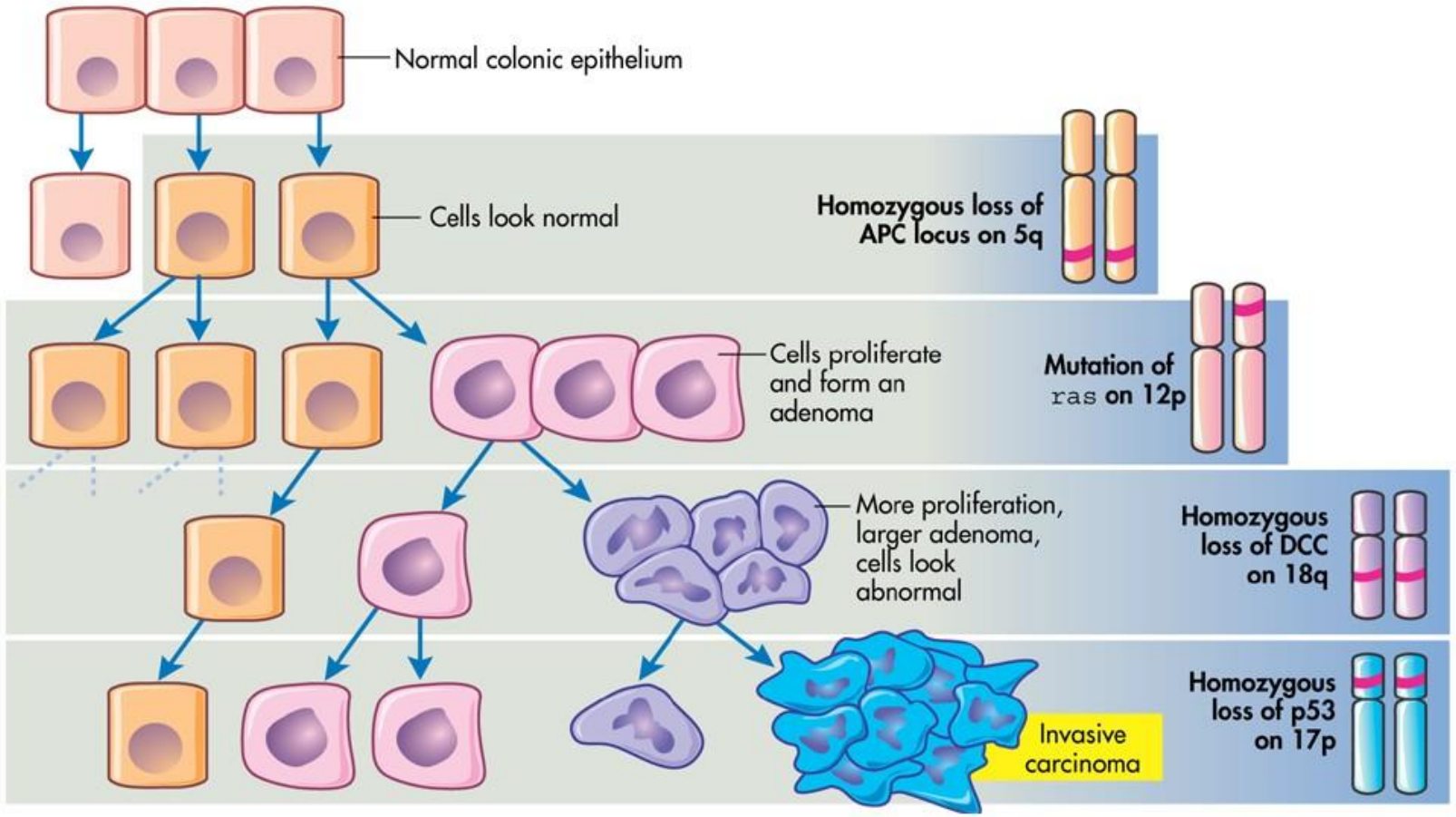


(Redrawn from Kumar V et al: *Robbins basic pathology*, ed 8, Philadelphia, 2007, Saunders, p 196.)

Principles of Carcinogenesis

Characteristic	Gain one mutation	Lose two mutations
Growth factor independence - oncogene		
Growth factor independence – TSG		
Genetic instability		
Autonomy - lose connections with cells		
Limitless replicative potential		
Avoid apoptosis		
Promote angiogenesis		
Produce enzymes to breakdown collagen		
Cytoskeletal changes to facilitate motility		

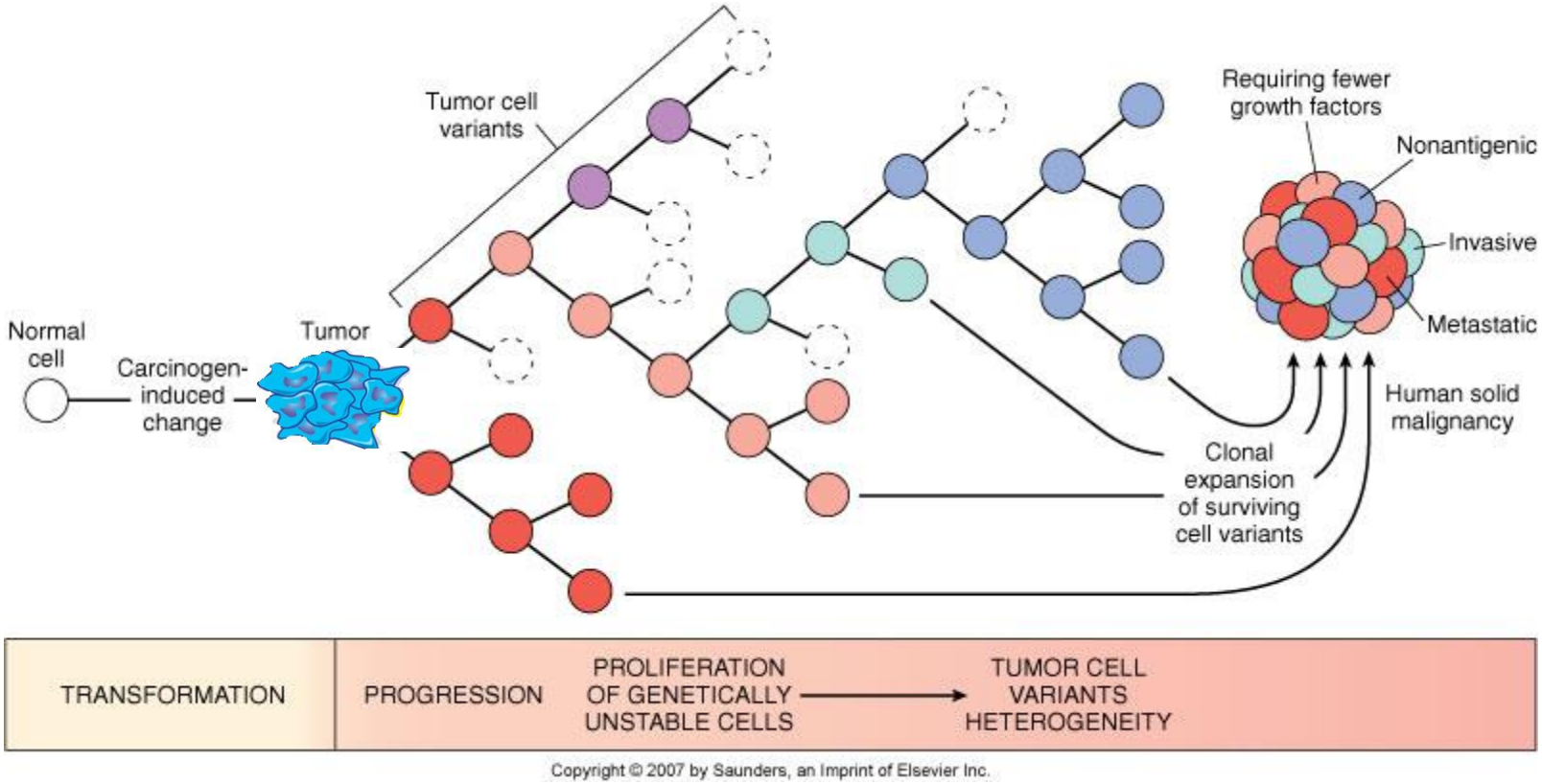
Principles of Carcinogenesis



Modified from Kumar V, Cotran RS, Robbins SL: *Basic pathology,* ed 6, Philadelphia, 1997, WB Saunders.

Mosby items and derived items copyright © 2004, 2000 by Mosby, Inc.

Tumour Heterogeneity



Copyright © 2007 by Saunders, an Imprint of Elsevier Inc.

Tumour progression and generation of **heterogeneity**. New subclones arise from the descendants of the original transformed cell by multiple mutations. With progression the tumour mass becomes enriched for variants that are more adept at evading host defenses and are likely to be more aggressive.

Etiology of Cancer

Host and Environmental Factors

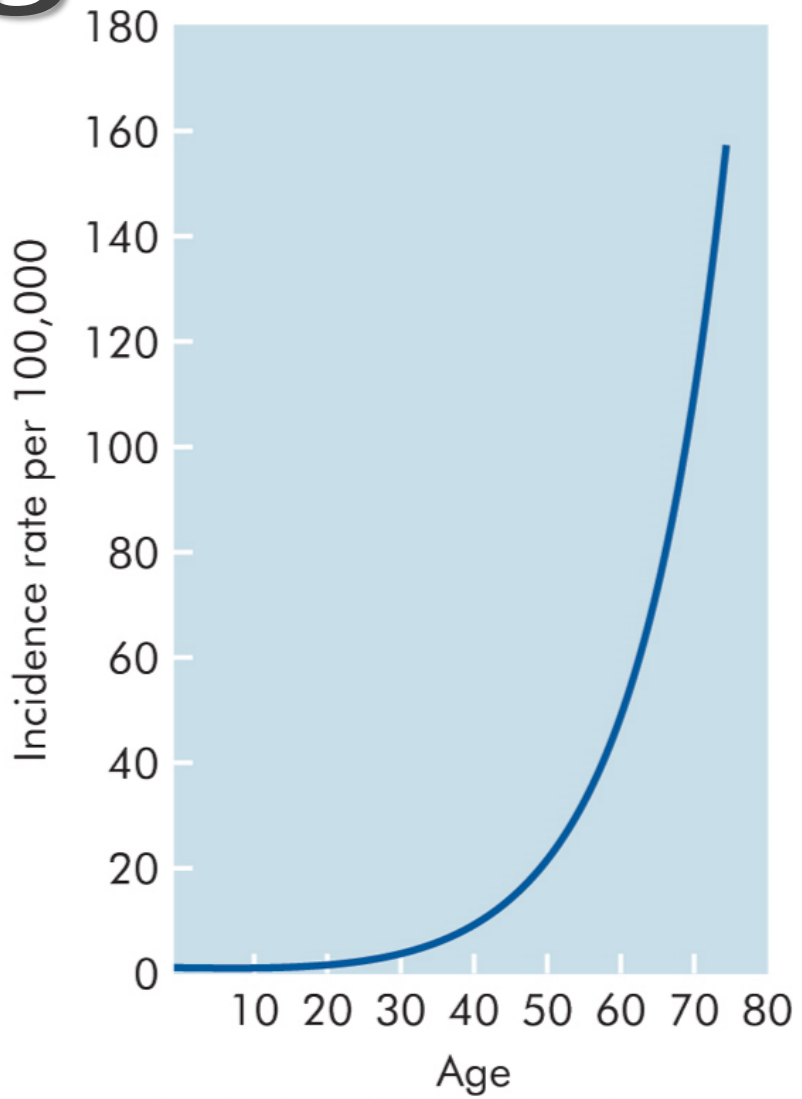
Chapter 8: p.180

Principles of Carcinogenesis

- Most human cancers arise ***spontaneously*** with no *known* prior exposure to carcinogens
- The frequency of mutation can be increased through exposure to carcinogens (**mutagens)**
  - ex. cigarette smoke

Principles of Carcinogenesis

- ***Mutations*** accumulate within cells, hence the number of cases of cancer increase with age



Modified from Alberts B et al: *Molecular biology of the cell*, ed 4, New York, 2002, Garland.

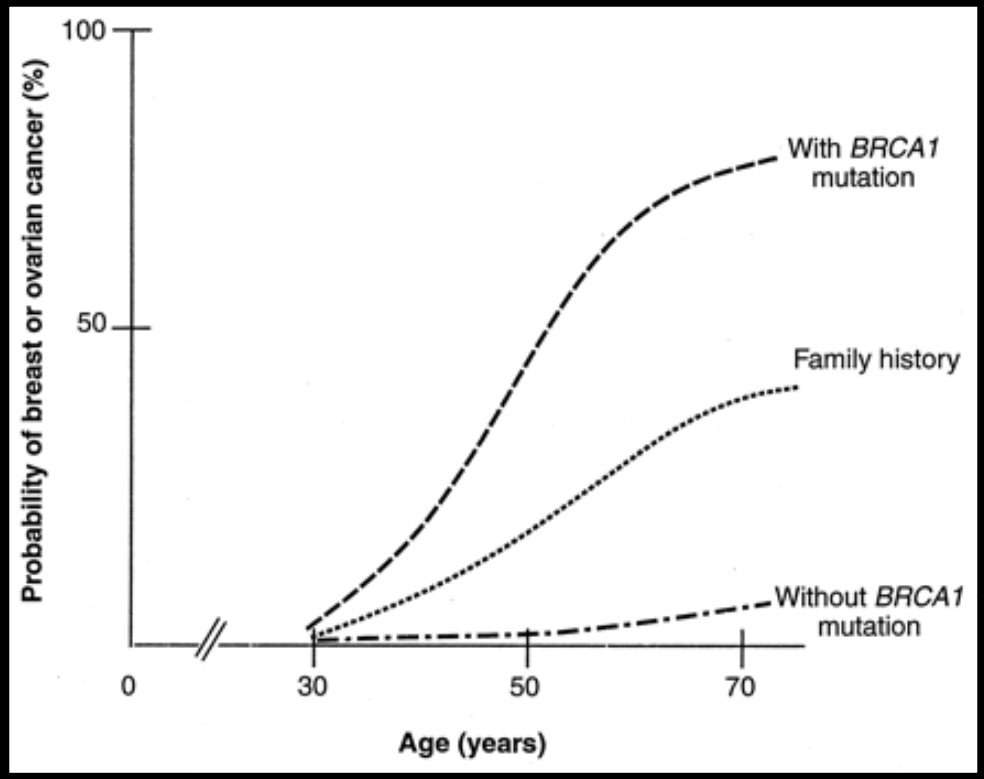
Principles of Carcinogenesis

- Genetic mutations in ***somatic cells*** are **NOT** inherited
  - ie. *acquired* epithelial cell mutations

- Genetic mutations in ***germinal cells*** are inherited (↑ incidence of specific cancers in some families)
  → ***familial predisposition***

Heredity -

- Individuals who inherit a *BRCA1 ( )* mutation have an 80% chance of developing breast cancer and ~50% chance of developing ovarian cancer in their lifetime



Heredity -

- Individuals who inherit an *APC* (
) mutation develop *familial adenomatous polypsis (FAP)* of the colon leading to hundreds of polyps, some become malignant
- 50% of patients develop colon cancer by 40
- of patients develop cancer by age 70
- Total colectomy before age 20 can allow a normal life

Environmental Impact

- **Different areas of the world experience different incidence rates of cancer**

- Study done on **first-generation** Japanese immigrants in Brazil
- Following over 2,000 cases of cancer, were the incident rates of stomach, skin, breast, and prostate cancers similar to native Japanese or Brazilians?

Shoichiro Tsugane *et al*, 1990. *Cancer Causes and Control.*

Environmental Impact

- Chemical carcinogens are usually dose dependent – the larger/longer the dose, the more likely cancer will develop
- *Delayed 5-30 years following exposure*
- Many create
- Direct-reactive agents are carcinogenic upon exposure
- Indirect-reactive agents (**procarcinogens**) require metabolic conversion

Cigarette Smoke

- Contains many procarcinogens
- Directly linked to lung and laryngeal cancers
- Associated with cancers of the mouth, nasal cavities, pharynx, esophagus, pancreas, liver, kidney, uterus, cervix, bladder, blood
- Related to **39% of ALL cancer deaths** and 85% of lung cancer deaths in Canada
- Second-hand smoke leads to over 500 cases of lung cancer each year

UV Radiation

- Ultraviolet radiation is a type of **ionizing radiation**, which removes electrons from molecules creating
- UV does not penetrate deeply, typically affecting skin
- Higher incidence in light-complexioned individuals
- Dose-dependent and delayed development
- Episodic exposure to intense sunlight during childhood is more likely to lead to cancer

INFECTION - Viruses

- Viruses are responsible for **15%** of human cancers
- Many are able to integrate into the host DNA, disrupting genes in the area
- Many to induce cell division (and thus more *proviruses*)
- **Indirect predisposition** due to healing and regeneration of tissue

Hepatitis B Virus

- This virus is spread through blood, semen and other bodily fluids
- Known to cause ~60% of all primary liver cancers
- HBV is able to into the host DNA – leading to mutations
- It causes chronic infections that can be severe, forcing significant regeneration of healthy tissue

Human Papillomavirus

- HPV is typically spread through sexual contact
- Many different strains of HPV, some are more likely to lead to *benign papillomas*, whereas others lead to *carcinomas*
  - some significantly increase oncogene expression
- Causes more than of all cervical and anal cancers, as well as the majority of vaginal, vulvar and penile cancers
- Integrates into host genome and causes chronic tissue damage

INFECTION - Bacteria

- Bacterial infections are known to increase the incidence of many human cancers
- Like viruses, they damage tissue forcing increased regeneration of cells
- They may result in direct/indirect formation of both **carcinogens** and

Helicobacter pylori

- *H. pylori* infects the stomach and is spread through saliva, vomit, stool, or contaminated food and water
- Approx. of the world's population is infected with this bacterium
- May disrupt gastric mucus production, causing autodigestion
  - chronic infection is a leading cause of stomach cancers

Clinical Manifestations

Chapter 8: p.184

Clinical Manifestations

- Variable, depending on location and extent of metastasis
  - first manifestations reflect primary site
- Early stages usually
- Consequence of cancer and/or treatment

Clinical Manifestations

- Local effects
- Pain
- Fatigue
- Cachexia
- Anemia
- Leukopenia/thrombocytopenia
- Infection
- Paraneoplastic syndrome

Local Effects

- As cancers grow they compress and damage blood vessels causing **ulceration**, necrosis and bleeding
  - reducing the overall function of the organ
  - cancer cells may produce enzymes and metabolic toxins that are destructive
  - as the cancer spreads, other tissues will be affected
- Presence near a
  commonly leads to symptomatic effusion
  - lung and breast cancers account for a significant portion of pleural effusions

Pain

- *Little to none in early stages*, but present in 60-80% of terminally ill persons
- Caused by:
  - **pressure**, obstruction, invasion of sensitive structures (compress nerve endings), stretching of visceral surfaces (ie. distend bowel), tissue destruction and inflammation
  -
- Compounded by **psychological factors** *(fear, anxiety, fatigue)*

Fatigue

- **symptom** of cancer
- Distinct from normal tiredness, not relieved by sleep
- May involve:
  - **sleep disturbances**
  - lack of energy
  - altered metabolism, including serotonin
  - **level of activity**
  - psychosocial factors

Cachexia

- Severe form of malnutrition
  - complex, in part due to hypermetabolic tumour
  - present in **of cancer patients**
- Syndrome includes:
  - ***weight loss*** *(muscle and fat)*
  - ***weakness*** *(asthenia)*
  - *anorexia (loss of appetite), early satiety (“full”)*
  - *taste alterations*
  - *anemia (low RBC count)*
  - *altered metabolism of nutrients*

Cachexia

- **Most manifestations** due to pain, depression, chemo- or radiation-therapy, altered taste, food aversions, **oral ulcers**
- Results in wasting/**emaciation**
  - loss of ~80% of adipose tissue and muscle
  - ↓ quality of life
- **Food input** may be the same or increased!
  - ↓ protein synthesis and altered metabolism
- Patients do not respond well to treatment
  - treat with high caloric intake

Anemia

- Due to:
  - **chronic bleeding**
  - impaired production
    -
    - *cytotoxic medical therapies (chemo. or radiation)*
    - ***bone marrow malignancies***
- Leads to decreased response to treatment and increased mortality
- Treat with **erythropoietin**

Leukopenia, Thrombocytopenia

- Decrease in WBCs, platelets
- Due to:
  - **chronic bleeding**
  - impaired production
    -
    - *cytotoxic medical therapies (chemo. or radiation)*
    - ***bone marrow malignancies***
- Treat with **growth factors**

Infection

- Significant complication that leads to **death**
- Risk ↑due to:
  - ↓ WBCs, ↓ immune function
  - immunosuppression due to chemo/radiation
  - ↓ resistance due to surgery
  - age
- **Increased incidence of nosocomial** (hospital-related) **infections**

Paraneoplastic Syndromes

- As cancer cells pick up mutations, some of these **turn on the production of hormones or other cell products** to *create manifestations not directly related to the disease*
  - ADH (leads to SIADH)
  - **ACTH** (leads to Cushing syndrome)
  - PTH (leads to hypercalcemia)
  - (venous thrombosis)
- Treatment may involve suppressing the production of the cellular product

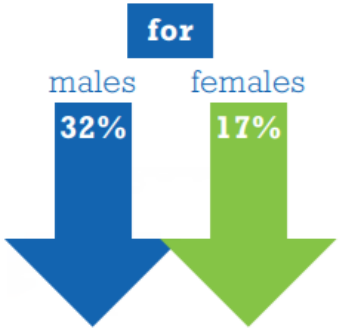
Screening, Diagnosis and Treatment

Chapter 8: p.188

Nearly **1 in 2 Canadians** is expected
to be diagnosed with cancer in their lifetime

About **1 in 4 Canadians**
is expected to die from cancer

1988 → 2017
Cancer death rates decreased



Critical Thinking: How is cancer diagnosed?

- The definitive approach for diagnosing cancer is through retrieval of cells ( ) followed by microscopic identification

Cancer Screening

- Treatment is more successful when the cancer is identified early
- Goal is to identify cancer in an otherwise **asymptomatic individual ( )**

How is screening performed? Examples?

- observation
- palpation
- imaging (x-ray, ultrasound, CT, MRI, endoscopy)

(*tumour markers*)

methods

Tumour Markers

- Antigens expressed on the surface of tumour cells *or* molecules released from normal cells in response to a tumour
  - detect *elevated levels of a protein or expression of fetal protein* (ie. PSA, CA-125, AFP)
  - usually specific to one or a few cancers, there are **no ‘ ’ markers**
  - levels fluctuate with size, extent of spread
  - limitations:
    - not for all cancers
    - usually expressed later in the course of the disease
    - may not be elevated in all cases of cancers
    - may be elevated from non-cancerous causes

Cytologic & Histologic Methods

- Since the definitive diagnosis of cancer is derived from microscopic analysis of tissue, retrieval and the study of cells is critical

Retrieval of Cells:

- Papanicolaou test
-

Study of Cells:

- light microscopy
- immunohistochemistry

Retrieval of Cells

- **Papanicolaou (Pap) Test**
  - prepared microscope slide of body secretions
  - remember, cancer cells may lack connections to neighbouring cells, thus are more likely to exfoliate and be found in surrounding fluids
  - can be performed on various fluids
    - **cervical secretions**
    - nipple drainage
    - pleural and peritoneal fluid
    - gastric washing

Retrieval of Cells

- **Tissue Biopsy**
  - removal of _____ specimen for microscopic study
    - needle biopsy
    - endoscopic
    - laparoscopic
    - surgical excisional

Simple cytology identified a malignant growth.

Is this a primary tumour?

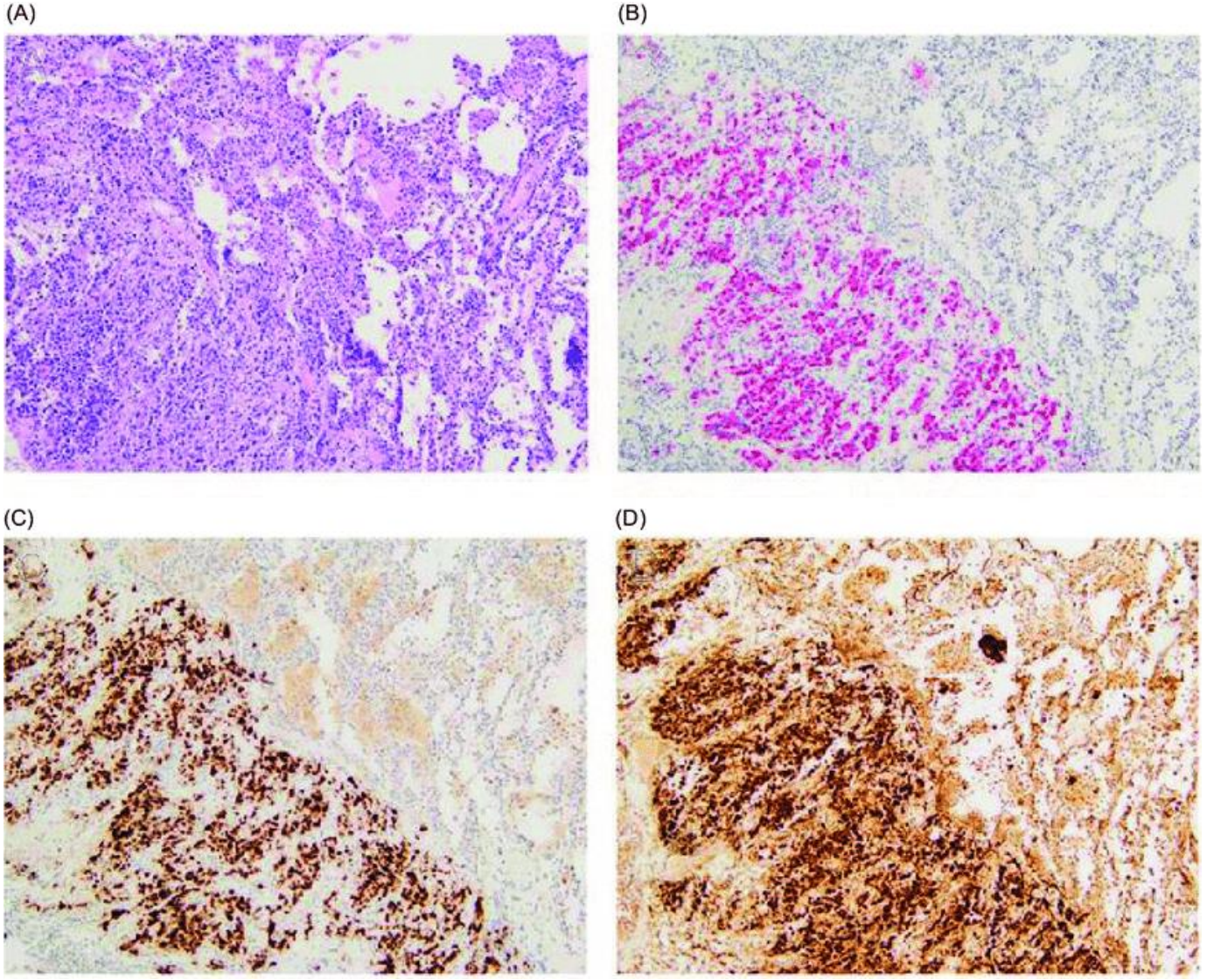
If not, *where did the cancer cells come from?*

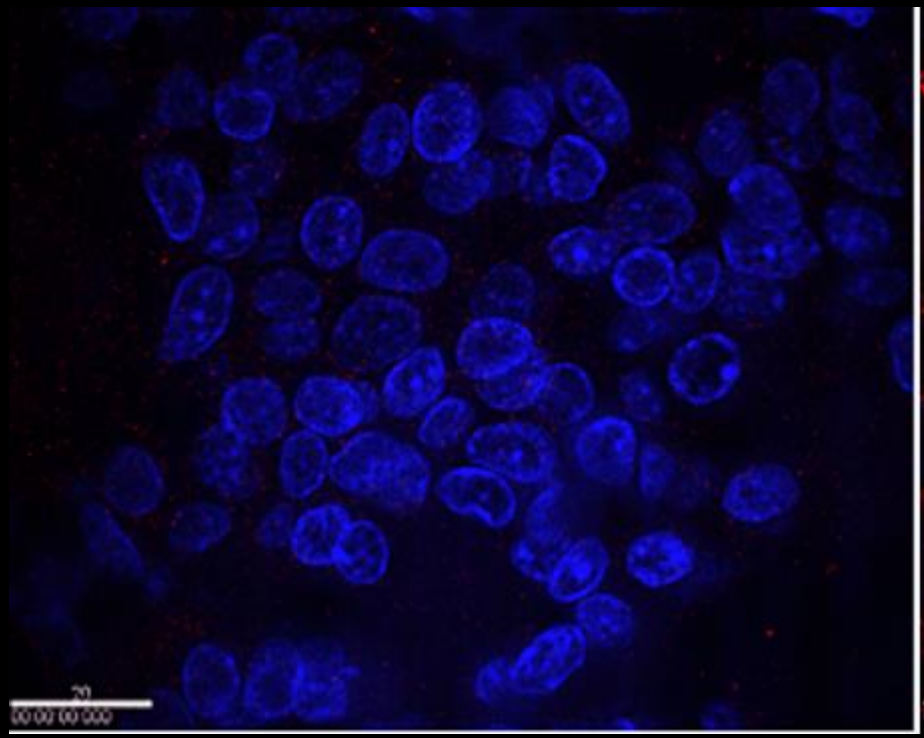
WHY DOES THIS MATTER?

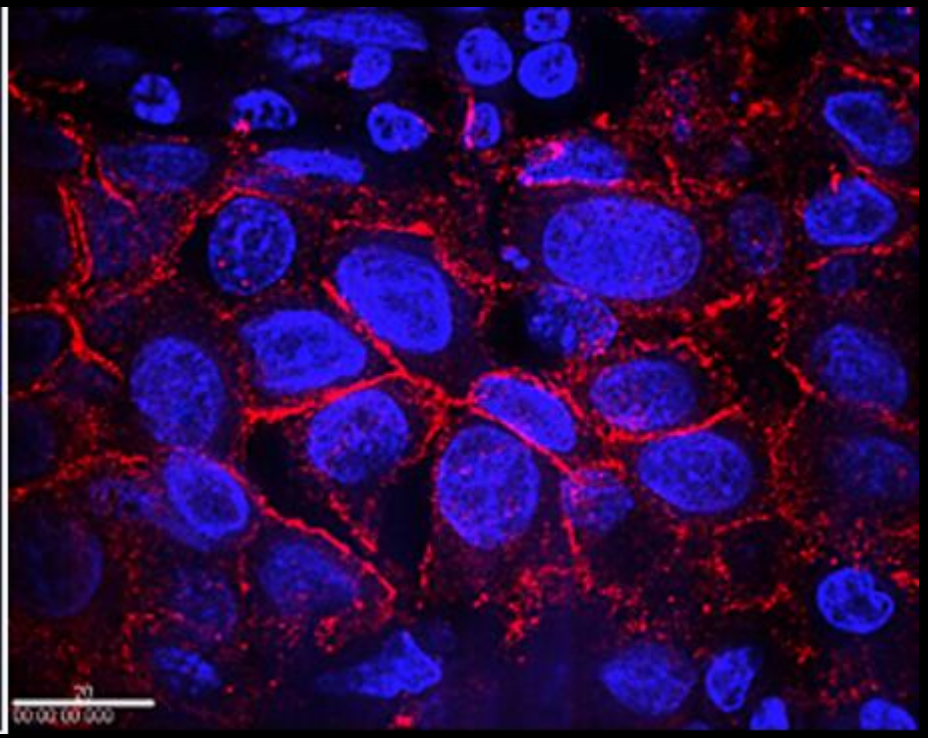
It will affect the treatment plan

- there may be _____
- some primary tumours respond differently to various treatments

In this case, a tumour was discovered in the lung. A typical H&E stain was used (A), but the origin of the malignant cells was unknown. Different antibodies were used (B-D) to determine the primary cell type (**kidney**).







15-25% of **breast cancer tumours** over-express HER2 (growth factor receptor), corresponding to aggressive disease and a poor prognosis

Sample on the right shows abundance of HER2 protein (**red**). The nucleus is labelled as a control (**blue**).

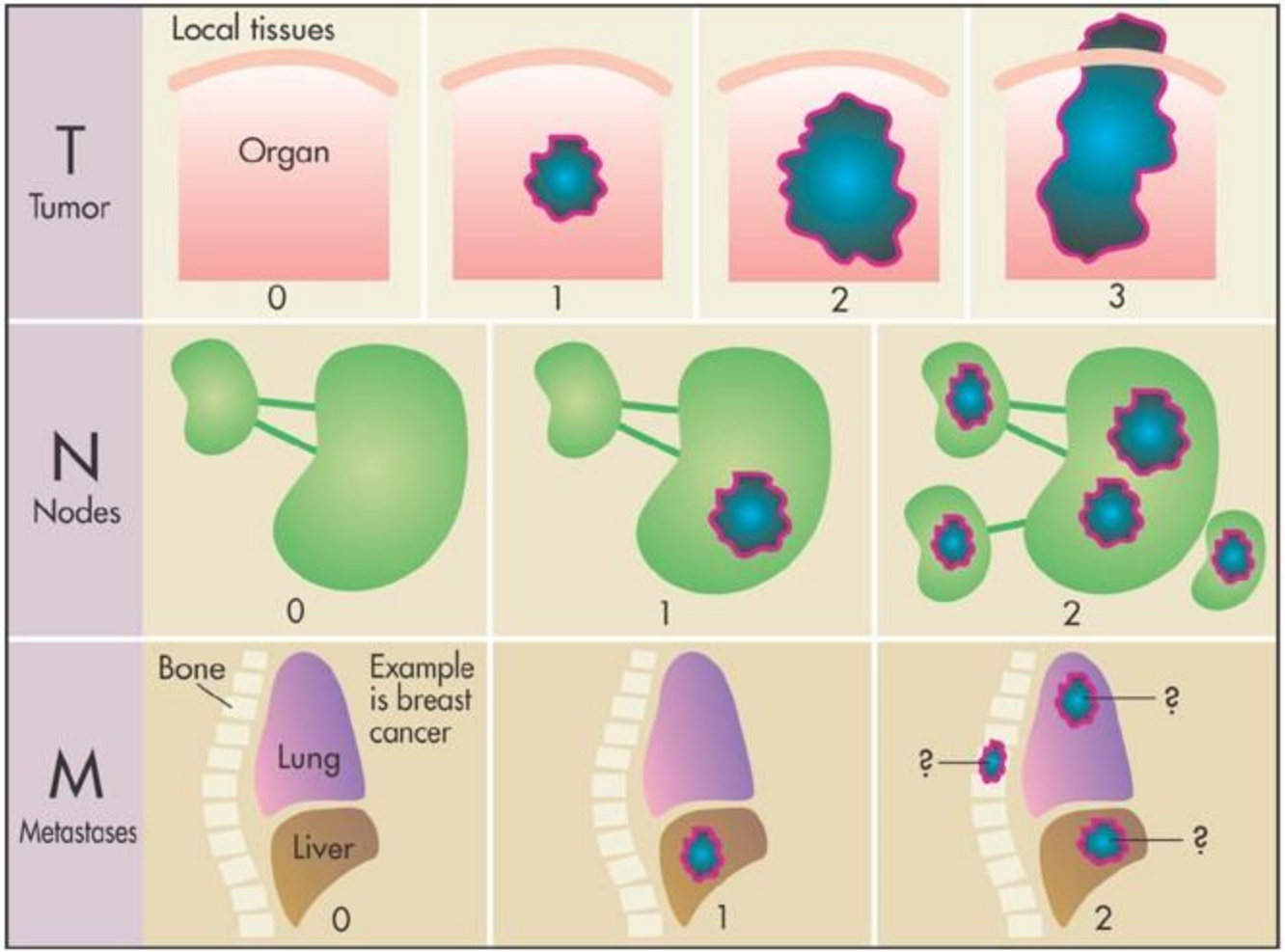
Treatment with the cancer drug trastuzumab would only be effective for the right patient (HER2 inhibitor)

Staging and Grading Tumours

Stage of a Cancer

- Indicates how much the cancer has spread (locally and systemically)
- The **biggest**
- Treatments are often based on the stage

Stage of a Cancer



T = Primary tumor; the number equals size of tumor and its local extent. The number can vary according to site.
T0 = Breast free of tumor
T1 = Lesion <2 cm in size
T2 = Lesion 2-5 cm
T3 = Skin and/or chest wall involved by invasion

N = Lymph node involvement; a higher number means more nodes are involved.
N0 = No axillary nodes involved
N1 = Mobile nodes involved
N2 = Fixed nodes involved

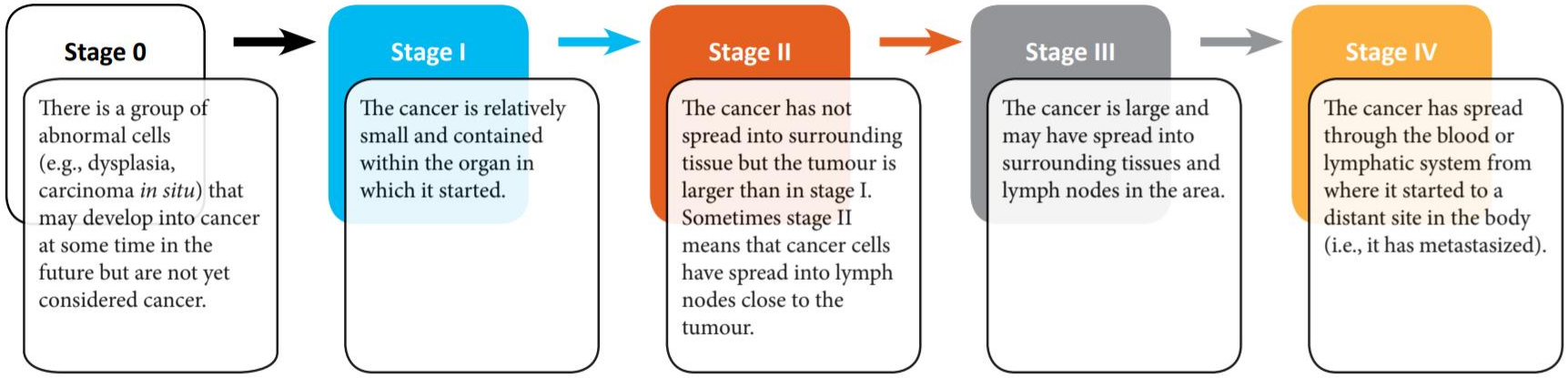
M = Extent of distant metastases.

M0 = No metastases
M1 = Demonstrable metastases
M2 = Suspected metastases

Mosby items and derived items copyright © 2004, 2000 by Mosby, Inc.

Stage of a Cancer

Canadian Cancer Society / Société canadienne du cancer



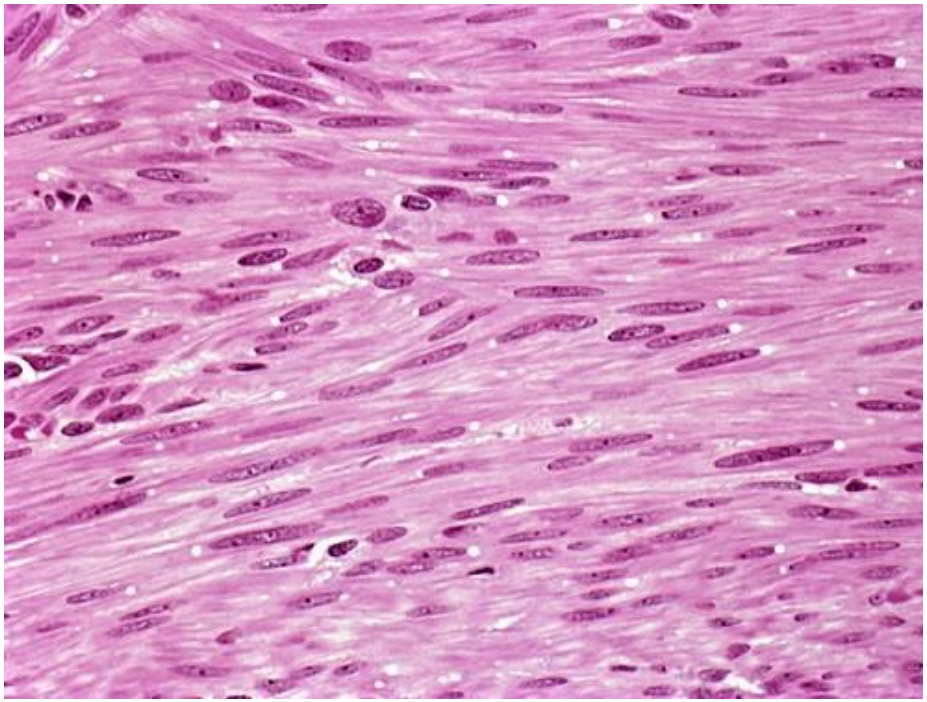
Stage 0	*In situ*
Stage I–II	Localized
Stage II–III	Regional
Stage IV	Distant

Histological Grade

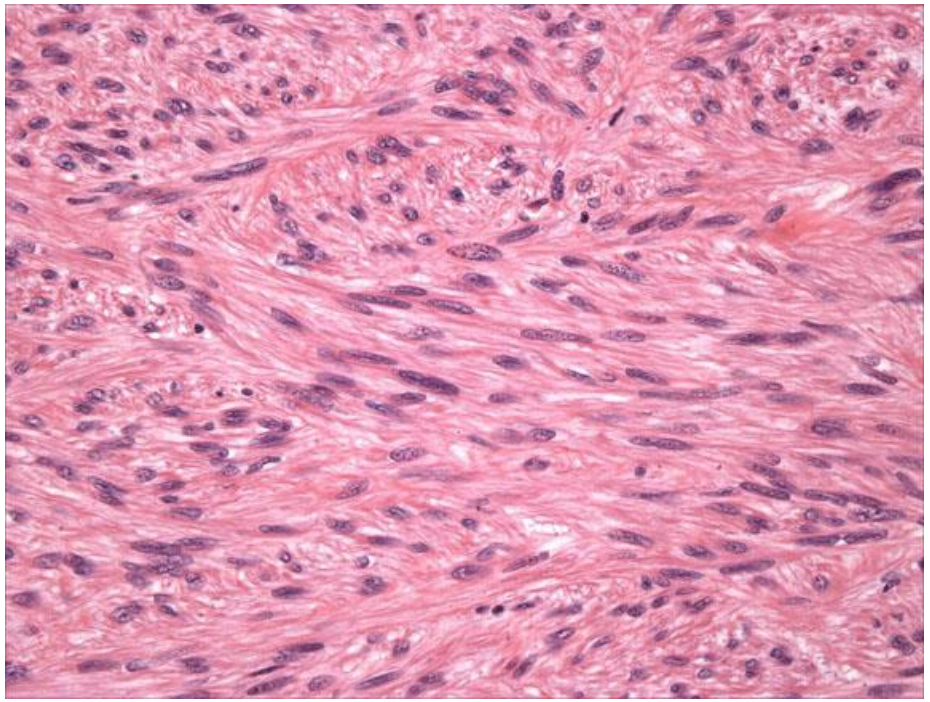
An important part of evaluating the ***aggressiveness of a tumour***.

- based on _____ - level of differentiation and number of proliferating cells (mitoses)
- **Low grade** = highly differentiated, appears like normal tissue
- **High grade** = poorly differentiated (anaplastic), pleiomorphic
- Grades 1 → 4

Histological Grade

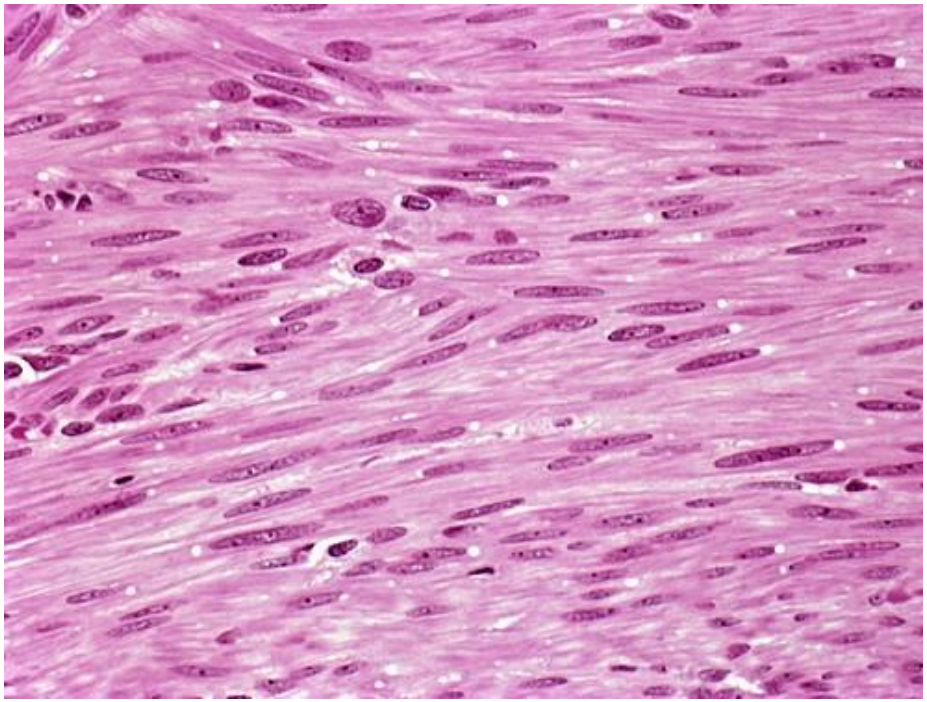


smooth muscle

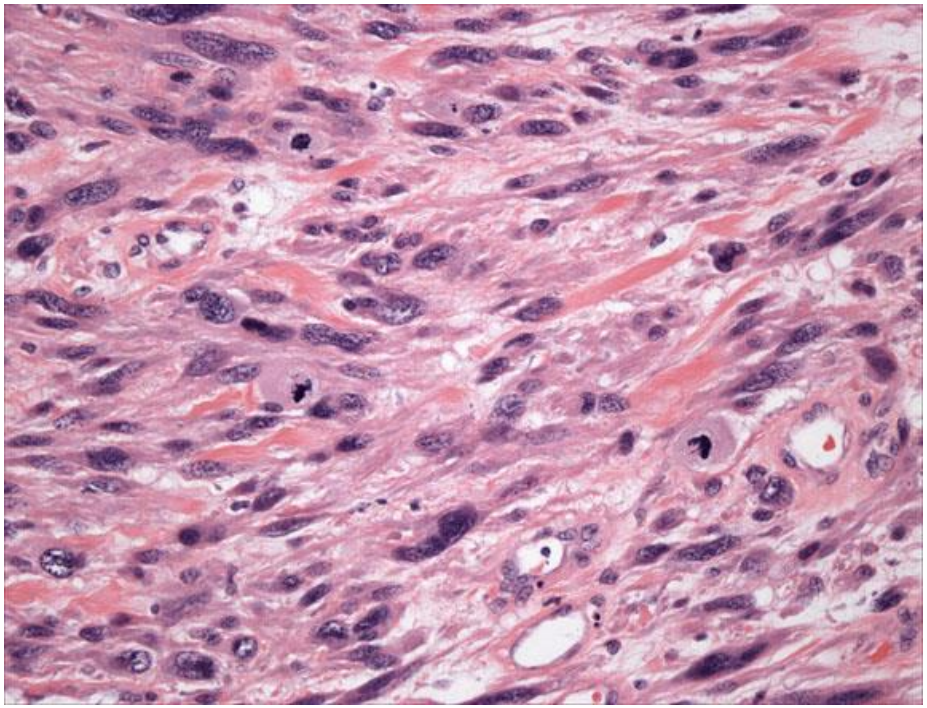


med/low grade leiomyosarcoma

Histological Grade



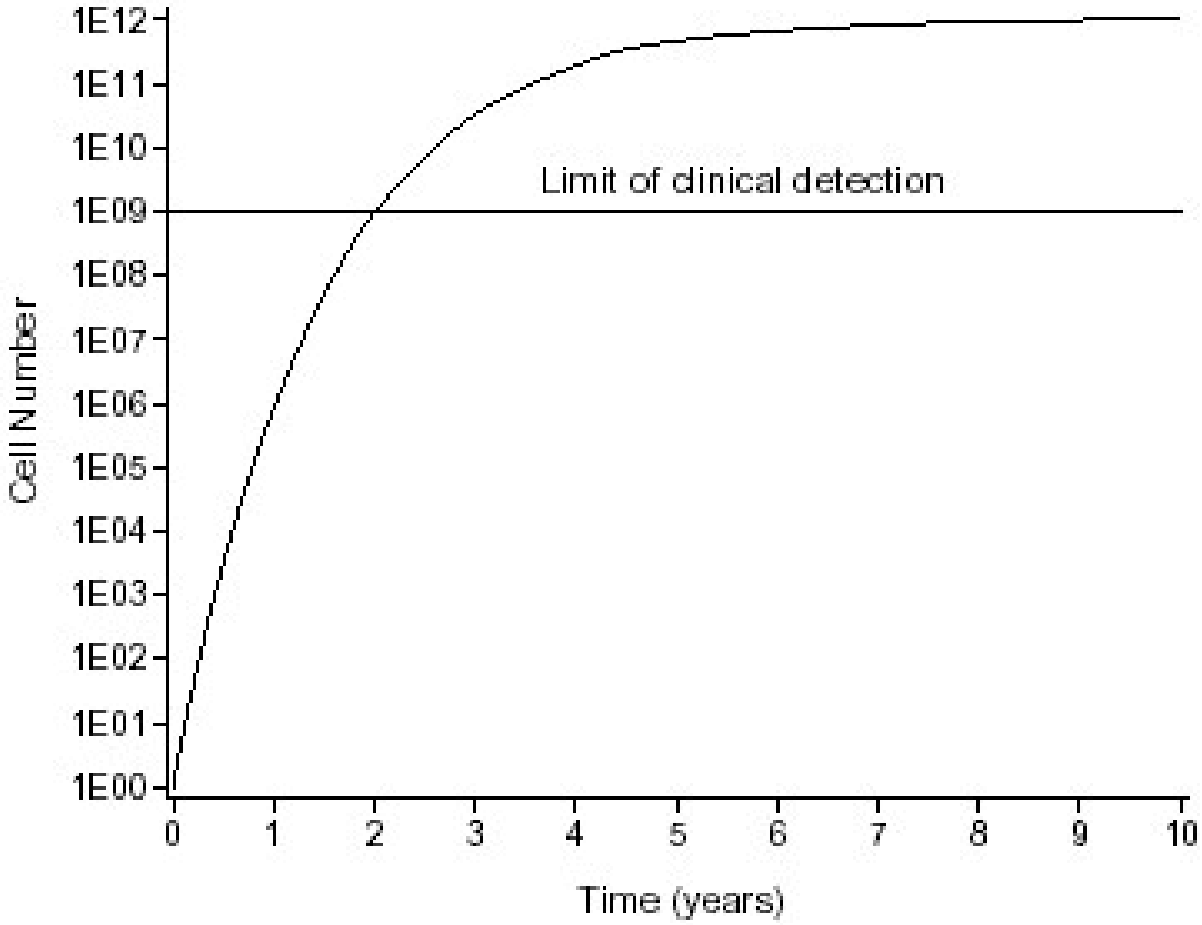
smooth muscle

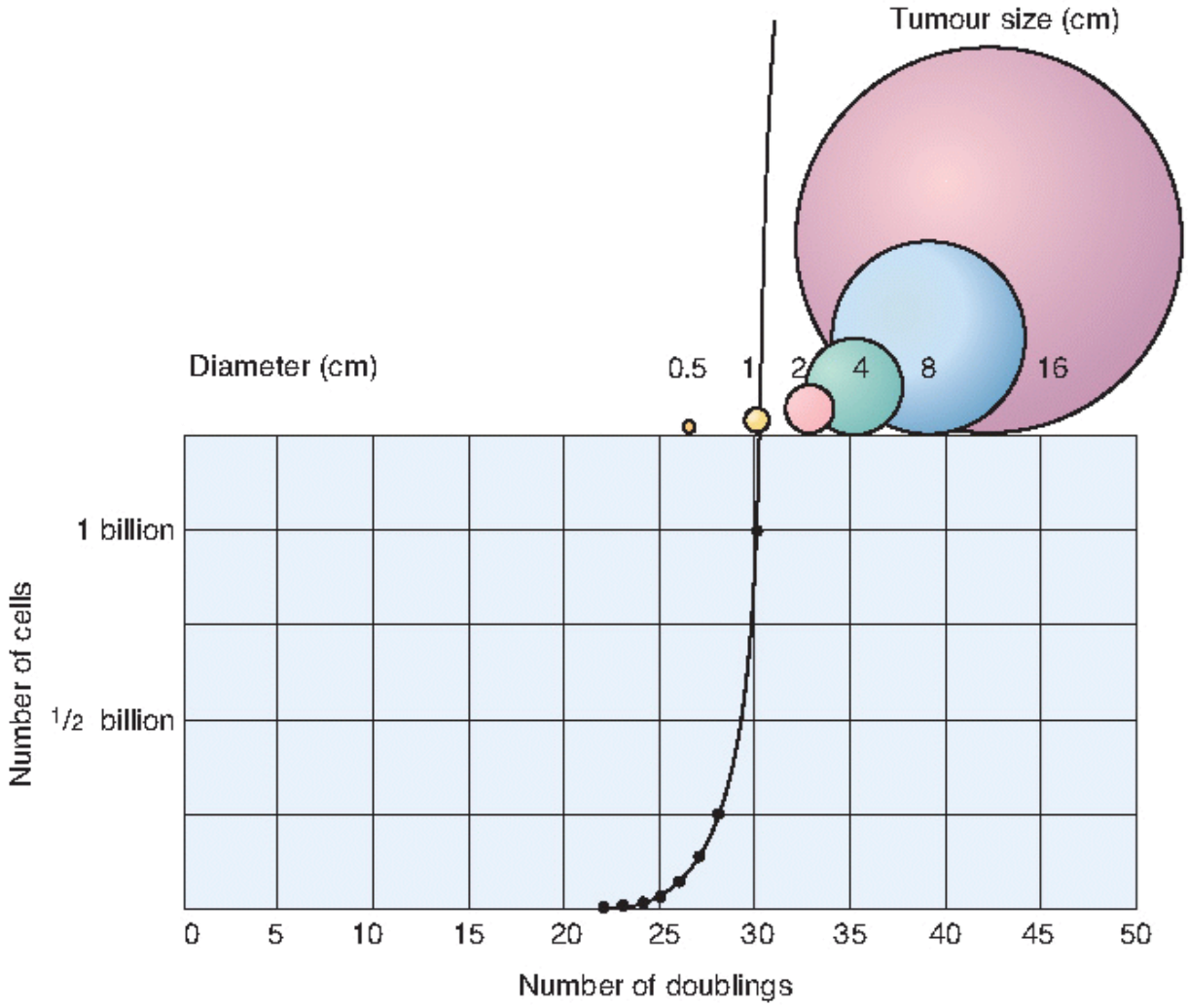


high grade leiomyosarcoma

- High grade tumours **behave very aggressively**
  - outcompete normal cells for nutrients
  - invade and destroy local tissues

Cancer Treatment


1E12
1E11
1E10
1E09
1E08
1E07
1E06
1E05
1E04
1E03
1E02
1E01
1E00
Cell Number
Limit of clinical detection
0
1
2
3
4
5
6
7
8
9
10
Time (years)


Tumour size (cm)
Diameter (cm)
0.5
1
2
4
8
16
1 billion
½ billion
Number of cells
0
5
10
15
20
25
30
35
40
45
50
Number of doublings

- **Sept. 2014** – diagnosed with
  (rare), aggressive
- 5 year survival rate is about 56% (10 yr is 39%)
  - prognosis drops significantly if it has spread
- **Staging**: 12cm x 12cm abdominal tumour
  2cm tumour found in left buttock
- **Had an abdominal CT scan in 2011**
  (tumour was too small to detect)
- **Died March 2016**

Treatment Options

- Surgery
- Radiation
- Chemotherapy
- Hormone Therapy
- Biotherapy

Adjuvant therapies

Surgery

Surgery

The roles of surgery in the management of cancer can be summarized as follows:

1. **Diagnosis**
2. **Prognosis**
3. **Prevention**
4. **Treatment**

Diagnosis and Prognosis

- Obtain tissue for **histologic diagnosis** as well as for **staging and grading (prognosis)**

- Ways of retrieving tissue include:
  - aspiration biopsy
  - needle (core) biopsy
  - incisional biopsy
  - endoscopic methods

Cancer Prevention

- Remove tissue from patients who have **congenital** (genetic) traits which place them at high risk for the development of cancer
- An oncologist and geneticist will advise on the risk of cancer, while a surgeon will advise on the risks and benefits of the proposed surgery

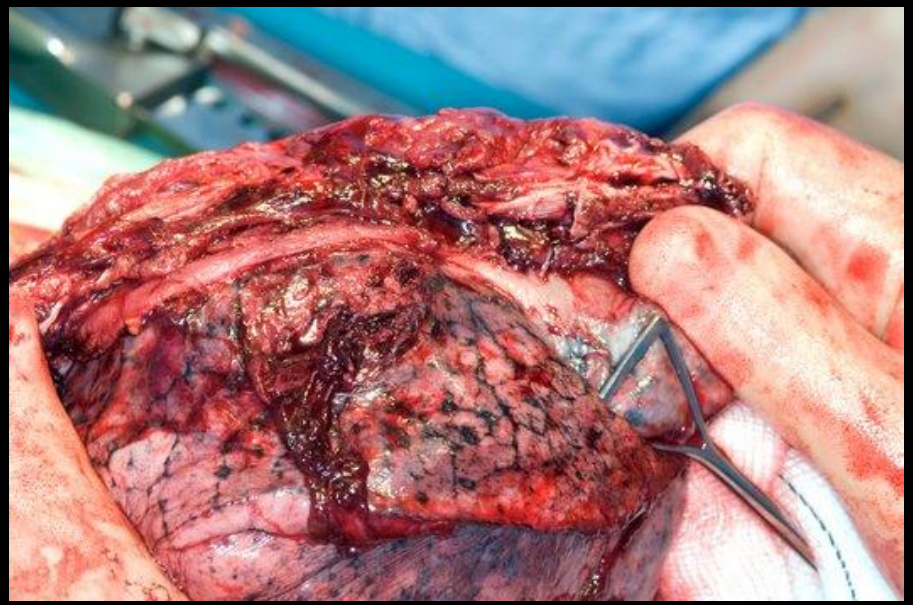
Cancer Prevention

Example. Familial adenomatous polyposis (FAP)

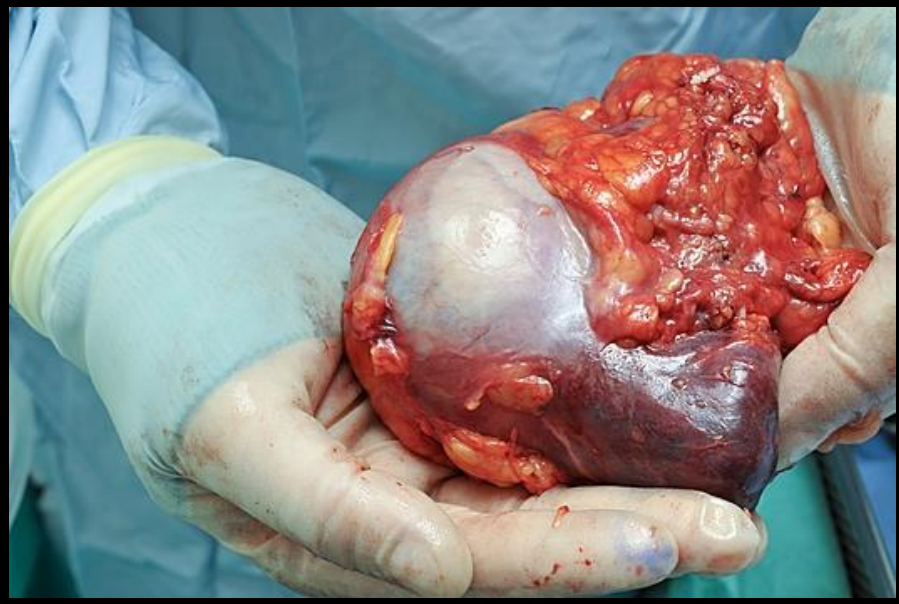
- autosomal dominant condition: 50% of patients develop adenocarcinoma of the colon by age 40
- 100% of patients develop colonic carcinoma by age 70
- total colectomy before age 20 can allow a normal life

Treatment of Cancer

- Surgery is considered the first line of treatment for **solid cancers** and can be curative if:
  - the tumour is locally contained
  - it has not metastasized
  - a large margin is removed around the tumour



lung cancer lobectomy



excised cancerous kidney

Treatment of Cancer

Surgical options include:

- **Removal of the cancer** with a surrounding margin of normal tissue to prevent local recurrence
- **surgery** (remove as much as possible – increasing chance of success of alternate therapies)
- Surgery for emergencies (bleeding, obstruction, or perforation of an organ)
- Surgery for **palliation** (for relief of pain or obstruction)
- Surgery for reconstruction or rehabilitation after the main tumour mass has been removed

Radiation

Radiation

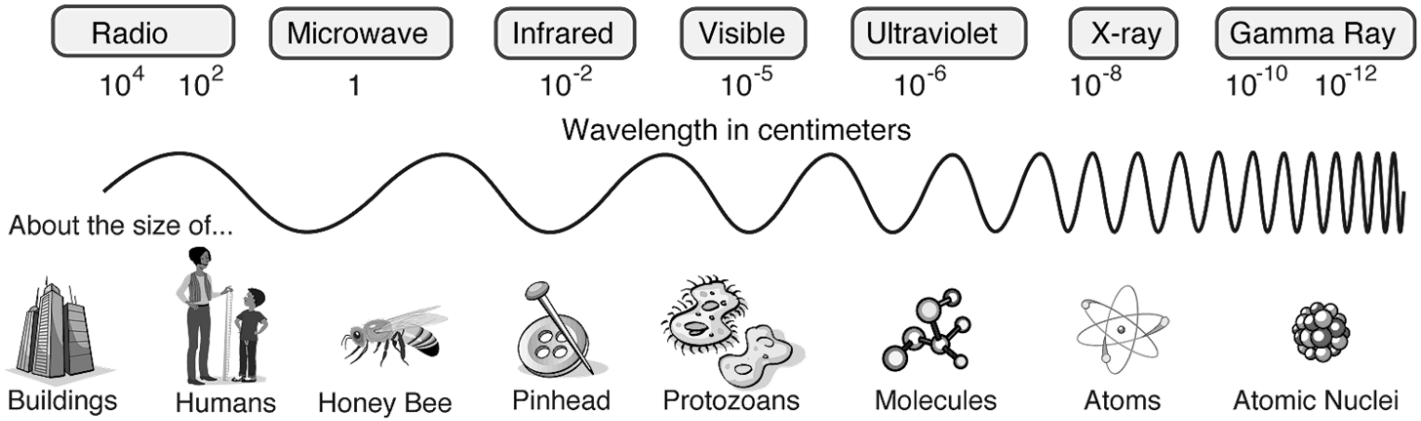
- **High energy radiation** can be **precisely aimed** at neoplastic tissue
  - energy waves absorbed by cells, ionizing molecules and leading to

How does that lead to cell death?

Radiation

- **Some cancers are more *radiosensitive* (responsive to radiation)**
  - fast-growing cancers generally more radiosensitive
  - some chemotherapy drugs increase radiosensitivity

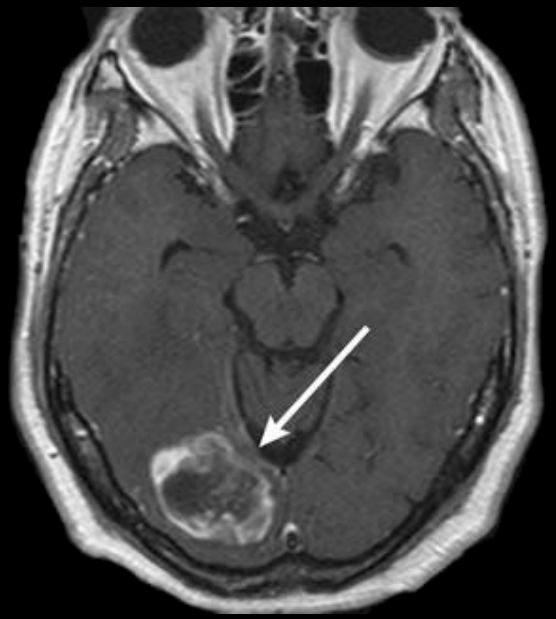
Cancers with greater blood supply are more radiosensitive. Why?



Patient under a linear accelerator that creates high energy (ionizing) X-rays that penetrate deeply

Radiation

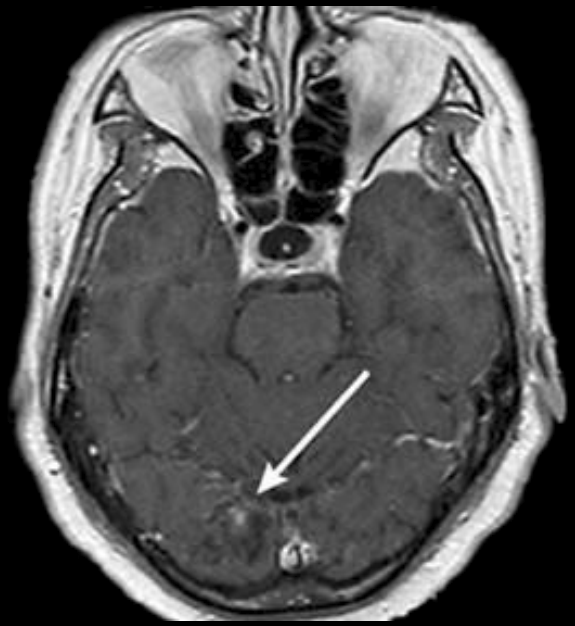
- Useful for **50-60%** of all cancer patients
- Benefits and side-effects are limited to the dividing cells in the area(s) being treated
-
  - give healthy cells opportunity to recover (total dose can be higher when divided into fractions)
  - ↑ chance cancer cell dividing
- Typically used:
  1. vital or inaccessible area
  2. in combination with surgery and/or chemo

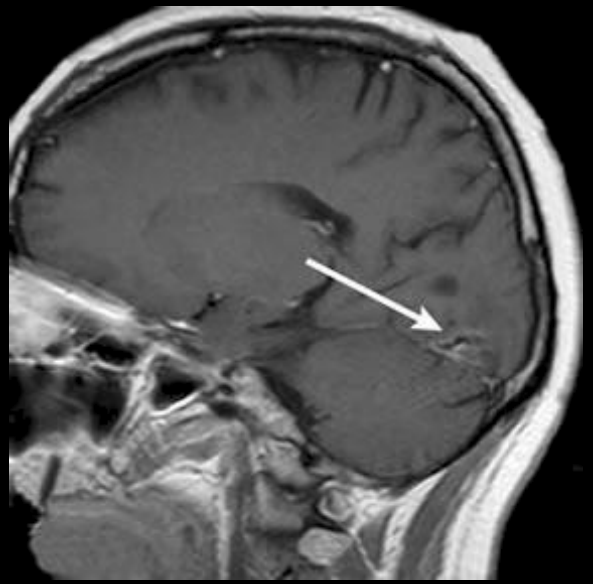


Axial View

Sagittal View

Before treatment





3 months post radiotherapy

Radiation

- Can not distinguish between cancer cells and rapidly dividing healthy cells
- Variable side-effects depending on location

GI:
- fatigue
- anorexia
- nausea
- vomiting
- diarrhea

Bone Marrow:
- fatigue
- infection
- bleeding
-

Skin:
- fatigue
- hair loss
- dryness
- mild burn

Radiation

- **Palliative treatment**
  - painful bone and soft tissue metastases
  - brain metastases
  - malignant spinal cord compression
  - airway obstruction, hemoptysis, ulcerating or bleeding lesions, etc.

60-80% of patients notice significant symptom improvement (even with 1-5 Rx's)

Chemotherapy

Chemotherapy

- Chemotherapy is a treatment used to treat forms of cancer that are:
  - widespread, **rapidly growing**
  - at high risk of micro-metastatic deposits (clinically undetectable)
  - hematological (leukemia/lymphoma)

capecitabine

- A **combination** of drugs are typically used

WHY?

Chemotherapy

- **Cytotoxic drugs** that target vital cellular machinery or metabolic pathways
  - required for normal cells as well
  - healthy cells better able to survive and repair
- Most chemotherapeutic agents interfere with **DNA or protein synthesis**

→inhibiting enzymes from be manufactured

→inhibiting DNA synthesis or mitosis

- more toxic to
- more effective in tumours with high *growth fractions*

Chemotherapy

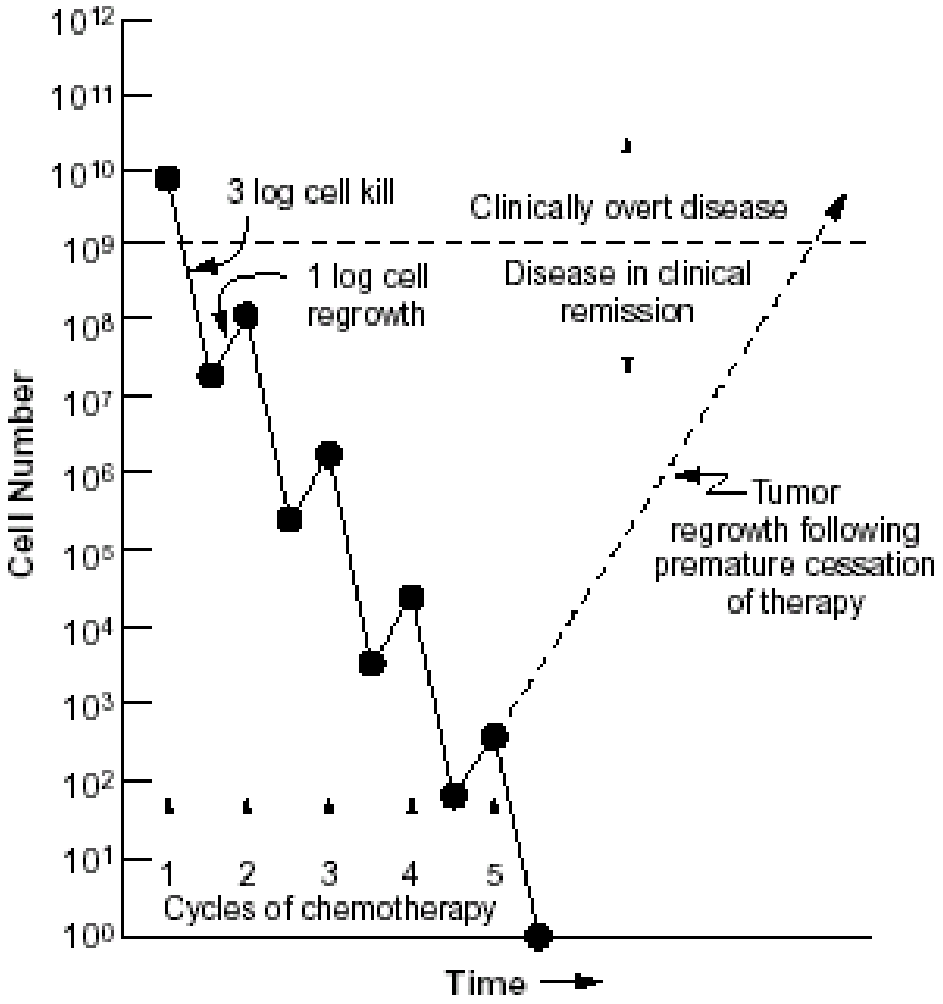
- **Therapeutic index** refers to the relative amount of the drug required to kill cancer cells compared to the dose that is harmful to normal cells
  - typically this value is (ie. the dose that kills cancer cells also kills normal cells)

Chemotherapy

- Chemotherapy cells by
  - dose of the drug kills the same **percentage** of cells rather than absolute number of cells
  - **multiple courses** of therapy are required to eradicate the tumour
- Often *some* cancer cells are not proliferating
  - not as responsive to chemotherapy
  - these cells can be stimulated later to regrow
  - responsible for a *tumour's persistence and growth*

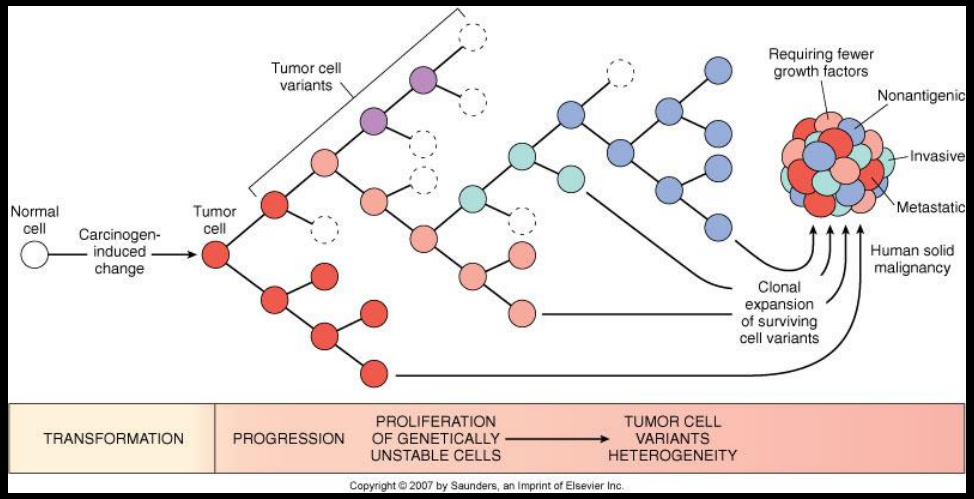
Chemotherapy kills cells by **exponential killing**

- the drug kills the same **percentage** of cells



Chemotherapy

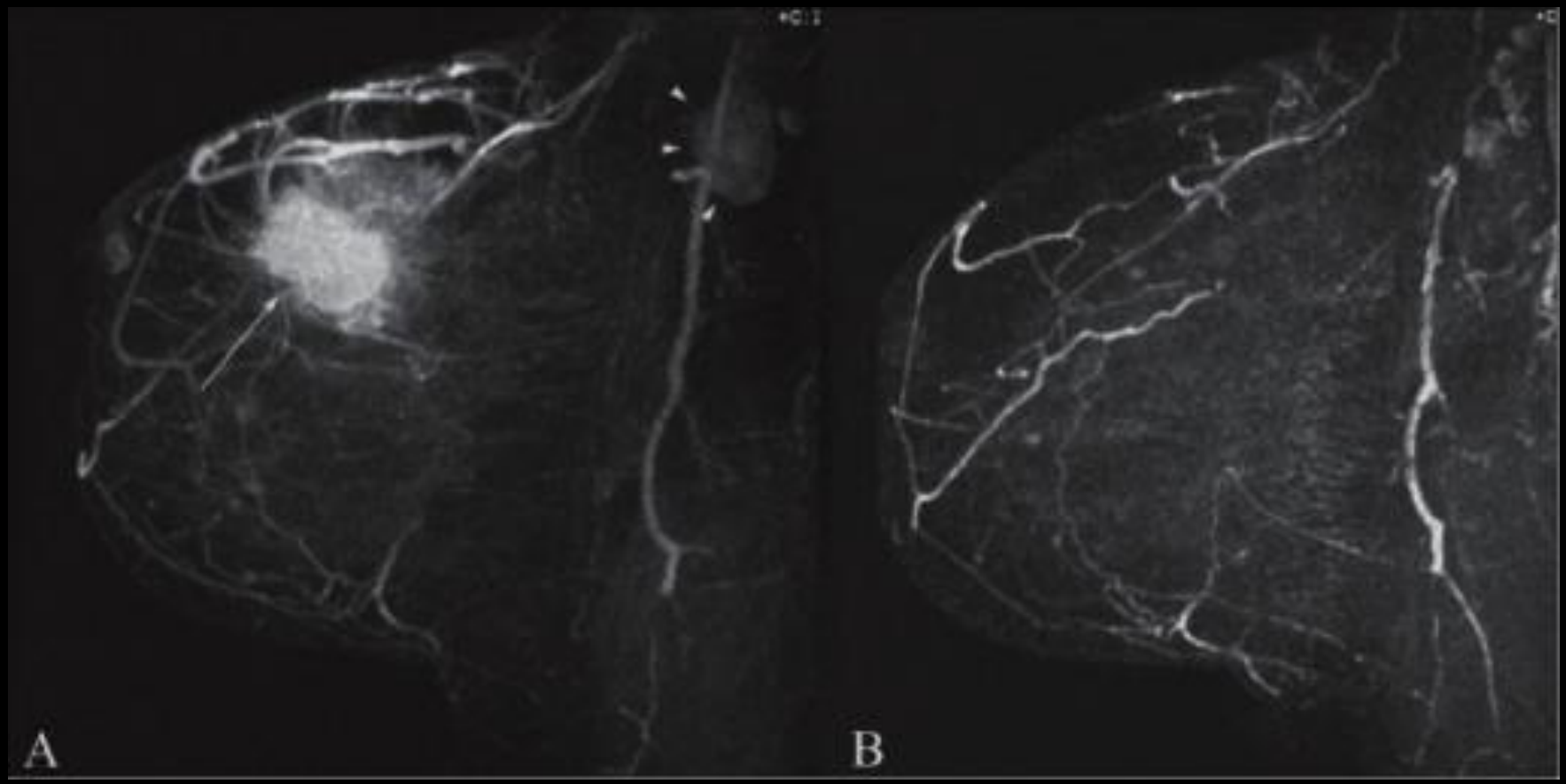
- Major issue facing chemotherapy is the development of resistance



- Variants may be more adept at evading host defenses and treatment (become aggressive)

Chemotherapy

- Combination chemotherapy can be very effective:



Adverse Effects of Chemotherapy

- In many cases, the treatment rivals the discomfort of the disease
- Compliance with therapy is directly related to the patient's perception of discomfort, which decreases with proper education
- Similar to radiation, chemotherapies target (but is systemic):
  - bone marrow
  - gastrointestinal tract
  - hair and skin
  - reproductive system

Bone Marrow

- Chemotherapy and radiation can lead to:

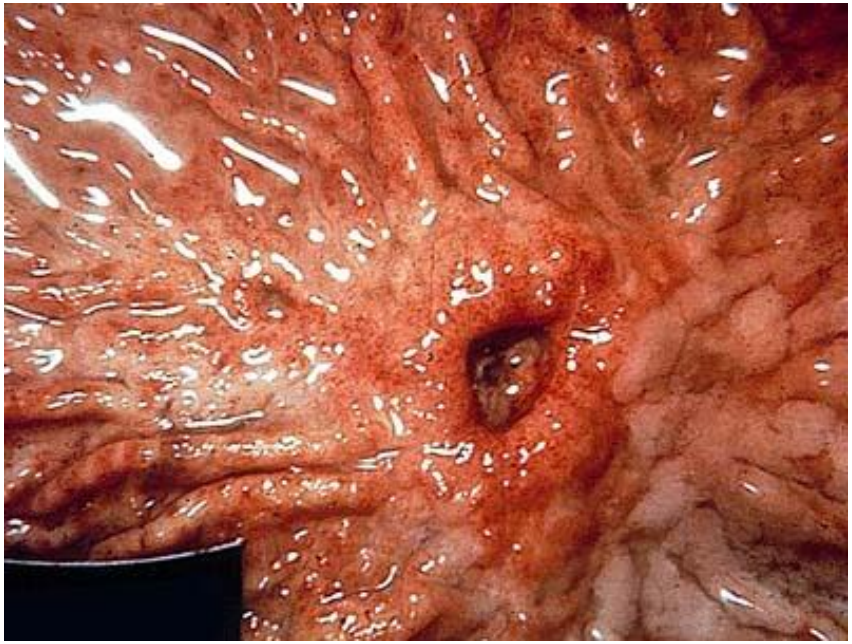
Expected outcomes?

- Can be restored through bone marrow/blood transfusions, **hematopoietic growth factors** and antibiotics can be used to help fight infections

GI Tract

- Both chemotherapy and radiation can lead to:

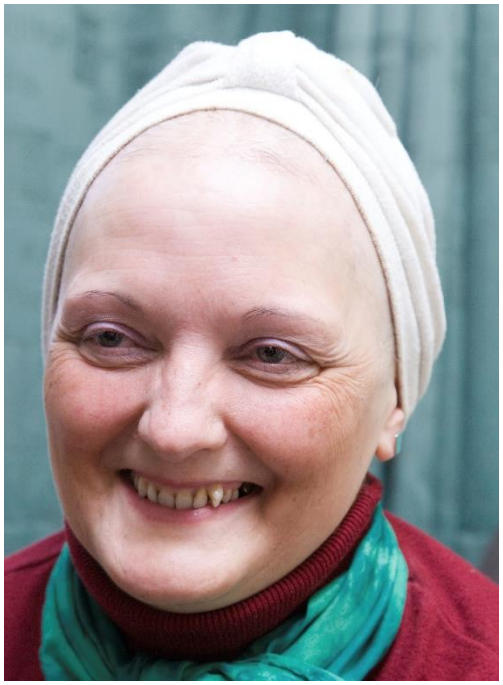
Expected outcomes?

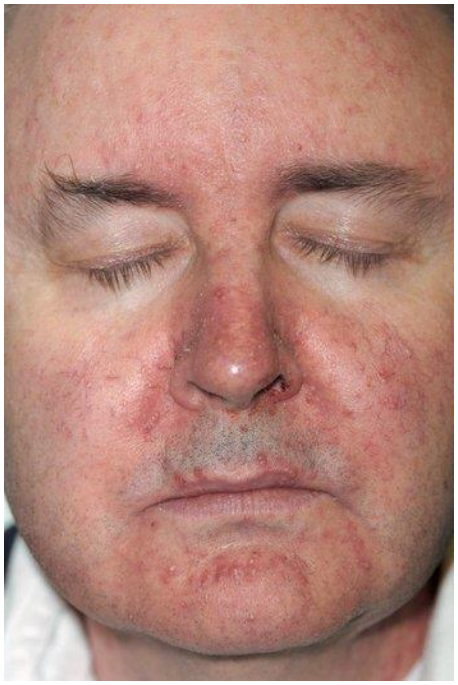


Hair and Skin

- Chemotherapy or radiation can lead to:

Expected outcomes?

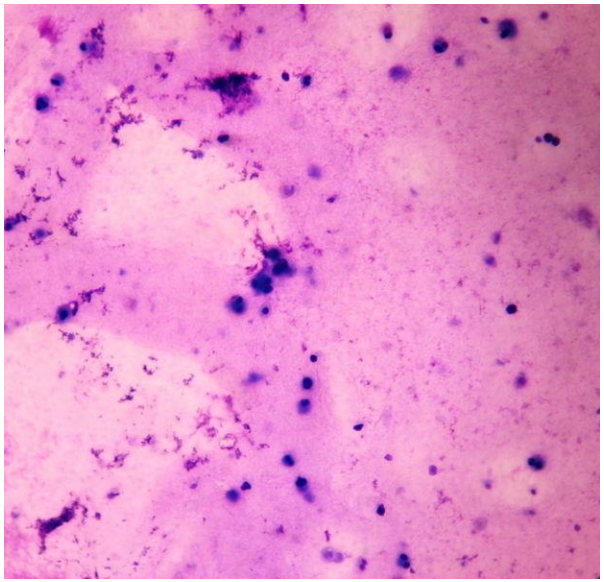




Reproductive Systems

- Chemotherapy and radiation can lead to:

Expected outcomes?



semen analysis revealing no sperm cells (azoospermia)

Hormonal Therapy

Why would this be given?

Hormone Therapy

- Some cancers are responsive or *dependent* on hormones for growth
  - breast and endometrial tissue increases growth with estrogen
- **Altering the hormonal environment** through suppression or receptor blockage may slow cancer growth
- Initially patients respond well to treatment, but cancer become resistant

Biotherapy

Biotherapy

- An attempt to increase the body's own immune response to the cancer
- Involves:

1. **Immunotherapy**
2. **Biologic Response Modifiers**

Immunotherapy

- Boost the natural immune system-mediated response against cancer (should be cancer-specific and long term)
- **Active immunotherapy:**
  - non-specific treatment with *attenuated bacterial strains* to stimulate immune system
  - inject specific to stimulate specific immune response
- **Passive immunotherapy:**
  - transfer of sensitized NK cells/ T-lymphocytes and *cytokines* (not very effective)
  - transfer **dendritic cells** presenting cancer antigens

Biologic Response Modifiers

- Three main types:

1. **Cytokines** (**interferons** and interleukins)
   - some have anti-cancer properties (next slide)
2.
   - manufactured to target tumour antigens
   - can be conjugated to toxin or drug
3. **Hematopoietic Growth Factors**
   - help replenish stock of WBCs (particularly following chemo/radiation)

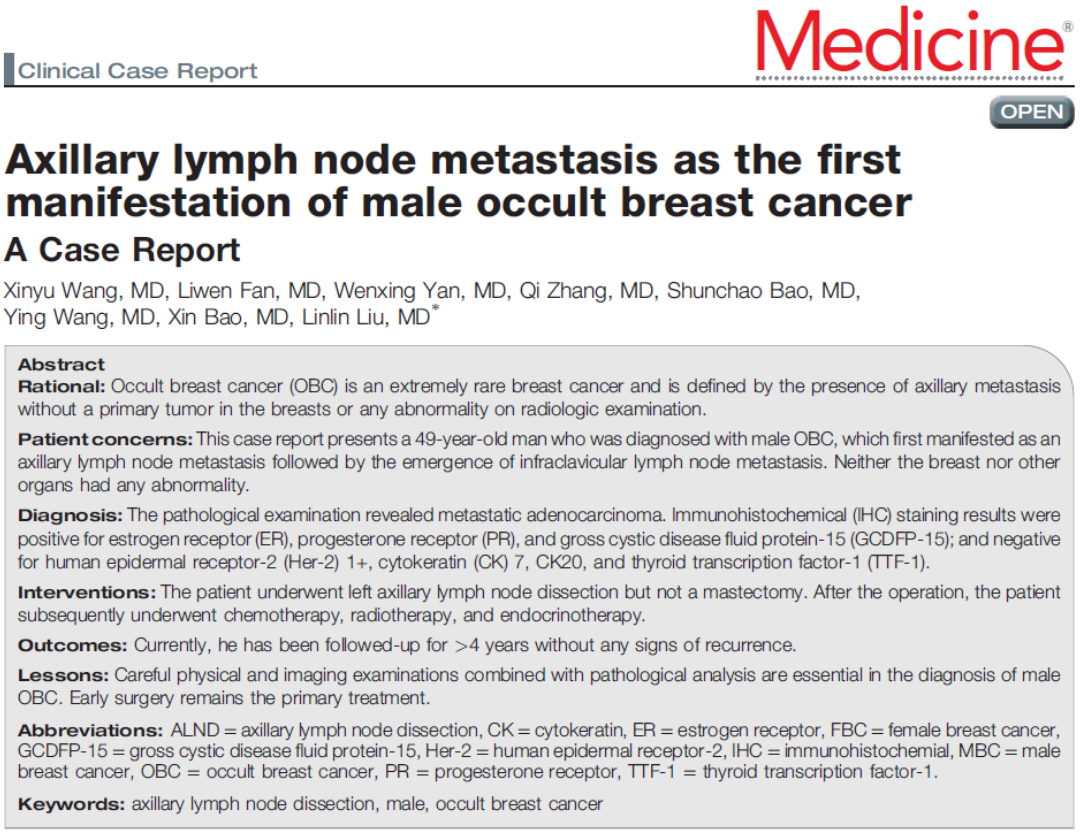
Interferons

- Interferons are used as an adjuvant therapy for a number of cancers due to their various properties:

- **anti-proliferative** (prolongs the cell cycle)
- **pro-apoptotic** (initiate cell death pathways)
- **immunostimulatic** (trigger immune response)
- (prevent new blood vessel formation)

Required Readings

- The next paper is posted:



Clinical Case Report

Medicine®

OPEN

Axillary lymph node metastasis as the first manifestation of male occult breast cancer

A Case Report

Xinyu Wang, MD, Liwen Fan, MD, Wenxing Yan, MD, Qi Zhang, MD, Shunchao Bao, MD, Ying Wang, MD, Xin Bao, MD, Linlin Liu, MD*

Abstract

Rational: Occult breast cancer (OBC) is an extremely rare breast cancer and is defined by the presence of axillary metastasis without a primary tumor in the breasts or any abnormality on radiologic examination.

Patient concerns: This case report presents a 49-year-old man who was diagnosed with male OBC, which first manifested as an axillary lymph node metastasis followed by the emergence of infraclavicular lymph node metastasis. Neither the breast nor other organs had any abnormality.

Diagnosis: The pathological examination revealed metastatic adenocarcinoma. Immunohistochemical (IHC) staining results were positive for estrogen receptor (ER), progesterone receptor (PR), and gross cystic disease fluid protein-15 (GCDFP-15); and negative for human epidermal receptor-2 (Her-2) 1+, cytokeratin (CK) 7, CK20, and thyroid transcription factor-1 (TTF-1).

Interventions: The patient underwent left axillary lymph node dissection but not a mastectomy. After the operation, the patient subsequently underwent chemotherapy, radiotherapy, and endocrinotherapy.

Outcomes: Currently, he has been followed-up for >4 years without any signs of recurrence.

Lessons: Careful physical and imaging examinations combined with pathological analysis are essential in the diagnosis of male OBC. Early surgery remains the primary treatment.

Abbreviations: ALND = axillary lymph node dissection, CK = cytokeratin, ER = estrogen receptor, FBC = female breast cancer, GCDFP-15 = gross cystic disease fluid protein-15, Her-2 = human epidermal receptor-2, IHC = immunohistochemial, MBC = male breast cancer, OBC = occult breast cancer, PR = progesterone receptor, TTF-1 = thyroid transcription factor-1.

Keywords: axillary lymph node dissection, male, occult breast cancer